

Gaussian Coefficient Calculus

**q -Binomial Kernels, Newton Bases, Bell Polynomials,
Parameter Jets, Inversion, and Asymptotics**

A consolidated companion volume to *Combinatorial Coefficient Calculus*

Consolidated from three research reports with detailed proofs

September 4, 2026

Abstract

This volume extends the basis-change and coefficient-extraction framework of *Combinatorial Coefficient Calculus* to Gaussian, or q -binomial, coefficients. It consolidates three independently written reports on the same subject into one document with a single notation, one statement of each shared result, and complete proofs. The development is organized around mathematical operations rather than a list of identities. Finite geometric products produce Gaussian kernels; logarithms of those products produce geometric power sums; ordinary Bell polynomials recover the coefficients; Newton interpolation on arbitrary nodes produces Gaussian, q -Stirling, and colored-node connection arrays; and ordinary formal composition and Lagrange inversion transport the resulting formulae to implicit series.

The shared core comprises the finite and reciprocal q -binomial theorems, Vandermonde and Chu–Vandermonde sums, Gaussian inversion with its subspace-lattice interpretation, the ramified product-to-Bell master formula, divisor-sum coefficients of Gaussian polynomials, Jackson differentiation and geometric interpolation, two explicitly distinguished q -Stirling normalizations, a weighted Bell–Riordan calculus, Lagrange inversion with finite-product input, all-order expansions at $q = 1$, a uniform deflation procedure at every root of unity, and Gaussian extrapolation filters on geometric grids. Around that core the volume develops the material that only one of the sources contained: a general node calculus with an affine-geometric master connection, colored and type- B q -Stirling arrays, a q -Dobinski identity, q -Worpitzky and Stirling–Eulerian transforms, a path form of the divided-difference chain rule, weighted Gaussian transforms and the failure of ordinary translation, the area-type q -Catalan functional equation, Good-determinant entries for coupled inverses, Dilcher divisor identities with their harmonic limit, positive linearization, weighted and differentiated finite sums, multinomial factorial-ratio criteria, three separated asymptotic regimes including a uniform double-scaling expansion, and the exact recovery of geometric-uniform convolution moments, which is the Fabius–Rvachev application.

Every theorem is proved or reduced explicitly to a proved result. Classical identities are identified as such; no priority is claimed for the synthesis. The three exact verification programs shipped with the sources are retained unchanged and are described in the closing chapter.

Relation to the canonical manuscript and to the sources

The canonical manuscript [1] develops two central mechanisms: changing to a basis adapted to an operator, and using Bell polynomials to recover coefficients from component data. It separates formal from analytic statements and keeps Bernoulli numbers distinct from Bell polynomials. This volume retains that organization and those distinctions.

Its first point of attachment is the manuscript’s Newton expansion: the additive nodes $0, 1, 2, \dots$ are replaced either by geometric nodes $a, aq, aq^2, \dots$, by q -integer nodes $[0]_q, [1]_q, [2]_q, \dots$, or by an arbitrary node list, for which the connection coefficients are complete and elementary symmetric functions of the nodes. Its second point of attachment is the exponential formula: a finite geometric alphabet has power sums $[N]_{q^r}$, so the ordinary Bell polynomials already used in the manuscript encode q -binomial identities without any q -Bell polynomial. Its third point of attachment is ordinary composition and reversion: a change to q -factorial coefficient normalization changes the matrices, not the meaning of substitution.

The three sources. This volume replaces three reports filed together on September 4, 2026: a book-length treatment organized by operations (Gaussian kernels, Bell polynomials, basis changes, parameter jets, inversion, and geometric filters); an article emphasizing the ramified master kernel, weighted transforms, the area-type q -Catalan equation, coupled inverses, and a normalization audit; and an article emphasizing the general node calculus, q -Eulerian polynomials, the divided-difference chain rule, asymptotic regimes, and the geometric-uniform moment application. Where two or three of them proved the same theorem, one statement and one proof are kept, chosen for completeness, and the others’ variant formulations are recorded as remarks or corollaries when they add a formula. Where their normalizations differed, the definitions in Chapter 1 fix the convention, and the provenance chapter records the dictionary between the sources’ symbols. Passages that were written by one source and kept verbatim keep their internal cross-references; the labels of shared results resolve to the single retained statement.

This is a standalone mathematical extension, not a replacement of the canonical manuscript and not a claim that its results have been formalized in Lean. It loads the corpus notation file `docs/fabius-notation.tex` for consistency with the other volumes, and defines the short macros it uses itself. The typography follows the manuscript’s Libertinus, olive, and coral conventions.

What “an extension” means here

The finite q -binomial theorem, Gaussian inversion, q -Chu–Vandermonde, Carlitz-type q -Stirling arrays, q -Worpitzky, divisor identities, and q -Lucas are established mathematics. They are red-erived to connect their hypotheses and normalizations. The main additional value is the common coefficient machinery: explicit all-order root-of-unity jets, full geometric-filter error formulae, the affine-geometric node connection, the path chain rule, and the uniform double-scaling expansion. No exhaustive novelty claim is made.

Contents

Abstract	i
Relation to the canonical manuscript and to the sources	ii
1 The coefficient-calculus setting	1
1.1 Rings, indices, and specialization	1
1.2 Bell and Bernoulli normalizations	1
1.3 Three distinct operations	2
1.4 The partition-profile kernel and additional notation	2
2 Gaussian polynomials and finite products	4
2.1 A division-free definition	4
2.2 Rectangle partitions, words, and flags	5
2.3 Multinomials, finite fields, and flags	5
2.4 The finite and reciprocal binomial theorems	6
2.5 A negative upper index	6
2.6 A formal infinite-binomial identity without a topology error	7
3 Convolution, inversion, and linearization	8
3.1 Two-block and many-block Vandermonde identities	8
3.2 Gaussian inversion and the subspace lattice	9
3.3 A positive linearization formula	10
3.4 Pochhammer convolution and terminating hypergeometric sums	10
3.5 Divided-power coordinates and weighted Gaussian transforms	11
3.6 Why ordinary translation fails	12
4 Symmetric functions and the Bell master formula	13
4.1 Elementary, complete, and power-sum coordinates	13
4.2 Arbitrary powers of geometric products	14
4.3 The ramified kernel	15
4.4 Composition, arbitrary powers, and parameter derivatives	15
4.5 Reciprocal powers by ordinary binomials	16
4.6 A determinant and the recovery of logarithmic data	16
5 Partial fractions and divisor identities	18
5.1 An interpolation formula for complete symmetric functions	18
5.2 A finite divisor-sum master identity	18
5.3 The ordinary harmonic limit	19
6 Newton bases on arbitrary nodes	20
6.1 The general node calculus	20
6.2 Geometric nodes and the Gaussian basis pair	21
6.3 An affine-geometric master connection	22
6.4 The q -integer lattice: the deformation of ordinary Newton expansion	22

7	Jackson differentiation and geometric Newton interpolation	24
7.1	The operator and its commutation rule	24
7.2	A geometric Taylor basis	24
7.3	Explicit stencils and divided differences	25
7.4	The iterated product rule	26
7.5	An exact Jackson chain rule through divided differences	26
8	Two q-Stirling arrays and normal ordering	28
8.1	The monic q -integer factorial basis	28
8.2	A Gaussian closed sum, with every power of q accounted for	29
8.3	Bell formulae for every diagonal	29
8.4	The normal-ordering normalization	30
8.5	Affine-geometric closed forms	30
8.6	Colored and type- B node specializations	31
8.7	A q -Dobinski identity	31
9	q-Eulerian polynomials and the Stirling–Eulerian bridge	33
10	Weighted Bell polynomials and Riordan composition	35
10.1	Gaussian convolution as a change of coefficient coordinates	35
10.2	Composition coefficients without an ambiguous “ q -Bell” convention	36
10.3	Weighted Riordan arrays	36
10.4	Why Jackson composition is a different problem	37
11	Lagrange inversion with geometric product data	38
11.1	The ordinary inversion theorem over a q -dependent field	38
11.2	A recurrence-free inverse for a product of shifted factorials	38
11.3	Optional-slot trees and a Gaussian divisibility identity	40
11.4	A genuinely shifted equation: the area-type q -Catalan series	41
11.5	A two-variable inverse with its Jacobian factor exposed	42
11.6	The Good determinant behind the formula	42
12	The coefficients of a Gaussian polynomial	44
12.1	A divisor-sum formula for each coefficient	44
12.2	A second finite formula by inclusion–exclusion	45
12.3	Finite Fourier extraction	45
13	Parameter derivatives and the expansion at one	46
13.1	Why the logarithmic parameter is the natural coordinate	46
13.2	Exact coefficient moments and probability	47
13.3	Arbitrary parameter points and Eulerian rational functions	48
14	Cyclotomic structure and all-order root-of-unity jets	50
14.1	Cyclotomic factorization and carries	50
14.2	The finite q -Lucas theorem	50
14.3	Deflating all apparent singularities	51
14.4	A second local prescription through the linear factors	53
15	Weighted sums and differentiated identities	54
15.1	Polynomial weights in the summation index	54
15.2	Alternating sums at a product zero	54
15.3	An all-order differentiation of central Vandermonde	55
16	Gaussian filters on geometric grids	57
16.1	Spectral design instead of recursive extrapolation	57
16.2	Noise amplification is bounded at a fixed geometric ratio	58
16.3	Fractional powers and arithmetic exponent lattices	59
16.4	Logarithmic errors and repeated spectral roots	59
16.5	Connection with the Fabius-function setting	60

16.6	Application: moments of geometric uniform convolutions	60
16.7	Moments from finite cumulant formulae	60
17	Multinomial coefficients and factorial ratios	63
17.1	The Gaussian multinomial extension	63
17.2	A polynomiality criterion for balanced factorial ratios	64
18	Three distinct asymptotic regimes	66
18.1	Fixed lower index and fixed base	66
18.2	Both sides large and fixed base	66
18.3	The double scaling $q = e^{-\tau/n}$	67
18.4	Explicit derivatives and multiplicative coefficients	68
19	Computation, verification, and integration	70
19.1	Choosing a representation for the task	70
19.2	Arithmetic cost and finite verification	70
19.3	The three retained verification programs	71
19.4	A numerical check of the double-scaling order	72
19.5	Normalization audit	72
19.6	Where the extension attaches	73
19.7	A theorem-level insertion map	73
19.8	Formal identities, analytic domains, and specialization	74
19.9	Directions not claimed as results here	74
19.10	Concluding perspective	75
A	A compact formula atlas	76
B	A formula-selection guide	78
C	Source provenance, notation dictionary, and reproducibility	79
C.1	Sources and consolidation	79
C.2	Notation dictionary	80
C.3	Formalization status	80
C.4	Files and commands	80
	Bibliography	81

Chapter 1

The coefficient-calculus setting

1.1 Rings, indices, and specialization

Unless stated otherwise, q is an indeterminate and calculations with quotients take place over $\mathbb{Q}(q)$, extended by the finitely many auxiliary parameters that occur. A formal series in z is an element of $K[[z]]$ for a commutative $\mathbb{Q}$ -algebra K . Composition $A(B(z))$ is defined when $B(0) = 0$; inversion under multiplication requires an invertible constant term. These statements have no convergence content.

All integer indices are nonnegative unless explicitly qualified. Empty products are one, sums over empty ranges are zero, and $0^0 = 1$ in coefficient formulae. Gaussian coefficients outside $0 \leq k \leq n$ are zero. Define

$$(a; q)_N = \prod_{j=0}^{N-1} (1 - aq^j), \quad [n]_q = \sum_{j=0}^{n-1} q^j, \quad [n]_q! = \prod_{j=1}^n [j]_q. \quad (1.1)$$

In particular, $[0]_q = 0$ and $[0]_q! = 1$. The polynomial interpretation of $[n]_q$ makes $[n]_1 = n$ immediate. The quotient $(1 - q^n)/(1 - q)$ is a useful representation, not the definition at $q = 1$. Our q -shifted factorial convention agrees with [4].

When a rational representation equals a polynomial, specialization means specializing that polynomial, after cancellation. In particular, the expression $(q; q)_n / ((q; q)_k (q; q)_{n-k})$ must not be evaluated as an uncanceled $0/0$ at a root of unity. Later we give explicit polynomial and local-series methods that do not make this mistake.

1.2 Bell and Bernoulli normalizations

The complete and partial exponential Bell polynomials are characterized by

$$\exp \left(\sum_{r \geq 1} x_r \frac{z^r}{r!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}, \quad (1.2)$$

$$\frac{1}{k!} \left(\sum_{r \geq 1} x_r \frac{z^r}{r!} \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{z^n}{n!}. \quad (1.3)$$

Thus $\mathcal{B}_0 = 1$, and

$$\mathcal{B}_n(x_1, \dots, x_n) = n! \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum r m_r = n}} \prod_{r=1}^n \frac{1}{m_r!} \left(\frac{x_r}{r!} \right)^{m_r}. \quad (1.4)$$

For $\mathcal{B}_{n,k}$, impose also $\sum m_r = k$. Expanding each exponential in (1.2) proves (1.4). In particular, a formula in Bell polynomials is a finite-sum formula, not a hidden recurrence.

Bernoulli numbers are denoted β_r , with

$$\frac{t}{e^t - 1} = \sum_{r \geq 0} \beta_r \frac{t^r}{r!}, \quad \beta_1 = -\frac{1}{2}. \quad (1.5)$$

We use $s(n, k)$ for signed ordinary Stirling numbers of the first kind and $S(n, k)$ for ordinary Stirling numbers of the second kind. Their q -versions will be defined independently, with the normalization stated in the definition.

1.3 Three distinct operations

There are three operations that should not be conflated. One may specialize the variables of a symmetric polynomial to a geometric progression. One may replace an ordinary derivative by the Jackson derivative. Or one may merely record ordinary series coefficients using $[n]_q!$ rather than $n!$. All three lead to Gaussian coefficients, but their product and composition rules are not interchangeable.

Master principle

The robust route is

finite alphabet $\longrightarrow$ power sums $\longrightarrow$ formal logarithm $\longrightarrow$ Bell coefficients.

Operator-adapted interpolation and formal reversion can then be attached to this route without guessing which ordinary factorials should acquire a subscript q .

1.4 The partition-profile kernel and additional notation

Several passages below, written originally in a slightly different notation, use the following abbreviations. We write

$$D_n = (q; q)_n = (1 - q)^n [n]_q!, \quad G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q$$

when the Gaussian polynomial is to be viewed as a function of its base, and, for a scalar Q , $G_{n,k}(Q)$ for its value at $q = Q$. A finite product of shifted factorials is often expanded through the following partition-profile kernel, which is the complete Bell polynomial in a different normalization.

For $n \geq 0$ let

$$\mathcal{P}(n) = \{(m_1, \dots, m_n) \in \mathbb{N}^n : \sum_{j=1}^n j m_j = n\}.$$

For $n = 0$ this consists of the empty tuple. Define

$$\mathcal{E}_n(L) = \sum_{\mathbf{m} \in \mathcal{P}(n)} \prod_{j=1}^n \frac{(L_j/j)^{m_j}}{m_j!}. \quad (1.6)$$

In terms of the complete exponential Bell polynomial $\mathcal{B}_n$ of (1.2),
Consequently

$$\mathcal{E}_n(L) = \frac{1}{n!} \mathcal{B}_n(0!L_1, 1!L_2, \dots, (n-1)!L_n), \quad \exp \left(\sum_{j \geq 1} \frac{L_j}{j} z^j \right) = \sum_{n \geq 0} \mathcal{E}_n(L) z^n. \quad (1.7)$$

Both statements follow by multiplying the individual exponential series. In particular, (1.6) is a definition by a finite sum, not by a recurrence. The first values provide a useful normalization check:

$$\begin{aligned}\mathcal{E}_0 &= 1, & \mathcal{E}_1 &= L_1, & \mathcal{E}_2 &= \frac{L_1^2 + L_2}{2}, & (1.8) \\ \mathcal{E}_3 &= \frac{L_1^3 + 3L_1L_2 + 2L_3}{6}, & \mathcal{E}_4 &= \frac{L_1^4 + 6L_1^2L_2 + 3L_2^2 + 8L_1L_3 + 6L_4}{24}.\end{aligned}$$

The q -dependence will usually enter through the arguments L_j , not through a new Bell-polynomial definition.

Chapter 2

Gaussian polynomials and finite products

2.1 A division-free definition

Definition 2.1. For $0 \leq k \leq n$, set

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{0 \leq i_1 < \dots < i_k < n} q^{i_1 + \dots + i_k - \binom{k}{2}}. \quad (2.1)$$

The $k = 0$ summand is the empty subset and contributes one.

Every exponent is nonnegative, because $i_j \geq j - 1$. Consequently this is a polynomial in $\mathbb{Z}_{\geq 0}[q]$, valid at every complex q .

Theorem 2.2 (Pascal, factorial, and symmetry formulae). *For $n \geq 1$, the Gaussian polynomials satisfy the first two recurrences below; the remaining identities hold for $0 \leq k \leq n$:*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q, \quad (2.2)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q, \quad (2.3)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}, \quad (2.4)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q, \quad \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (2.5)$$

The quotient identities hold in $\mathbb{Q}(q)$. For $0 \leq k \leq n$, the polynomial has degree $k(n-k)$, constant coefficient one, and leading coefficient one.

Proof. Separate the subsets in (2.1) according as they contain $n-1$. Removing that element decreases the exponent by $n-k$, proving (2.2). The factorial quotient has the same boundary values and satisfies the same recurrence: after putting the two terms over a common denominator the remaining numerator is $(1 - q^{n-k}) + q^{n-k}(1 - q^k) = 1 - q^n$. Induction therefore proves (2.4). Its symmetry gives (2.3) by replacing k with $n-k$.

The smallest subset $\{0, \dots, k-1\}$ has exponent zero, and the largest $\{n-k, \dots, n-1\}$ has exponent $k(n-k)$. Each is unique. Reflecting indices by $i_j \mapsto n-1-i_{k+1-j}$ replaces an exponent d by $k(n-k) - d$, proving reciprocity and palindromicity. $\square$

A subset gives a partition inside a k -by- $(n - k)$ rectangle by taking $\lambda_j = i_{k+1-j} - (k - j)$. These integers decrease weakly, lie between zero and $n - k$, and sum to the exponent in (2.1). The construction is reversible. This proves the rectangular-partition interpretation directly. At $q = 1$, every subset has weight one, so $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_1 = \binom{n}{k}$.

Example 2.3.

$$\left[\begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right]_q = 1 + q + 2q^2 + q^3 + q^4, \quad \left[\begin{smallmatrix} 6 \\ 3 \end{smallmatrix} \right]_q = 1 + q + 2q^2 + 3q^3 + 3q^4 + 3q^5 + 3q^6 + 2q^7 + q^8 + q^9.$$

The coefficient sums are respectively 6 and 20; the symmetry centers are respectively 2 and $9/2$.

2.2 Rectangle partitions, words, and flags

The subset sum in Equation (2.1) has two further finite models, and the multinomial generalization has a finite-field model. They are used later for positivity arguments and for the q -Eulerian transforms.

Theorem 2.4 (Three polynomial models). *For $0 \leq k \leq n$,*

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q = \sum_{0 \leq i_1 < \dots < i_k < n} q^{i_1 + \dots + i_k - \binom{k}{2}}, \quad (2.6)$$

$$= \sum_{0 \leq b_1 \leq \dots \leq b_k \leq n-k} q^{b_1 + \dots + b_k}. \quad (2.7)$$

Equivalently, it enumerates partitions fitting in a $k \times (n - k)$ rectangle by their size. It also enumerates binary words with k ones and $n - k$ zeros by the number of pairs consisting of a zero to the left of a one.

Proof. Let the first sum be $H_{n,k}$. Subsets not containing $n - 1$ contribute $H_{n-1,k}$; deleting $n - 1$ from the others decreases the exponent by $(n - 1) - (k - 1) = n - k$. Thus H satisfies Equation (2.3) with the same boundary values as G . The substitution $b_j = i_j - (j - 1)$ gives the second sum. Reversing the b_j gives a partition inside the rectangle. In the associated binary word, the j th one has exactly $i_j - (j - 1)$ zeros before it. Summing gives the stated word statistic. $\square$

Reversal of the word changes that statistic to $k(n - k)$ minus itself, which explains reciprocity without division. For example,

$$\left[\begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right]_q = 1 + q + 2q^2 + q^3 + q^4. \quad (2.8)$$

The two objects of weight 2 are the rectangle partitions $(2, 0)$ and $(1, 1)$.

2.3 Multinomials, finite fields, and flags

For $n_1 + \dots + n_r = n$, write

$$\left[\begin{smallmatrix} n \\ n_1, \dots, n_r \end{smallmatrix} \right]_q = \frac{[n]_q!}{\prod_{i=1}^r [n_i]_q!} = \prod_{i=1}^r \left[\begin{smallmatrix} n - \sum_{j < i} n_j \\ n_i \end{smallmatrix} \right]_q. \quad (2.9)$$

The right side proves polynomiality. Assigning the positions of letter 1, then letter 2, and so on shows that it enumerates words with these multiplicities by the usual inversion statistic; reversal also gives the complementary convention.

Proposition 2.5 (Subspace and flag counts). *If Q is a prime power, $G_{n,k}(Q)$ counts the k -dimensional subspaces of $\mathbb{F}_Q^n$. The multinomial in Equation (2.9), evaluated at $q = Q$, counts flags whose successive dimension increments are $n_1, \dots, n_r$.*

Proof. There are $(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1})$ ordered independent k -tuples in $\mathbb{F}_Q^n$. Each k -subspace has $(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$ ordered bases. Dividing and canceling the powers of Q gives Equation (2.4). Choose each next subspace in the quotient by the preceding one to obtain the flag product. $\square$

The indeterminate q is not itself a field size. The finite-field interpretation applies only at prime powers; the polynomial identities hold independently of it.

2.4 The finite and reciprocal binomial theorems

Theorem 2.6. *For every $N \geq 0$,*

$$(z; q)_N = \sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (2.10)$$

For $N \geq 1$, as a formal series in z ,

$$\frac{1}{(z; q)_N} = \sum_{r \geq 0} \begin{bmatrix} N + r - 1 \\ r \end{bmatrix}_q z^r. \quad (2.11)$$

When $N = 0$, the reciprocal is one.

Proof. To obtain z^k in the finite product, choose the $-zq^i$ term in precisely k factors. Its weight is $(-1)^k q^{\sum i}$; comparison with (2.1) proves (2.10).

Write $H_N(z) = 1/(z; q)_N$. Then $(1 - q^{N-1}z)H_N(z) = H_{N-1}(z)$. If $c_{N,r} = [z^r]H_N$, it follows that $c_{N,r} = c_{N-1,r} + q^{N-1}c_{N-1,r-1}$. The candidate $\begin{bmatrix} N + r - 1 \\ r \end{bmatrix}_q$ has this recurrence by (2.2), with $c_{1,r} = 1$ and $c_{N,0} = 1$. Induction on $N + r$ proves the reciprocal identity. $\square$

These classical identities are the finite and negative-power backbone of the extension; compare the standard conventions in [4, 5].

2.5 A negative upper index

The reciprocal identity (2.11) suggests extending the upper index downward. Only the upper index is extended here; the lower index stays nonnegative.

For $M \geq 1$ and $r \geq 0$ define a negative upper entry by the finite product

$$\begin{bmatrix} -M \\ r \end{bmatrix}_q := \prod_{j=0}^{r-1} \frac{1 - q^{-M-j}}{1 - q^{j+1}}.$$

Factoring each numerator gives the explicit conversion

$$\begin{bmatrix} -M \\ r \end{bmatrix}_q = (-1)^r q^{-Mr - \binom{r}{2}} \begin{bmatrix} M + r - 1 \\ r \end{bmatrix}_q. \quad (2.12)$$

Only the upper index has been extended here. This avoids the competing conventions that arise when both indices are negative; a broader treatment is available in [16].

2.6 A formal infinite-binomial identity without a topology error

Theorem 2.7. *There is a unique $R_a(z) \in \mathbb{Q}(q, a)[[z]]$ with $R_a(0) = 1$ satisfying*

$$(1 - z)R_a(z) = (1 - az)R_a(qz). \quad (2.13)$$

It is

$$R_a(z) = \sum_{r \geq 0} \frac{(a; q)_r}{(q; q)_r} z^r. \quad (2.14)$$

For complex $|q| < 1$ and $|z| < 1$, it also equals

$$R_a(z) = \frac{(az; q)_\infty}{(z; q)_\infty}. \quad (2.15)$$

Proof. Comparison of coefficients in (2.13) gives $(1 - q^r)c_r = (1 - aq^{r-1})c_{r-1}$, with $c_0 = 1$. This proves both uniqueness and (2.14). For $|q| < 1$, the infinite products converge locally uniformly in their arguments, since the tails are bounded by convergent geometric sums. The denominator has no zero for $|z| < 1$. Their ratio satisfies the same functional equation, is analytic at zero, and has constant term one. Its Taylor coefficients are therefore the coefficients already computed. Unless the series terminates, the coefficient ratio tends to one, so the series converges absolutely for $|z| < 1$; termination makes convergence immediate. $\square$

Normalization and domain

With q merely an indeterminate, the product $\prod_{j \geq 0} (1 - zq^j)$ is not defined by z -adic convergence: its linear coefficient changes at every step. Equation (2.14) or the functional equation is the formal definition. Equation (2.15) is an additional analytic realization.

Chapter 3

Convolution, inversion, and linearization

3.1 Two-block and many-block Vandermonde identities

Theorem 3.1. For $m, n, r \geq 0$,

$$\begin{bmatrix} n+m \\ r \end{bmatrix}_q = \sum_{k=\max(0, r-m)}^{\min(n, r)} q^{(n-k)(r-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} m \\ r-k \end{bmatrix}_q. \quad (3.1)$$

More generally, if $n_1 + \cdots + n_s = N$, then

$$\begin{bmatrix} N \\ r \end{bmatrix}_q = \sum_{\substack{0 \leq k_i \leq n_i \\ \sum k_i = r}} q^{\sum_{i < j} (n_i - k_i) k_j} \prod_{i=1}^s \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q. \quad (3.2)$$

Proof. Split $(-z; q)_{n+m} = (-z; q)_n (-zq^n; q)_m$ and use (2.10) with $-z$ in place of z . After dividing the coefficient of z^r by $q^{\binom{r}{2}}$, the exponent is

$$\binom{k}{2} + \binom{r-k}{2} + n(r-k) - \binom{r}{2} = (n-k)(r-k).$$

The same computation for consecutive blocks of lengths n_i gives $\sum_i \binom{k_i}{2} + \sum_{i < j} n_i k_j - \binom{r}{2} = \sum_{i < j} (n_i - k_i) k_j$, proving the general case. $\square$

In particular, symmetry and the substitution $k \mapsto n - k$ give

$$\sum_{k=0}^n q^{k^2} \begin{bmatrix} n \\ k \end{bmatrix}_q^2 = \begin{bmatrix} 2n \\ n \end{bmatrix}_q. \quad (3.3)$$

The exponent k^2 is essential; removing it does not give a Gaussian analogue of the ordinary central-binomial identity.

Corollary 3.2 (Three useful companions). For $M, N \geq 1$ and $r \geq 0$,

$$\begin{bmatrix} M+N+r-1 \\ r \end{bmatrix}_q = \sum_{k=0}^r q^{M(r-k)} \begin{bmatrix} M+k-1 \\ k \end{bmatrix}_q \begin{bmatrix} N+r-k-1 \\ r-k \end{bmatrix}_q, \quad (3.4)$$

$$\delta_{r0} = \sum_{k=0}^{\min(N, r)} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q \begin{bmatrix} N+r-k-1 \\ r-k \end{bmatrix}_q. \quad (3.5)$$

For $0 \leq k \leq n$,

$$\sum_{j=k}^n q^{j-k} \begin{bmatrix} j \\ k \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q. \quad (3.6)$$

Proof. The first formula compares coefficients in $1/(z; q)_{M+N} = 1/((z; q)_M (zq^M; q)_N)$. The second compares coefficients in $(z; q)_N / (z; q)_N = 1$. For the third, rearrange the Pascal identity as $q^{j-k} \begin{bmatrix} j \\ k \end{bmatrix}_q = \begin{bmatrix} j+1 \\ k+1 \end{bmatrix}_q - \begin{bmatrix} j \\ k+1 \end{bmatrix}_q$ and telescope. $\square$

3.2 Gaussian inversion and the subspace lattice

Theorem 3.3 (Gaussian binomial inversion). *For arbitrary sequences over $\mathbb{Q}(q)$,*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k. \quad (3.7)$$

Both identities also make sense over $\mathbb{Z}[q]$ when the sequences do.

Proof. The coefficient of a_j after substitution on the right is

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

Cancel factorials to obtain $\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q$. The remaining sum is $(1; q)_{n-j}$, after reversing its index and using symmetry. It is zero unless $n = j$, when it is one. The two triangular matrices therefore multiply to the identity. Their unit diagonal makes either one-sided inverse also the inverse on every finite truncation. $\square$

The same inverse has a geometric interpretation. For a prime power Q , the number of k -dimensional subspaces of $\mathbb{F}_Q^n$ is $\begin{bmatrix} n \\ k \end{bmatrix}_Q$. Indeed, count ordered linearly independent k -tuples and divide by the number of ordered bases of $\mathbb{F}_Q^k$:

$$\frac{\prod_{j=0}^{k-1} (Q^n - Q^j)}{\prod_{j=0}^{k-1} (Q^k - Q^j)} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

An interval of rank difference r in the subspace lattice is itself a subspace lattice of dimension r . Its incidence-algebra Möbius function is $(-1)^r Q^{\binom{r}{2}}$, because its defining convolution with the constant-one function is exactly $(1; Q)_r = 0$ for $r > 0$. This gives a direct counterpart of the canonical manuscript's use of incidence algebras for ordinary inversion.

Corollary 3.4 (Gaussian annihilator). *For an indeterminate x ,*

$$\sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q x^j = (x; q)_N. \quad (3.8)$$

In particular the sum at $x = 1$ is zero for $N > 0$. If $q \neq 0$ and $x = q^{-r}$ with $0 \leq r < N$, it is again zero.

Proof. This is Equation (2.10); in the two specializations one factor of the product is zero. $\square$

3.3 A positive linearization formula

Theorem 3.5. For $0 \leq k, \ell \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n \\ \ell \end{bmatrix}_q = \sum_{r=\max(k,\ell)}^{\min(n,k+\ell)} q^{(r-k)(r-\ell)} \begin{bmatrix} n \\ r \end{bmatrix}_q \begin{bmatrix} r \\ k \end{bmatrix}_q \begin{bmatrix} k \\ k+\ell-r \end{bmatrix}_q. \quad (3.9)$$

Proof. First let $q = Q$ be a prime power and count ordered pairs (V, W) of subspaces of dimensions k, ℓ . Set $U = V + W$, $\dim U = r$, and $J = V \cap W$, $\dim J = j = k + \ell - r$. Choose U , then $V \subseteq U$, then $J \subseteq V$, giving the three Gaussian factors. In U/J , the space V/J has dimension $k - j = r - \ell$. The possible W/J are complements of this space, of dimension $\ell - j = r - k$. Relative to one fixed complement, every other complement is the graph of a unique linear map into V/J . There are $Q^{(r-k)(r-\ell)}$ such maps. This proves the formula for every prime power. Both sides are polynomials in q ; infinitely many equal values prove the polynomial identity. $\square$

For example,

$$[n]_q^2 = [n]_q + q(1+q) \begin{bmatrix} n \\ 2 \end{bmatrix}_q. \quad (3.10)$$

This is a Gaussian factorial-moment identity. A general power analogue follows from q -Stirling basis change in Chapter 8.

3.4 Pochhammer convolution and terminating hypergeometric sums

Theorem 3.6. For $N \geq 0$,

$$(ab; q)_N = \sum_{j=0}^N \begin{bmatrix} N \\ j \end{bmatrix}_q a^j (b; q)_j (a; q)_{N-j}. \quad (3.11)$$

Consequently, wherever the displayed denominators are nonzero,

$$\sum_{j=0}^N \frac{(q^{-N}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} q^j = a^N \frac{(c/a; q)_N}{(c; q)_N}, \quad (3.12)$$

$$\sum_{j=0}^N \frac{(q^{-N}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} \left(\frac{cq^N}{a} \right)^j = \frac{(c/a; q)_N}{(c; q)_N}. \quad (3.13)$$

The last two formulae are the terminating q -Chu–Vandermonde identities.

Proof. The functional equation (2.13) implies $R_{ab}(z) = R_a(z)R_b(az)$: both sides have constant term one and satisfy $(1-z)R(z) = (1-abz)R(qz)$. Comparison of z^N in (2.14), followed by multiplication by $(q; q)_N$, proves (3.11). This proof is entirely formal.

Divide that formula by $(a; q)_N$. Direct reversal of finite products gives

$$\begin{bmatrix} N \\ j \end{bmatrix}_q \frac{(a; q)_{N-j} a^j}{(a; q)_N} = \frac{(q^{-N}; q)_j}{(q; q)_j (q^{1-N}/a; q)_j} q^j. \quad (3.14)$$

For completeness, use

$$(q^{-N}; q)_j = (-1)^j q^{-Nj + \binom{j}{2}} \frac{(q; q)_N}{(q; q)_{N-j}}$$

and

$$(aq^{N-j}; q)_j = (-a)^j q^{Nj - j(j+1)/2} (q^{1-N}/a; q)_j$$

in the left side. The powers reduce to q^j , proving (3.14). Now set the original $a = q^{1-N}/c$ and rename the original b as a . The product reversal $(x; q)_N = (-x)^N q^{\binom{N}{2}} (q^{1-N}/x; q)_N$ turns $(ab; q)_N / (a; q)_N$ into the right side of (3.12).

Finally replace q, a, c in (3.12) by q^{-1}, a^{-1}, c^{-1} . Reversing each finite product changes the summand's extra power into $(cq^N/a)^j$; the prefactor a^{-N} on the right cancels the reversal factor a^N . This yields (3.13). All calculations first take place in a rational-function field; any admissible specialization then follows. $\square$

The identities are classical; [6] records their hypergeometric normalizations. The proof above explains why they are coefficient-convolution identities before they are hypergeometric summations.

3.5 Divided-power coordinates and weighted Gaussian transforms

Gaussian inversion, Theorem 3.3, is the $c = 1$ case of a one-parameter family of transforms. The family is best seen through the two q -exponentials, which also govern the q -factorial coordinates used for composition in Chapter 10.

Define

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{[n]_q!}, \quad E_q(z) = \sum_{n \geq 0} q^{\binom{n}{2}} \frac{z^n}{[n]_q!}. \quad (3.15)$$

Using Theorem 2.7 with $a = 0$ and the finite theorem followed by its coefficientwise limit gives

$$e_q(z) = \frac{1}{((1-q)z; q)_\infty}, \quad E_q(z) = (-(1-q)z; q)_\infty, \quad e_q(z)E_q(-z) = 1. \quad (3.16)$$

For completeness, the coefficientwise limit used for the second equality is $q^{\binom{n}{2}}/D_n$: in the finite theorem, $\left[\begin{smallmatrix} N \\ n \end{smallmatrix} \right]_q = \prod_{j=0}^{n-1} (1 - q^{N-j})/D_n$ tends q -adically to $1/D_n$. Thus (3.16) also holds purely formally over $\mathbb{Q}(q)$. The functions in (3.15) use q -factorials; a convention with D_n in the denominator rescales their argument by $1 - q$.

For sequences a, b put

$$(a \star_q b)_n = \sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q a_k b_{n-k}, \quad \mathcal{G}_q(a; z) = \sum_{n \geq 0} a_n \frac{z^n}{[n]_q!}. \quad (3.17)$$

Multiplication of formal series immediately proves

$$\mathcal{G}_q(a \star_q b; z) = \mathcal{G}_q(a; z) \mathcal{G}_q(b; z). \quad (3.18)$$

Hence $\star_q$ is associative and commutative and has identity $\epsilon_n = \delta_{n0}$. Alternatively, associativity follows directly from

Theorem 3.7 (Weighted Gaussian inversion). *Let c be a scalar. The two transforms*

$$b_n = \sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q c^{n-k} a_k, \quad (3.19)$$

$$a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q c^{n-k} b_k \quad (3.20)$$

are mutually inverse. These are polynomial identities, valid over every commutative ring after assigning a value to q and c .

Proof. Over $\mathbb{Q}(q)$, the forward transform multiplies the generating series by $e_q(cz)$ and the inverse multiplies by $E_q(-cz)$. Their product is 1. For a denominator-free proof, the coefficient of a_j after both transforms is

$$c^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q = c^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_q (1; q)_{n-j}.$$

This is 1 when $n = j$ and 0 otherwise. Interchanging the transforms gives the same sum, proving both inverse compositions. Since the entire argument is a polynomial identity with integer coefficients, arbitrary specialization follows. $\square$

3.6 Why ordinary translation fails

Let T_c denote (3.19). Define the homogeneous polynomial

$$H_r(c, d; q) = \sum_{j=0}^r \begin{bmatrix} r \\ j \end{bmatrix}_q c^j d^{r-j}.$$

Associativity of convolution gives the exact composition law

$$(T_c T_d a)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q H_{n-k}(c, d; q) a_k. \quad (3.21)$$

For example,

$$H_2(c, d; q) = c^2 + (1 + q)cd + d^2,$$

which is not $(c + d)^2$ for generic commuting c, d . Thus $T_c T_d \neq T_{c+d}$ in general, and T_{-c} is not the inverse of T_c . The missing factor in the naive inverse is precisely $q^{\binom{n-k}{2}}$. This contrasts with the ordinary weighted-translation theorem in the canonical manuscript. The correct replacement is either (3.21), or the multiplicative cocycle of Theorem 3.6 when shifted-factorial kernels are used. There is, however, a genuine Gaussian binomial theorem for noncommuting variables.

Proposition 3.8 (Quantum-plane binomial theorem). *In an associative algebra with a central parameter q and $YX = qXY$,*

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k. \quad (3.22)$$

Proof. Right-multiply by $X + Y$ and use $Y^k X = q^k X Y^k$. The new coefficient is $q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$. The initial case is $n = 0$. $\square$

This theorem is consistent with the failure of ordinary scalar translation in Section 3.5: the Gaussian cross term is produced by a commutation relation, not by the ordinary expansion of commuting $c + d$.

Chapter 4

Symmetric functions and the Bell master formula

4.1 Elementary, complete, and power-sum coordinates

For a finite alphabet $X = (x_1, \dots, x_N)$, define

$$e_r(X) = \sum_{i_1 < \dots < i_r} x_{i_1} \cdots x_{i_r}, \quad h_r(X) = \sum_{i_1 \leq \dots \leq i_r} x_{i_1} \cdots x_{i_r}, \quad p_r(X) = \sum_i x_i^r.$$

Both degree-zero functions are one. Direct multiplication gives

$$\sum_{r \geq 0} e_r(X) z^r = \prod_i (1 + x_i z), \quad (4.1)$$

$$\sum_{r \geq 0} h_r(X) z^r = \prod_i (1 - x_i z)^{-1}. \quad (4.2)$$

Taking formal logarithms gives, for $r \geq 0$,

$$e_r(X) = \frac{1}{r!} \mathcal{B}_r(p_1, -1!p_2, 2!p_3, \dots, (-1)^{r-1}(r-1)!p_r), \quad (4.3)$$

$$h_r(X) = \frac{1}{r!} \mathcal{B}_r(p_1, 1!p_2, 2!p_3, \dots, (r-1)!p_r). \quad (4.4)$$

These identities follow from (1.2); no additional combinatorial normalization is involved. Also $\sum_{j=0}^r (-1)^j e_j(X) h_{r-j}(X) = \delta_{r0}$, because the two generating functions with opposite signs multiply to one.

Theorem 4.1 (Principal specialization). *For $X_N = (1, q, \dots, q^{N-1})$,*

$$p_r(X_N) = [N]_{q^r}, \quad e_k(X_N) = q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q, \quad h_k(X_N) = \begin{bmatrix} N+k-1 \\ k \end{bmatrix}_q \quad (N \geq 1). \quad (4.5)$$

Proof. The first formula is a geometric sum. The other two follow by comparing (4.1) and (4.2) with (2.10) and (2.11). $\square$

Thus a Gaussian coefficient is an ordinary symmetric polynomial at geometric nodes. In particular,

$$q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q = \frac{1}{k!} \mathcal{B}_k \left(((-1)^{r-1}(r-1)! [N]_{q^r})_{r=1}^k \right), \quad (4.6)$$

$$\begin{bmatrix} N+k-1 \\ k \end{bmatrix}_q = \frac{1}{k!} \mathcal{B}_k \left(((r-1)! [N]_{q^r})_{r=1}^k \right). \quad (4.7)$$

For instance,

$$\begin{bmatrix} N+1 \\ 2 \end{bmatrix}_q = \frac{[N]_q^2 + [N]_q^2}{2}, \quad q \begin{bmatrix} N \\ 2 \end{bmatrix}_q = \frac{[N]_q^2 - [N]_q^2}{2}.$$

The apparent halves cancel because these are symmetric-function identities over $\mathbb{Z}[q]$.

4.2 Arbitrary powers of geometric products

Theorem 4.2 (Geometric-product coefficient formula). *Let $N_\ell \geq 0$, $d_\ell \geq 1$, and let a_ℓ, α_ℓ be commuting parameters in a $\mathbb{Q}$ -algebra. Define formally*

$$F(z) = \prod_{\ell=1}^L (a_\ell z; q^{d_\ell})_{N_\ell}^{-\alpha_\ell}, \quad P_r = \sum_{\ell=1}^L \alpha_\ell a_\ell^r [N_\ell]_{q^{d_\ell}}. \quad (4.8)$$

Powers with nonintegral exponents mean $\exp(-\alpha_\ell \log(\cdot))$ with constant term one. Then

$$[z^n]F(z) = \frac{1}{n!} \mathcal{B}_n(P_1, 1!P_2, \dots, (n-1)!P_n) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum r m_r = n}} \prod_{r=1}^n \frac{P_r^{m_r}}{r^{m_r} m_r!}. \quad (4.9)$$

Proof. Each factor has a formal logarithm, and

$$\log F(z) = \sum_{\ell=1}^L \alpha_\ell \sum_{j=0}^{N_\ell-1} \sum_{r \geq 1} \frac{(a_\ell q^{d_\ell j} z)^r}{r} = \sum_{r \geq 1} P_r \frac{z^r}{r}.$$

At every fixed degree this rearranges only finitely many contributions from the finite alphabets. Apply (1.2) with $x_r = (r-1)!P_r$, then (1.4). $\square$

The first four coefficients are

$$\begin{aligned} f_1 &= P_1, & f_2 &= \frac{P_1^2 + P_2}{2}, \\ f_3 &= \frac{P_1^3 + 3P_1P_2 + 2P_3}{6}, \\ f_4 &= \frac{P_1^4 + 6P_1^2P_2 + 3P_2^2 + 8P_1P_3 + 6P_4}{24}. \end{aligned}$$

Negative integral α_ℓ encode numerator products, positive integral ones encode repeated reciprocal products, and arbitrary parameters encode binomial powers. Several bases q^{d_ℓ} require no new machinery.

Corollary 4.3 (A two-base convolution). *For $M, N \geq 1$,*

$$\begin{aligned} \sum_{j=0}^n a^j b^{n-j} \begin{bmatrix} M+j-1 \\ j \end{bmatrix}_q \begin{bmatrix} N+n-j-1 \\ n-j \end{bmatrix}_{q^d} \\ = \frac{1}{n!} \mathcal{B}_n \left(((r-1)!(a^r[M]_{q^r} + b^r[N]_{q^{dr}}))_{r=1}^n \right). \end{aligned} \quad (4.10)$$

Proof. Expand the two reciprocal products using (2.11) and multiply. Alternatively, apply Theorem 4.2 directly to their product. Equating the two coefficients proves the identity. $\square$

4.3 The ramified kernel

Theorem 4.2 allows each factor its own base q^{d_ℓ} . A factor may also depend on a power z^d of the series variable. The following statement, in the partition-profile notation of Section 1.4, records the resulting ramification rule, which cannot be omitted; it also covers infinite products through their logarithms.

For a finite $N \geq 0$ define

$$S_N(t) = \sum_{j=0}^{N-1} t^j. \quad (4.11)$$

This polynomial, rather than $(1 - t^N)/(1 - t)$, is the preferred expression at $t = 1$. When $N = \infty$, write $S_\infty(t) = (1 - t)^{-1}$ with the formal or analytic interpretation of Theorem 2.7.

Theorem 4.4 (Ramified shifted-factorial coefficient calculus). *Let d_r, h_r be positive integers, $N_r \in \mathbb{N} \cup \{\infty\}$, and let a_r, α_r be commuting parameters, $1 \leq r \leq s$. Set*

$$P(z) = \prod_{r=1}^s (a_r z^{d_r}; q^{h_r})_{N_r}^{\alpha_r}, \quad (4.12)$$

and define, by finite sums for each $j \geq 1$,

$$L_j = - \sum_{\substack{1 \leq r \leq s \\ d_r | j}} d_r \alpha_r a_r^{j/d_r} S_{N_r} \left(q^{h_r j/d_r} \right). \quad (4.13)$$

Then for every $n \geq 0$ and every scalar β ,

$$[z^n] P(z)^\beta = \mathcal{E}_n(\beta L) = \sum_{\mathbf{m} \in \mathcal{P}(n)} \prod_{j=1}^n \frac{(\beta L_j/j)^{m_j}}{m_j!}. \quad (4.14)$$

Here infinite factors are defined by their logarithms over $\mathbb{Q}(q)$; finite factors need no such qualification.

Proof. For one finite factor,

$$\log(a z^d; q^h)_N = - \sum_{\ell \geq 1} \frac{a^\ell}{\ell} S_N(q^{h\ell}) z^{d\ell}.$$

The same formula defines an infinite factor with S_∞ . In degree j , a factor contributes only if $d \mid j$; writing $\ell = j/d$ makes its contribution $-da^{j/d} S_N(q^{hj/d})/j$. Summing the exponents α_r yields $\log P = \sum_{j \geq 1} L_j z^j/j$. Equation (1.7) proves (4.14). All coefficient extractions use only $L_1, \dots, L_n$. $\square$

4.4 Composition, arbitrary powers, and parameter derivatives

Corollary 4.5 (Composition with a unit product). *If $H(u) = \sum_{k \geq 0} h_k u^k$ and P is as in Theorem 4.4, then*

$$[z^n] H(P(z) - 1) = \sum_{k=0}^n h_k \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} \mathcal{E}_n(jL). \quad (4.15)$$

If $M(\mathbf{m}) = \sum_j m_j$, then

$$[z^n] P(z)^\beta (\log P(z))^r = \sum_{\mathbf{m} \in \mathcal{P}(n)} (M(\mathbf{m}))_r \beta^{M(\mathbf{m})-r} \prod_{j=1}^n \frac{(L_j/j)^{m_j}}{m_j!}, \quad (4.16)$$

where terms with $M(\mathbf{m}) < r$ are zero, so no negative power of β is used.

Proof. Expand $(P - 1)^k = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} P^j$. Since $P - 1$ has positive z -order, only $k \leq n$ can contribute. For the second formula differentiate (4.14) r times in β ; its degree in β is at most n . $\square$

For example, put

$$P(z) = \frac{(-z^2; q^2)_M^\gamma}{(az; q)_N^\alpha}.$$

The logarithmic inputs are explicitly

$$L_j = \alpha a^j S_N(q^j) - \mathbf{1}_{2|j} 2\gamma (-1)^{j/2} S_M(q^j). \quad (4.17)$$

Thus all powers, logarithmic factors, and compositions of this mixed even/ordinary product are evaluated by one finite kernel. This example also makes clear why the ramification factor d_r in (4.13) cannot be omitted.

4.5 Reciprocal powers by ordinary binomials

For a single reciprocal factor, the Bell form of Theorem 4.2 has an elementary companion. Write λ for the exponent.

For a formal parameter λ , Theorem 4.2 gives

$$[z^n](z; q)_N^{-\lambda} = \sum_{\sum j m_j = n} \prod_{j=1}^n \frac{\lambda^{m_j} [N]_{q^j}^{m_j}}{j^{m_j} m_j!}. \quad (4.18)$$

It is polynomial in λ of degree at most n . With λ a positive integer, one may instead expand each factor ordinarily:

$$[z^n](z; q)_N^{-\lambda} = \sum_{d_0 + \dots + d_{N-1} = n} q^{\sum_{j=0}^{N-1} j d_j} \prod_{j=0}^{N-1} \binom{\lambda + d_j - 1}{d_j}. \quad (4.19)$$

The proof is multiplication of $(1 - q^j z)^{-\lambda}$; hence Equations (4.18) and (4.19) form an explicit identity between a partition sum and an ordinary-binomial weighted sum. More general mixed ratios require only the signed power sums L_j of Theorem 4.2. In particular, negative α_ℓ introduce numerator factors without changing the coefficient mechanism.

4.6 A determinant and the recovery of logarithmic data

Proposition 4.6. *If $F(z) = \exp(\sum_{r \geq 1} P_r z^r / r) = \sum f_n z^n$, then*

$$f_n = \frac{1}{n!} \det \begin{pmatrix} P_1 & -1 & 0 & \cdots & 0 \\ P_2 & P_1 & -2 & \cdots & 0 \\ P_3 & P_2 & P_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -(n-1) \\ P_n & P_{n-1} & \cdots & P_2 & P_1 \end{pmatrix}. \quad (4.20)$$

Conversely, put $x_j = j! f_j$. Then

$$P_n = \frac{1}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}). \quad (4.21)$$

Proof. The equation $F' = (\sum_{r \geq 1} P_r z^{r-1})F$ gives $nf_n = \sum_{r=1}^n P_r f_{n-r}$. If D_n is the determinant above, expansion along its last row gives

$$D_n = \sum_{r=1}^n \frac{(n-1)!}{(n-r)!} P_r D_{n-r}, \quad D_0 = 1.$$

The superdiagonal products supply the factorial ratio, and the Laplace signs cancel their signs. Therefore $D_n/n!$ satisfies the same recurrence and initial value as f_n . For the inverse formula, expand $\log F = \sum_{k \geq 1} (-1)^{k-1} (F-1)^k/k$, use (1.3), and compare z^n with P_n/n . $\square$

This is precisely the moment–cumulant mechanism in the canonical manuscript, with geometric power sums serving as the logarithmic data.

Chapter 5

Partial fractions and divisor identities

5.1 An interpolation formula for complete symmetric functions

Lemma 5.1. *If $x_1, \dots, x_N$ are distinct and nonzero, then*

$$\prod_{i=1}^N \frac{1}{1 - x_i z} = \sum_{i=1}^N \frac{x_i^{N-1}}{\prod_{j \neq i} (x_i - x_j)} \frac{1}{1 - x_i z}. \quad (5.1)$$

Consequently,

$$h_r(x_1, \dots, x_N) = \sum_{i=1}^N \frac{x_i^{r+N-1}}{\prod_{j \neq i} (x_i - x_j)}. \quad (5.2)$$

Proof. Multiply the proposed rational identity by $\prod_j (1 - x_j z)$. Both sides are polynomials of degree at most $N - 1$; at $z = 1/x_i$, only one summand on the right survives, and it equals one. Agreement at N distinct points proves the identity. Expanding each geometric denominator and comparing coefficients proves (5.2). $\square$

The divided-difference expression on the right has removable singularities when nodes coincide; the left side supplies the polynomial continuation. This is a simple instance of why a rational summation formula can encode a polynomial.

5.2 A finite divisor-sum master identity

Theorem 5.2 (Dilcher-type identity and Bell form). *For $N \geq 1$, $r \geq 0$, and indeterminate q , let $u_j = q^j / (1 - q^j)$. Then*

$$\sum_{j=1}^N (-1)^{j-1} \frac{q^{\binom{j}{2} + rj}}{(1 - q^j)^r} \left[\begin{matrix} N \\ j \end{matrix} \right]_q = h_r(u_1, \dots, u_N) \quad (5.3)$$

$$\begin{aligned} &= \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} \frac{q^{i_1 + \dots + i_r}}{(1 - q^{i_1}) \dots (1 - q^{i_r})} \\ &= \frac{1}{r!} \mathcal{B}_r \left(\left((s-1)! \sum_{j=1}^N \frac{q^{js}}{(1 - q^j)^s} \right)_{s=1}^r \right). \end{aligned} \quad (5.4)$$

For $r = 0$, every right-hand side is interpreted as one.

Proof. Apply Theorem 5.1 to $q, q^2, \dots, q^N$. For the node q^j , the residue coefficient is

$$\frac{1}{\prod_{i \neq j} (1 - q^{i-j})} = \frac{(-1)^{j-1} q^{\binom{j}{2}}}{(q; q)_{j-1} (q; q)_{N-j}}.$$

Multiplication by $(q; q)_N$ therefore yields

$$\frac{(q; q)_N}{(zq; q)_N} = \sum_{j=1}^N (-1)^{j-1} q^{\binom{j}{2}} (1 - q^j) \begin{bmatrix} N \\ j \end{bmatrix}_q \frac{1}{1 - zq^j}. \quad (5.5)$$

Set $z = 1 + t$. The left side becomes $\prod_{j=1}^N (1 - u_j t)^{-1}$. The coefficient of t^r on the right is exactly the left side of (5.3). The complete-symmetric and Bell descriptions now follow from (4.2) and (4.4). $\square$

This is the finite identity associated with Dilcher's divisor-function identities [13]. The partial-fraction proof simultaneously explains the alternating single sum, the weakly ordered multiple sum, and the Bell polynomial. For $r = 1$, it reduces to

$$\sum_{j=1}^N (-1)^{j-1} \frac{q^{j(j+1)/2}}{1 - q^j} \begin{bmatrix} N \\ j \end{bmatrix}_q = \sum_{j=1}^N \frac{q^j}{1 - q^j}.$$

For $r = 2$, the right side is $\frac{1}{2}((\sum u_j)^2 + \sum u_j^2)$, and for $r = 3$, it is $\frac{1}{6}((\sum u_j)^3 + 3(\sum u_j)(\sum u_j^2) + 2 \sum u_j^3)$.

5.3 The ordinary harmonic limit

Multiply (5.3) by $(1 - q)^r$ and let $q \rightarrow 1$. Since $(1 - q)u_j \rightarrow 1/j$, the limit is

$$\sum_{j=1}^N (-1)^{j-1} \binom{N}{j} \frac{1}{j^r} = h_r \left(1, \frac{1}{2}, \dots, \frac{1}{N} \right) = \frac{1}{r!} \mathcal{B}_r(H_N^{(1)}, 1!H_N^{(2)}, \dots, (r-1)!H_N^{(r)}), \quad (5.6)$$

where $H_N^{(s)} = \sum_{j=1}^N j^{-s}$. This limit is termwise because all sums are finite. Thus the ordinary binomial-harmonic identities of coefficient calculus are literal degenerations of the Gaussian identities, not merely analogous-looking formulae.

Chapter 6

Newton bases on arbitrary nodes

Gaussian coefficients are Newton connection coefficients for geometric nodes, in the same sense in which ordinary Stirling numbers are the connection coefficients for the nodes $0, 1, 2, \dots$. This chapter develops the node calculus in general, then specializes it to geometric, affine-geometric, and q -integer nodes. The affine-geometric case is the engine behind the closed q -Stirling formulae of Chapter 8.

6.1 The general node calculus

Let $a_0, a_1, \dots$ be elements of a commutative ring and set

$$P_0^a(x) = 1, \quad P_k^a(x) = \prod_{j=0}^{k-1} (x - a_j). \quad (6.1)$$

Since these polynomials are monic of successive degrees, there are unique arrays S_a and s_a such that

$$x^n = \sum_{k=0}^n S_a(n, k) P_k^a(x), \quad P_n^a(x) = \sum_{k=0}^n s_a(n, k) x^k. \quad (6.2)$$

This common framework for binomial, Stirling, and Gaussian coefficients is classical; Voigt's short paper [19] is particularly close in spirit to the basis-change program of the canonical manuscript.

For a finite list $A = (a_0, \dots, a_k)$, the elementary and complete homogeneous polynomials are defined by

$$e_m(A) = \sum_{i_1 < \dots < i_m} a_{i_1} \cdots a_{i_m}, \quad h_m(A) = \sum_{i_1 \leq \dots \leq i_m} a_{i_1} \cdots a_{i_m}.$$

Both have value one at $m = 0$; $e_m = 0$ above the list length.

Theorem 6.1 (Universal finite node formulae). *For $0 \leq k \leq n$,*

$$S_a(n, k) = h_{n-k}(a_0, \dots, a_k), \quad (6.3)$$

$$s_a(n, k) = (-1)^{n-k} e_{n-k}(a_0, \dots, a_{n-1}). \quad (6.4)$$

The arrays are inverse lower-triangular matrices. In particular,

$$\sum_{j=k}^n s_a(n, j) S_a(j, k) = \delta_{n,k}. \quad (6.5)$$

Their recurrences are

$$S_a(n+1, k) = S_a(n, k-1) + a_k S_a(n, k), \quad (6.6)$$

$$s_a(n+1, k) = s_a(n, k-1) - a_n s_a(n, k). \quad (6.7)$$

Proof. The identity $xP_k = P_{k+1} + a_k P_k$ gives Equation (6.6). Separating monomials in $h_m(a_0, \dots, a_k)$ according as they contain a_k gives the same recurrence for Equation (6.3), including the boundary values. Formula Equation (6.4) follows by expanding the product, and multiplication by $x - a_n$ gives Equation (6.7). Substituting one basis expansion into the other proves matrix inversion by uniqueness of monomial coefficients. $\square$

All these identities are valid even when nodes coincide. Only the next representation requires distinct nodes.

Theorem 6.2 (Divided differences and a nonrecursive spectral sum). *Over a field, assume $a_0, \dots, a_k$ are distinct. Then*

$$S_a(n, k) = \sum_{j=0}^k \frac{a_j^n}{\prod_{0 \leq i \leq k, i \neq j} (a_j - a_i)}. \quad (6.8)$$

More generally, the coefficient of P_k^a in the Newton interpolant to values $f(a_0), \dots, f(a_M)$ is

$$f[a_0, \dots, a_k] = \sum_{j=0}^k \frac{f(a_j)}{\prod_{i \neq j} (a_j - a_i)}. \quad (6.9)$$

Thus its degree- M interpolating polynomial is $\sum_{k=0}^M f[a_0, \dots, a_k] P_k^a(x)$.

Proof. The Lagrange interpolant on the first $k+1$ nodes is

$$L_k(x) = \sum_{j=0}^k f(a_j) \frac{\prod_{i \neq j} (x - a_i)}{\prod_{i \neq j} (a_j - a_i)}.$$

Its leading coefficient is the sum in Equation (6.9). In a Newton expansion, every term with index greater than k vanishes on these nodes, and every term with index less than k has smaller degree. Hence that leading coefficient is precisely the k th Newton coefficient. Apply this to $f(x) = x^n$ for Equation (6.8). $\square$

6.2 Geometric nodes and the Gaussian basis pair

Theorem 6.3 (The Gaussian Newton pair). *Put $P_k(x) = \prod_{j=0}^{k-1} (x - q^j)$. Then*

$$x^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q P_k(x), \quad (6.10)$$

$$P_n(x) = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q x^k. \quad (6.11)$$

Proof. By Equation (6.3), the first coefficient is $h_{n-k}(1, q, \dots, q^k)$, which is $G_{n,k}(q)$ by Equation (2.11). The second is the elementary symmetric coefficient in the same geometric list, and Equation (2.10) gives exactly the stated phase. Equivalently, the first recurrence is Equation (2.2) with $a_k = q^k$. $\square$

This is the precise sense in which Gaussian coefficients are Stirling-type coefficients: the nodes $0, 1, 2, \dots$ produce ordinary Stirling numbers, while $1, q, q^2, \dots$ produce Gaussian polynomials. At $q = 1$ these geometric nodes all equal one, so Equation (6.10) becomes the ordinary binomial expansion $x^n = \sum_k \binom{n}{k} (x-1)^k$, not the ordinary Stirling expansion.

For generic nonzero q , the explicit geometric-grid formula is

$$f[1, q, \dots, q^k] = \frac{q^{-\binom{k}{2}}}{(q; q)_k} \sum_{j=0}^k (-1)^j q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q f(q^j). \quad (6.12)$$

Indeed,

$$\prod_{i \neq j} (q^j - q^i) = (-1)^j q^{jk-j(j+1)/2} (q; q)_j (q; q)_{k-j}, \quad (6.13)$$

obtained by separating $i < j$ from $i > j$. Substitution in Equation (6.9) proves Equation (6.12). This is a finite interpolation formula; an infinite Newton expansion would need separate convergence assumptions.

6.3 An affine-geometric master connection

Theorem 6.4 (Affine-geometric nodes). *For nodes $a_j = A + Bq^j$, the connection coefficients are*

$$S_{A+Bq^\bullet}(n, k) = \sum_{d=k}^n \binom{n}{d} A^{n-d} B^{d-k} \begin{bmatrix} d \\ k \end{bmatrix}_q. \quad (6.14)$$

This is a polynomial identity in A, B, q , including $B = 0$.

Proof. First take B invertible and write $x = A + By$. Then $P_k^a(x) = B^k P_k(y)$. Expand $(A + By)^n$ ordinarily, then expand each y^d by Equation (6.10). The coefficient of $B^k P_k(y)$ is Equation (6.14). Both sides are polynomials, so the identity descends from $\mathbb{Q}(A, B, q)$ to $\mathbb{Z}[A, B, q]$ and can be specialized at $B = 0$. $\square$

This theorem is a compact engine for several families that otherwise look unrelated. The next section makes its q -arithmetic specializations explicit.

6.4 The q -integer lattice: the deformation of ordinary Newton expansion

Define the monic polynomials

$$F_k(x) = \prod_{i=0}^{k-1} (x - [i]_q), \quad F_0(x) = 1. \quad (6.15)$$

Their normalized versions

$$N_k(x; q) = \frac{F_k(x)}{q^{\binom{k}{2}} [k]_q!}$$

satisfy $N_k([m]_q; q) = \begin{bmatrix} m \\ k \end{bmatrix}_q$ for $m \geq 0$. To see this, use $[m]_q - [i]_q = q^i [m-i]_q$ and multiply over i . This basis belongs to the broader setting of quantum integer-valued polynomials [18]; its normalization is stated here explicitly.

Theorem 6.5 (Finite q -Newton expansion). *For every polynomial p of degree at most d over R ,*

$$p(x) = \sum_{k=0}^d c_k N_k(x; q), \quad (6.16)$$

$$c_k = \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q p([j]_q). \quad (6.17)$$

At $q = 1$ this becomes $p(x) = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}$.

Proof. The distinct degrees and unit leading coefficients of $N_0, \dots, N_d$ make them a basis. Evaluate its expansion at $x = [m]_q$ for $0 \leq m \leq d$: $p([m]_q) = \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q c_k$. Gaussian inversion gives (6.17). At $q = 1$, the factors in N_k become $(x - i)$, their denominator becomes $k!$, and the sum for c_k is the ordinary finite-difference formula. All denominators in this specialization are nonzero. $\square$

Chapter 7

Jackson differentiation and geometric Newton interpolation

7.1 The operator and its commutation rule

Let $T_q f(x) = f(qx)$, and let X denote multiplication by x . Define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x}. \quad (7.1)$$

For polynomials, this has no singularity at $x = 0$, because the numerator has zero constant term. It acts by $D_q x^m = [m]_q x^{m-1}$, and hence

$$D_q X = 1 + qX D_q, \quad D_q T_q = qT_q D_q. \quad (7.2)$$

These identities follow on every monomial and hence on every polynomial. This is the convention for the Jackson derivative used throughout; compare [4].

Theorem 7.1 (The shifted product rule). *For $n \geq 0$,*

$$D_q^n (fg)(x) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}x)(D_q^{n-k} g)(x). \quad (7.3)$$

Proof. The first-order formula is $D_q(fg) = (D_q f)g + (T_q f)D_q g$, obtained by adding and subtracting $f(qx)g(x)$ in the numerator. Suppose the formula is true at order n . Differentiating a summand gives a contribution with k unchanged and a contribution with k increased by one. In the latter, differentiation of $(D_q^k f)(q^{n-k}x)$ produces a factor q^{n-k} . Thus the new coefficient at index k is $\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$, by (2.2). The shifts are exactly those in the stated formula. $\square$

Normalization and domain

The argument shift $q^{n-k}x$ is part of the theorem. Replacing the ordinary binomial coefficient by a Gaussian coefficient in the usual Leibniz rule, while leaving all factors evaluated at x , is false.

7.2 A geometric Taylor basis

Define

$$(x-a)_q^{[k]} = \prod_{j=0}^{k-1} (x - aq^j). \quad (7.4)$$

This is a polynomial basis in x , not a power of $x - a$.

Theorem 7.2 (Jackson–Newton expansion). *For every polynomial p of degree at most d ,*

$$p(x) = \sum_{k=0}^d \frac{(D_q^k p)(a)}{[k]_q!} (x - a)_q^{[k]}. \quad (7.5)$$

The identity first holds over $\mathbb{Q}(q, a)$, and specializes wherever its coefficients are defined. At $q = 1$ it becomes ordinary Taylor expansion at a .

Proof. The polynomial $D_q(x - a)_q^{[k]}$ has degree $k - 1$ and leading coefficient $[k]_q$. At each $x = aq^j$, $0 \leq j \leq k - 2$, its two numerator terms vanish. Therefore

$$D_q(x - a)_q^{[k]} = [k]_q (x - a)_q^{[k-1]}.$$

This argument at generic a, q proves a polynomial identity and hence also covers $a = 0$. Expand p in this monic basis, apply D_q^k , and set $x = a$. Terms with index less than k vanish under differentiation; terms with larger index retain a factor $x - a$; the index k term yields $[k]_q!$ times its coefficient. This proves the formula. $\square$

For example,

$$x^2 = a^2 + (1 + q)a(x - a) + (x - a)(x - aq),$$

and

$$x^3 = a^3 + [3]_q a^2(x - a) + [3]_q a(x - a)(x - aq) + (x - a)(x - aq)(x - aq^2).$$

At a sample node $x = aq^n$, factors with $k > n$ vanish, and direct factorization gives the discrete Gaussian transform

$$p(aq^n) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a(q - 1))^k q^{\binom{k}{2}} (D_q^k p)(a). \quad (7.6)$$

Indeed, $\prod_{j < k} (aq^n - aq^j) = [a(q - 1)]^k q^{\binom{k}{2}} [k]_q! \begin{bmatrix} n \\ k \end{bmatrix}_q$. This identifies Gaussian inversion with the extraction of geometric Newton coefficients.

7.3 Explicit stencils and divided differences

Theorem 7.3. *For $n \geq 0$,*

$$D_q^n f(x) = \frac{1}{(1 - q)^n x^n} \sum_{j=0}^n (-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q f(q^j x). \quad (7.7)$$

At distinct nonzero nodes $a, aq, \dots, aq^n$,

$$f[a, aq, \dots, aq^n] = \frac{(D_q^n f)(a)}{[n]_q!}. \quad (7.8)$$

The left side denotes the leading coefficient of the degree-at-most- n interpolating polynomial through the given data.

Proof. The commutation relation $T_q X^{-1} = q^{-1} X^{-1} T_q$ gives

$$D_q^n = (1 - q)^{-n} X^{-n} \prod_{r=0}^{n-1} (1 - q^{-r} T_q).$$

Expand the finite product with base q^{-1} , and use reciprocity (2.5). Its T_q^j coefficient becomes $(-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q$, proving (7.7). Let p interpolate f at these nodes. Formula (7.7) shows that $(D_q^n f)(a) = (D_q^n p)(a)$. The latter equals $[n]_q!$ times the leading coefficient of p , either by monomial differentiation or by Theorem 7.2. This proves (7.8). $\square$

Corollary 7.4 (An exact Gaussian coefficient extractor). *If p has degree at most n , then*

$$[x^n]p(x) = \frac{1}{(q; q)_n} \sum_{j=0}^n (-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q p(q^j). \quad (7.9)$$

The numerator annihilates x^r for every $0 \leq r < n$.

Proof. Apply (7.7) at $x = 1$ and divide by $[n]_q!$. Alternatively, the numerator on x^r is $(q^{r-n+1}; q)_n$, by (2.10). One factor is zero for $0 \leq r < n$, while for $r = n$ the product is $(q; q)_n$. $\square$

This is the geometric counterpart of the additive finite-difference stencil in the source. Its cancelling objects are monomials evaluated at geometric nodes.

7.4 The iterated product rule

Repeated application of Theorem 7.1 to the first factor and the remaining product gives, for r factors,

$$D_q^n \left(\prod_{i=1}^r f_i \right) (x) = \sum_{n_1 + \dots + n_r = n} \frac{[n]_q!}{\prod_i [n_i]_q!} \prod_{i=1}^r (D_q^{n_i} f_i) (q^{n_{i+1} + \dots + n_r} x). \quad (7.10)$$

The successive Gaussian coefficients telescope to the displayed Gaussian multinomial, and the argument shifts accumulate from the right: the i th factor is evaluated at $q^{n_{i+1} + \dots + n_r} x$. The same coefficient occurs in the noncommutative binomial formula of Theorem 3.8.

7.5 An exact Jackson chain rule through divided differences

The preceding distinctions become especially important for composition. There is no general identity $D_q(f \circ g) = (D_q f) \circ g \cdot D_q g$. For $f(u) = u^2$ and $g(x) = x^2$, its two sides would be $[4]_q x^3$ and $[2]_q^2 x^3$, which differ for generic q . A correct finite chain rule uses divided differences. The general construction belongs to the divided-difference chain-rule literature [15]; the following proof also explains its relation to coefficient calculus.

Theorem 7.5 (Path form of the divided-difference chain rule). *Let $x_0, \dots, x_n$ be distinct and put $y_i = g(x_i)$. Suppose first that the y_i are distinct. For $n \geq 1$,*

$$(f \circ g)[x_0, \dots, x_n] = \sum_{k=1}^n \sum_{0=i_0 < i_1 < \dots < i_k=n} f[y_{i_0}, \dots, y_{i_k}] \prod_{r=1}^k g[x_{i_{r-1}}, \dots, x_{i_r}]. \quad (7.11)$$

Whenever confluent limits exist, the same formula extends to coinciding y_i by replacing divided differences by their confluent values. For $x_i = aq^i$ and $a \neq 0$, multiplication of the right side by $[n]_q!$ gives $(D_q^n(f \circ g))(a)$.

Proof. We give the finite matrix argument in detail. Let T be the $(n+1) \times (n+1)$ upper bidiagonal matrix whose diagonal is $x_0, \dots, x_n$ and whose first superdiagonal consists of ones. Direct back substitution in $(zI - T)R = I$ shows that

$$R_{ij} = \frac{1}{\prod_{h=i}^j (z - x_h)} \quad (i \leq j).$$

For a polynomial h , applying polynomial functional calculus, or expanding this rational expression by partial fractions, therefore gives

$$h(T)_{ij} = h[x_i, \dots, x_j]. \quad (7.12)$$

For completeness, the partial-fraction coefficient of $1/(z - x_r)$ is $\prod_{i \leq h \leq j, h \neq r} (x_r - x_h)^{-1}$; multiplying by $h(x_r)$ and summing is exactly the explicit divided-difference formula.

More generally, let A be upper triangular with diagonal a_i and strict upper part N . The finite expansion

$$(zI - A)^{-1} = \sum_{k=0}^n ((zI - \text{diag } A)^{-1} N)^k (zI - \text{diag } A)^{-1}$$

shows that its $(0, n)$ entry is a sum over increasing paths $0 = i_0 < \dots < i_k = n$. The contribution of a path is

$$\frac{A_{i_0 i_1} \cdots A_{i_{k-1} i_k}}{(z - a_{i_0}) \cdots (z - a_{i_k})}.$$

The same partial-fraction calculation gives

$$f(A)_{0n} = \sum_{\text{paths}} f[a_{i_0}, \dots, a_{i_k}] A_{i_0 i_1} \cdots A_{i_{k-1} i_k}.$$

Take $A = g(T)$, use Equation (7.12), and use $f(g(T)) = (f \circ g)(T)$. This proves Equation (7.11) for polynomials. For arbitrary data with distinct y_i , interpolate g on the x_i and f on the y_i ; every term is unchanged. Passage to confluent limits proves the stated extension when the necessary smoothness or analyticity is available. Finally apply Equation (7.8) to the geometric nodes. $\square$

For $n = 2$ the two paths give the particularly transparent identity

$$\begin{aligned} (f \circ g)[x_0, x_1, x_2] &= f[y_0, y_2]g[x_0, x_1, x_2] \\ &\quad + f[y_0, y_1, y_2]g[x_0, x_1]g[x_1, x_2]. \end{aligned} \tag{7.13}$$

As $q \rightarrow 1$, geometric nodes coalesce and divided differences become ordinary derivatives divided by factorials. Thus this is a finite-node precursor of ordinary Faà di Bruno, rather than an unjustified replacement of all factorials by q -factorials.

Chapter 8

Two q -Stirling arrays and normal ordering

8.1 The monic q -integer factorial basis

Define

$$P_k(x) = \prod_{j=0}^{k-1} (x - [j]_q), \quad P_n(x) = \sum_{k=0}^n s_q(n, k) x^k, \quad x^n = \sum_{k=0}^n S_q(n, k) P_k(x). \quad (8.1)$$

These are changes between two monic polynomial bases. They define s_q, S_q unambiguously over $\mathbb{Z}[q]$, even when specialized nodes coincide.

Theorem 8.1. *The arrays are mutually inverse, and*

$$s_q(n+1, k) = s_q(n, k-1) - [n]_q s_q(n, k), \quad (8.2)$$

$$S_q(n+1, k) = S_q(n, k-1) + [k]_q S_q(n, k), \quad (8.3)$$

$$s_q(n, k) = (-1)^{n-k} e_{n-k}([0]_q, \dots, [n-1]_q), \quad (8.4)$$

$$S_q(n, k) = h_{n-k}([0]_q, \dots, [k]_q). \quad (8.5)$$

Both arrays have entry one at $(0, 0)$ and vanish outside the triangular range.

Proof. Multiplication of $P_n(x)$ by $x - [n]_q$ gives (8.2). The identity $xP_k = P_{k+1} + [k]_q P_k$ gives (8.3). The coefficient formula (8.4) follows by choosing roots from the product defining P_n . For (8.5), use

$$h_r([0]_q, \dots, [k]_q) = h_r([0]_q, \dots, [k-1]_q) + [k]_q h_{r-1}([0]_q, \dots, [k]_q).$$

With $r = n - k$, this is precisely the recurrence for S_q , with the same boundary values. Finally, substituting either basis change into the other and using uniqueness of coefficients proves matrix inversion. $\square$

At $q = 1$, these are the ordinary signed first-kind and second-kind Stirling numbers. At $q = 0$, all positive-index nodes equal one, so for $n \geq 1$,

$$S_0(n, k) = \binom{n-1}{k-1}, \quad s_0(n, k) = (-1)^{n-k} \binom{n-1}{k-1} \quad (1 \leq k \leq n).$$

This follows from $P_n(x) = x(x-1)^{n-1}$ and from the complete-symmetric formula. It illustrates the usefulness of a polynomial definition at degenerate bases.

8.2 A Gaussian closed sum, with every power of q accounted for

Theorem 8.2. Over $\mathbb{Q}(q)$, for $0 \leq k \leq n$,

$$S_q(n, k) = \frac{q^{-\binom{k}{2}}}{[k]_q!} \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q [j]_q^n. \quad (8.6)$$

The expression has the polynomial continuation specified by (8.1).

Proof. The coefficient of the degree- k Newton basis polynomial in the interpolation of x^n at $[0]_q, \dots, [k]_q$ is

$$\sum_{j=0}^k \frac{[j]_q^n}{\prod_{0 \leq i \leq k, i \neq j} ([j]_q - [i]_q)}.$$

It equals the coefficient in the full Newton expansion because terms of degree larger than k vanish at all these nodes. For $i < j$, $[j]_q - [i]_q = q^i[j - i]_q$, whereas for $i > j$ it is $-q^j[i - j]_q$. Hence the denominator is

$$(-1)^{k-j} q^{jk-j(j+1)/2} [j]_q! [k-j]_q!.$$

Its inverse, multiplied by $[k]_q!$, has exponent $\binom{j+1}{2} - jk = \binom{k-j}{2} - \binom{k}{2}$. This proves (8.6). $\square$

There is no guesswork in the factor $q^{-\binom{k}{2}}$: it is the normalization of the monic q -integer Newton basis. Omitting it produces a different array.

8.3 Bell formulae for every diagonal

For $k \geq 0$ and $r \geq 1$, let

$$U_r(k) = \sum_{j=0}^k [j]_q^r = \frac{1}{(1-q)^r} \sum_{a=0}^r (-1)^a \binom{r}{a} [k+1]_q^a. \quad (8.7)$$

The $a = 0$ term means $k+1$. The equality follows by expanding $(1-q^j)^r$ binomially and summing the geometric powers q^{aj} . Equations (8.4)–(8.5) and the Bell formulae give

$$S_q(n, k) = \frac{1}{(n-k)!} \mathcal{B}_{n-k} \left(((r-1)! U_r(k))_{r=1}^{n-k} \right), \quad (8.8)$$

$$s_q(n, k) = \frac{(-1)^{n-k}}{(n-k)!} \mathcal{B}_{n-k} \left(((-1)^{r-1} (r-1)! U_r(n-1))_{r=1}^{n-k} \right). \quad (8.9)$$

These are recurrence-free finite formulae involving only binomial coefficients, q -integers, factorials, and the finite sum (1.4). For example,

$$S_q(n, n-1) = \sum_{j=1}^{n-1} [j]_q = \sum_{a=0}^{n-2} (n-1-a) q^a,$$

and

$$S_q(n, n-2) = \frac{1}{2} (U_1(n-2)^2 + U_2(n-2)).$$

The latter uses the alphabet ending at $[n-2]_q$, not at $[n-1]_q$.

8.4 The normal-ordering normalization

Define

$$\tilde{S}_q(n, k) = q^{\binom{k}{2}} S_q(n, k). \quad (8.10)$$

Then

$$\tilde{S}_q(n+1, k) = [k]_q \tilde{S}_q(n, k) + q^{k-1} \tilde{S}_q(n, k-1). \quad (8.11)$$

Both this and (8.3) occur in q -Stirling literature; the restricted-growth-word viewpoint and related identities are discussed in [11]. The defining recurrence must accompany any comparison of conventions.

Theorem 8.3 (Normal ordering and factorial moments). *On polynomials,*

$$(XD_q)^n = \sum_{k=0}^n \tilde{S}_q(n, k) X^k D_q^k. \quad (8.12)$$

For integers $m, n \geq 0$,

$$[m]_q^n = \sum_{k=0}^{\min(m, n)} \tilde{S}_q(n, k) [k]_q! \left[\begin{matrix} m \\ k \end{matrix} \right]_q. \quad (8.13)$$

Proof. Induction from (7.2) gives $D_q X^k = [k]_q X^{k-1} + q^k X^k D_q$. Left multiplication of the right side of (8.12) by XD_q therefore produces recurrence (8.11), with the correct initial value. This proves normal ordering. Apply both sides to x^m . The left side is $[m]_q^n x^m$, and $X^k D_q^k x^m = [k]_q! \left[\begin{matrix} m \\ k \end{matrix} \right]_q x^m$; terms with $k > m$ vanish. Comparison proves (8.13). $\square$

For $n = 2$, normal ordering reads $(XD_q)^2 = XD_q + qX^2D_q^2$. Thus $\tilde{S}_q(2, 2) = q$, while $S_q(2, 2) = 1$. This tiny example is an effective audit of the two normalizations.

8.5 Affine-geometric closed forms

The nodes $[j]_q$ are affine-geometric: $[j]_q = A + Bq^j$ with $A = (1 - q)^{-1}$ and $B = -(1 - q)^{-1}$. Theorem 6.4 therefore gives finite mixed ordinary/Gaussian sums for both arrays, in addition to the symmetric-function forms of Theorem 8.1 and the spectral sum of Theorem 8.2.

Corollary 8.4 (Affine-geometric closed forms). *For $0 \leq k \leq n$,*

$$S_q(n, k) = (1 - q)^{k-n} \sum_{d=k}^n (-1)^{d-k} \binom{n}{d} \left[\begin{matrix} d \\ k \end{matrix} \right]_q, \quad (8.14)$$

$$s_q(n, k) = (1 - q)^{k-n} \sum_{d=k}^n (-1)^{d-k} \binom{d}{k} q^{\binom{n-d}{2}} \left[\begin{matrix} n \\ d \end{matrix} \right]_q. \quad (8.15)$$

All apparent singularities at $q = 1$ are removable: both sides are polynomials in q .

Proof. Substituting $A = (1 - q)^{-1}$, $B = -(1 - q)^{-1}$ in (6.14) gives $A^{n-d} B^{d-k} = (-1)^{d-k} (1 - q)^{k-n}$, which is (8.14). For the inverse array put $y = 1 - (1 - q)x$, so that $x - [j]_q = -(y - q^j)/(1 - q)$ and $P_n(x) = (-1)^n (1 - q)^{-n} \prod_{j=0}^{n-1} (y - q^j)$ in the notation of (8.1). Insert (6.11) for the product, expand $y^d = \sum_k \binom{d}{k} (-1 - q)^k x^k$ ordinarily, and compare the coefficients of x^k : the sign is $(-1)^n (-1)^{n-d} (-1)^k = (-1)^{d-k}$ and the power of $1 - q$ is $k - n$. This is (8.15). Polynomiality of both arrays, from (8.1), proves the cancellation claims. $\square$

The first few rows are

$n \backslash k$	0	1	2		3		4
0	1						
1	0	1					
2	0	1	1				
3	0	1	$2 + q$		1		
4	0	1	$3 + 3q + q^2$		$3 + 2q + q^2$	1	

(8.16)

In particular $S_q(n, 1) = 1$ for $n \geq 1$. The limit $q \rightarrow 1$ recovers ordinary Stirling numbers, while $q = 0$ gives $S_0(n, k) = \binom{n-1}{k-1}$ for $1 \leq k \leq n$: all positive-index nodes have become one, and Equation (8.5) counts weak compositions of $n - k$ into k positions.

8.6 Colored and type- B node specializations

Corollary 8.5 (Nodes $[r + mj]_q$). *Fix integers $m \geq 1$ and $r \geq 0$, and define $S_q^{(m,r)}(n, k)$ using the nodes $a_j = [r + mj]_q$. Then*

$$S_q^{(m,r)}(n, k) = (1 - q)^{k-n} \sum_{d=k}^n \binom{n}{d} (-q^r)^{d-k} \left[\begin{matrix} d \\ k \end{matrix} \right]_{q^m}. \quad (8.17)$$

It is a polynomial with nonnegative integer coefficients. For $(m, r) = (2, 1)$ its defining recurrence is

$$S_q^{(2,1)}(n+1, k) = S_q^{(2,1)}(n, k-1) + [2k+1]_q S_q^{(2,1)}(n, k), \quad (8.18)$$

the type- B convention discussed in [14].

Proof. Write $[r + mj]_q = (1 - q)^{-1} - q^r(1 - q)^{-1}(q^m)^j$ and use Theorem 6.4. Nonnegativity follows from the complete homogeneous formula Equation (6.3); Equation (8.18) is Equation (6.6). $\square$

No signed-permutation interpretation is needed to prove Equation (8.17). It gives a finite Gaussian formula for the entire node family, including its boundary values $S_q^{(m,r)}(n, 0) = [r]_q^n$.

8.7 A q -Dobinski identity

The factorial-moment identity (8.13) has a generating-function form in which the normal-ordering normalization $\tilde{S}_q$ is forced by the q -Weyl relation, not chosen by analogy with the ordinary Dobinski formula.

Theorem 8.6 (q -Dobinski). *Let $T_{n,q}(z) = \sum_{k=0}^n \tilde{S}_q(n, k) z^k$ and let $e_q(z) = \sum_{j \geq 0} z^j / [j]_q!$ be the q -exponential of (3.15). Then*

$$T_{n,q}(z) = \frac{1}{e_q(z)} \sum_{j \geq 0} [j]_q^n \frac{z^j}{[j]_q!}. \quad (8.19)$$

This holds formally in z over $\mathbb{Q}(q)$; for $0 < q < 1$ it also holds analytically when $|(1 - q)z| < 1$.

Proof. Multiply (8.13) by $z^m / [m]_q!$ and sum over $m \geq 0$:

$$\sum_{m \geq 0} [m]_q^n \frac{z^m}{[m]_q!} = \sum_{k=0}^n \tilde{S}_q(n, k) [k]_q! \sum_{m \geq k} \frac{z^m}{[k]_q! [m - k]_q!} = \sum_{k=0}^n \tilde{S}_q(n, k) z^k e_q(z),$$

because $\left[\begin{matrix} m \\ k \end{matrix} \right]_q / [m]_q! = 1 / ([k]_q! [m - k]_q!)$ and the k -sum is finite, so the interchange is an identity of formal series. Dividing by the unit $e_q(z)$ gives (8.19). Analytically, $[j]_q$ is bounded by $(1 - q)^{-1}$ for $0 < q < 1$, so both series converge absolutely in the stated disk. $\square$

At $q = 1$ the identity is Dobinski's formula $\sum_k S(n, k) z^k = e^{-z} \sum_j j^n z^j / j!$. For $n = 2$ it reads $z + qz^2 = e_q(z)^{-1} \sum_j [j]_q^2 z^j / [j]_q!$, which is the normal-ordering audit $\tilde{S}_q(2, 2) = q$ once more.

Chapter 9

q -Eulerian polynomials and the Stirling–Eulerian bridge

For a permutation $\pi = (\pi_1, \dots, \pi_n)$, let $\text{Des}(\pi) = \{i : \pi_i > \pi_{i+1}\}$, $\text{des}(\pi) = |\text{Des}(\pi)|$, and $\text{maj}(\pi) = \sum_{i \in \text{Des}(\pi)} i$. Define

$$A_n(t, q) = \sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} q^{\text{maj}(\pi)} = \sum_{j=0}^{n-1} A_{n,j}(q) t^j \quad (n \geq 1). \quad (9.1)$$

The major index, rather than an unspecified inversion statistic, is important in the identities below.

Theorem 9.1 (q -Worpitzky and a finite Eulerian sum). *For $n \geq 1$,*

$$\sum_{m \geq 0} [m+1]_q^n t^m = \frac{A_n(t, q)}{(t; q)_{n+1}}. \quad (9.2)$$

Consequently, for $m \geq 0$,

$$[m+1]_q^n = \sum_{j=0}^{n-1} A_{n,j}(q) \begin{bmatrix} m+n-j \\ n \end{bmatrix}_q, \quad (9.3)$$

$$A_{n,j}(q) = \sum_{r=0}^j (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n+1 \\ r \end{bmatrix}_q [j-r+1]_q^n. \quad (9.4)$$

Proof. Expand $[m+1]_q^n$ as the weight sum of vectors $(a_1, \dots, a_n) \in \{0, \dots, m\}^n$, with weight $q^{\sum a_i}$. Sort the labels by decreasing a_i , resolving equal entries by increasing label. This gives a unique permutation π such that $a_{\pi_1} \geq \dots \geq a_{\pi_n}$, with a strict inequality at every descent of π . Put $d = \text{des}(\pi)$ and subtract from the i th sorted entry the number of descents at positions at least i . The resulting sequence is $m-d \geq b_1 \geq \dots \geq b_n \geq 0$, and its weight is shifted by $\text{maj}(\pi)$. Conversely, this construction reconstructs exactly the vectors producing π . Write $e_0 = m-d-b_1$, $e_i = b_i - b_{i+1}$ for $1 \leq i < n$, and $e_n = b_n$. Then $m = d + \sum_{i=0}^n e_i$ and $\sum b_i = \sum_{i=1}^n i e_i$. The generating function for these gaps is

$$t^d q^{\text{maj}(\pi)} \prod_{i=0}^n \frac{1}{1 - t q^i}.$$

Sum over permutations to get Equation (9.2). Extract t^m using Equation (2.11) for Equation (9.3); multiply by $(t; q)_{n+1}$ and use Equation (2.10) for Equation (9.4). $\square$

Theorem 9.2 (Gaussian Stirling–Eulerian transforms). *For $1 \leq k \leq n$,*

$$q^{\binom{k}{2}} [k]_q! S_q(n, k) = \sum_{\ell=1}^k q^{k(k-\ell)} A_{n, \ell-1}(q) \begin{bmatrix} n-\ell \\ k-\ell \end{bmatrix}_q. \quad (9.5)$$

Conversely,

$$A_n(t, q) = \sum_{k=1}^n q^{\binom{k}{2}} [k]_q! S_q(n, k) t^{k-1} (tq^{k+1}; q)_{n-k}. \quad (9.6)$$

Proof. Set $U_k = q^{\binom{k}{2}} [k]_q! S_q(n, k)$. The Newton expansion at $x = [m+1]_q$ gives $[m+1]_q^n = \sum_{k=1}^n U_k G_{m+1, k}(q)$. Since

$$\sum_{m \geq 0} G_{m+1, k}(q) t^m = \frac{t^{k-1}}{(t; q)_{k+1}},$$

comparison with Equation (9.2) proves Equation (9.6). To invert it, we first establish the finite identity

$$\sum_{r=0}^N \begin{bmatrix} N \\ r \end{bmatrix}_q q^{r(r-1)} u^r (uq^r; q)_{N-r} = 1. \quad (9.7)$$

The coefficient of u^j on the left equals

$$\begin{bmatrix} N \\ j \end{bmatrix}_q q^{\binom{j}{2}} \sum_{r=0}^j (-1)^{j-r} q^{\binom{r}{2}} \begin{bmatrix} j \\ r \end{bmatrix}_q,$$

which is $\delta_{j,0}$ by Theorem 3.4. Set $N = n - \ell$ and $u = tq^{\ell+1}$ in Equation (9.7) and multiply by $t^{\ell-1}$. This expresses $t^{\ell-1}$ as

$$\sum_{k=\ell}^n q^{k(k-\ell)} \begin{bmatrix} n-\ell \\ k-\ell \end{bmatrix}_q t^{k-1} (tq^{k+1}; q)_{n-k}.$$

Insert this expansion for each monomial of $A_n(t, q)$ in Equation (9.6). These basis polynomials have distinct lowest powers, so their coefficients are unique. The resulting coefficient of the k th basis polynomial is Equation (9.5). $\square$

Identity Equation (9.5) is classical; its normalization and history are recorded in [14]. The proof here exhibits it as another triangular coefficient transform, directly parallel to the ordinary Stirling–Eulerian relations in the canonical manuscript. For example $A_3(t, q) = 1 + (2q + 2q^2)t + q^3t^2$, and at $(n, k) = (3, 2)$ both sides of Equation (9.5) equal $2q + 3q^2 + q^3$.

Chapter 10

Weighted Bell polynomials and Riordan composition

10.1 Gaussian convolution as a change of coefficient coordinates

Set $w_n = [n]_q!$, and record a sequence by

$$A(z) = \sum_{n \geq 0} a_n \frac{z^n}{w_n}.$$

Ordinary multiplication of series is then the Gaussian convolution

$$(a \star_q b)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k b_{n-k}. \quad (10.1)$$

Associativity and commutativity follow from ordinary series multiplication. More explicitly, for s factors the convolution coefficient is

$$\sum_{i_1 + \dots + i_s = n} \frac{[n]_q!}{[i_1]_q! \dots [i_s]_q!} a_{i_1}^{(1)} \dots a_{i_s}^{(s)}. \quad (10.2)$$

The factorial ratio is the Gaussian multinomial. It is polynomial because it is the product $\begin{bmatrix} n \\ i_1 \end{bmatrix}_q \begin{bmatrix} n - i_1 \\ i_2 \end{bmatrix}_q \dots$. Thus associativity extends polynomially to every specialization of q , even when the divided-power representation itself is not available.

Proposition 10.1 (Explicit convolution inverse). *If a_0 is invertible, the inverse b for $\star_q$ satisfies $b_0 = a_0^{-1}$, and for $n > 0$,*

$$b_n = \frac{1}{a_0} \sum_{r=1}^n \frac{(-1)^r}{a_0^r} \sum_{\substack{i_1 + \dots + i_r = n \\ i_j \geq 1}} \frac{[n]_q!}{[i_1]_q! \dots [i_r]_q!} \prod_{j=1}^r a_{i_j}. \quad (10.3)$$

Proof. Write $A(z) = a_0(1 + U(z))$, where $U(0) = 0$. The formal geometric expansion $A(z)^{-1} = a_0^{-1} \sum_{r \geq 0} (-U(z))^r$ is coefficientwise finite. Multiplying its degree- n coefficient by w_n gives the displayed formula. $\square$

In particular, with

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{[n]_q!}, \quad E_q(z) = \sum_{n \geq 0} q^{\binom{n}{2}} \frac{z^n}{[n]_q!},$$

Gaussian inversion is the coefficient identity $e_q(z)E_q(-z) = 1$.

10.2 Composition coefficients without an ambiguous “ q -Bell” convention

Let $B(z) = \sum_{j \geq 1} b_j z^j / w_j$. Define

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n}{w_k} [z^n] B(z)^k. \quad (10.4)$$

This is a specified weighted coefficient family; it is not a claim that all objects called q -Bell polynomials in the literature have this meaning.

Theorem 10.2. For $0 \leq k \leq n$,

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n k!}{w_k} \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum m_j = k, \sum j m_j = n}} \prod_{j=1}^n \frac{1}{m_j!} \left(\frac{b_j}{w_j} \right)^{m_j}. \quad (10.5)$$

Equivalently,

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n k!}{w_k n!} \mathcal{B}_{n,k} \left(\frac{1! b_1}{w_1}, \frac{2! b_2}{w_2}, \dots \right). \quad (10.6)$$

If $A(z) = \sum_{k \geq 0} a_k z^k / w_k$, then

$$w_n [z^n] A(B(z)) = \sum_{k=0}^n a_k \mathfrak{B}_{n,k}^{[q]}(b). \quad (10.7)$$

Proof. Expand $B(z)^k$ by the ordinary multinomial theorem. A multiplicity vector (m_j) contributes $k! \prod_j (b_j / w_j)^{m_j} / m_j!$. The two constraints count the factors and their total degree. This proves (10.5). Compare it with the defining finite sum for $\mathcal{B}_{n,k}$ to obtain (10.6). Finally, expand $A(B)$ in powers of B and use (10.4). $\square$

At $q = 1$, $w_n = n!$, and $\mathfrak{B}_{n,k}^{[q]}$ reduces to the ordinary exponential Bell polynomial $\mathcal{B}_{n,k}$. At $q = 0$, $w_n = 1$, and it becomes the ordinary coefficient $[z^n] B(z)^k$. This cleanly separates coefficient normalization from the choice of a deformation of set partitions. The first nontrivial examples expose a frequent normalization error:

$$\mathfrak{B}_{3,2}^{[q]} = \frac{2[3]_q}{[2]_q} b_1 b_2, \quad (10.8)$$

$$\mathfrak{B}_{4,2}^{[q]} = \frac{2[4]_q}{[2]_q} b_1 b_3 + \frac{[4]_q [3]_q}{([2]_q)^2} b_2^2, \quad (10.9)$$

$$\mathfrak{B}_{4,3}^{[q]} = \frac{3[4]_q}{[2]_q} b_1^2 b_2. \quad (10.10)$$

Also $\mathfrak{B}_{n,1}^{[q]} = b_n$ and $\mathfrak{B}_{n,n}^{[q]} = b_1^n$. The rational coefficient $2[3]_q / [2]_q$ is not $[3]_q$. It need not be a polynomial in q , and it may have a pole at a root of unity. This is expected: the coordinate system itself uses inverses of q -factorials.

10.3 Weighted Riordan arrays

For $A(0) \neq 0$ and $B(0) = 0$ with invertible linear coefficient, put

$$\mathcal{M}_q(A, B)_{n,k} = \frac{w_n}{w_k} [z^n] A(z) B(z)^k. \quad (10.11)$$

Theorem 10.3. *These lower triangular matrices obey*

$$\mathcal{M}_q(A, B)\mathcal{M}_q(C, D) = \mathcal{M}_q(AC(B), D(B)), \quad (10.12)$$

$$\mathcal{M}_q(A, B)^{-1} = \mathcal{M}_q\left(\frac{1}{A(\overline{B})}, \overline{B}\right), \quad (10.13)$$

where $\overline{B}$ is the ordinary compositional inverse of B .

Proof. The (n, k) entry of the product equals

$$\frac{w_n}{w_k} \sum_j [z^n] A(z) B(z)^j [t^j] C(t) D(t)^k.$$

For fixed n , the sum is finite because $B(0) = 0$. Summing before taking $[z^n]$ produces $A(z)C(B(z))D(B(z))^k$, proving the product law. Inserting $D = \overline{B}$ and $C = 1/A(\overline{B})$ gives $\mathcal{M}_q(1, z)$, the identity matrix. The same substitution in the other order gives the other inverse identity. $\square$

This is diagonal transport of ordinary Riordan calculus. It extends the canonical manuscript's Riordan framework while making clear which operation is unchanged: composition is still ordinary substitution of formal power series. As a useful check,

$$\mathcal{M}_q(e_q(z), z)_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

and its inverse is $\mathcal{M}_q(E_q(-z), z)$. This recovers Gaussian inversion, including the factor $(-1)^{n-k} q^{\binom{n-k}{2}}$. If polynomials $P_n(x)$ are defined by $g(z)e_q(xz) = \sum P_n(x)z^n/[n]_q!$, then $D_q P_n(x) = [n]_q P_{n-1}(x)$: apply D_q in x to the generating function and compare coefficients. This supplies the corresponding q -Appell lowering property with no ambiguity about normalization.

10.4 Why Jackson composition is a different problem

The first-order chain rule for the Jackson derivative is

$$D_q(F \circ G)(x) = F[G(x), G(qx)] D_q G(x), \quad (10.14)$$

where $F[u, v] = (F(u) - F(v))/(u - v)$, with its polynomial continuation on the diagonal. This follows simply by factoring the numerator in (7.1). It is not usually $(D_q F)(G(x))D_q G(x)$. For example, with $F(u) = u^2$ and $G(x) = x^2$, the actual answer is $[4]_q x^3$, whereas that guess gives $[2]_q^2 x^3$. The weighted Bell family in this chapter describes ordinary composition and must not be silently substituted into a higher Jackson chain rule.

The exact higher-order rule is the path formula of Section 7.5: the n th Jackson derivative of a composition is a sum over increasing paths of products of divided differences, not a substitution of $[j]_q!$ for $j!$ in Faà di Bruno's formula.

Chapter 11

Lagrange inversion with geometric product data

11.1 The ordinary inversion theorem over a q -dependent field

Theorem 11.1 (Formal Lagrange–Bürmann). *Let $\Phi(0) = 1$. There is a unique $y(z) \in zK[[z]]$ satisfying $y = z\Phi(y)$. For $n \geq 1$,*

$$[z^n]H(y(z)) = \frac{1}{n}[t^{n-1}]H'(t)\Phi(t)^n. \quad (11.1)$$

In particular, for $s \geq 1$ and $n \geq s$,

$$[z^n]y(z)^s = \frac{s}{n}[t^{n-s}]\Phi(t)^n. \quad (11.2)$$

Proof. The equation determines the coefficient of degree n of y from lower degrees, starting with $[z]y = 1$. This proves existence and uniqueness. The inverse substitution is $z = t/\Phi(t)$. For a Laurent series, residue means the coefficient of degree -1 . The residue of a formal derivative is zero, and the substitution rule $\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du$ follows on monomials: for powers other than -1 , the integrand is a derivative; for the power -1 , the logarithmic derivative has residue one. Thus

$$\begin{aligned} [z^n]H(y(z)) &= \text{Res}_t H(t) \left(\frac{\Phi(t)^n}{t^{n+1}} - \frac{\Phi(t)^{n-1}\Phi'(t)}{t^n} \right) dt \\ &= \frac{1}{n} \text{Res}_t H'(t) \frac{\Phi(t)^n}{t^n} dt. \end{aligned}$$

The second equality is obtained by taking the zero residue of the derivative of $H(t)\Phi(t)^n t^{-n}$. It is (11.1); taking $H(t) = t^s$ proves (11.2). $\square$

This is the ordinary theorem used in the source, now over a field containing q ; see also the proof-oriented survey [17].

11.2 A recurrence-free inverse for a product of shifted factorials

Theorem 11.2. *Suppose $\Phi(t)$ is the product (4.8), with logarithmic power sums P_r . The solution of $y = z\Phi(y)$ has coefficients*

$$[z^n]y(z) = \frac{1}{n!} \mathcal{B}_{n-1} \left((n(r-1)!P_r)_{r=1}^{n-1} \right) \quad (n \geq 1). \quad (11.3)$$

More generally,

$$[z^n]y(z)^s = \frac{s}{n(n-s)!} \mathcal{B}_{n-s} \left((n(r-1)!P_r)_{r=1}^{n-s} \right), \quad n \geq s \geq 1. \quad (11.4)$$

Proof. The logarithm of $\Phi(t)^n$ has power sums nP_r . Apply Theorem 4.2 inside (11.2). The case $s = 1$ has denominator $n(n-1)! = n!$. $\square$

This solves, coefficient by coefficient and without auxiliary recurrence, the inverse problem for

$$f(x) = x \prod_{\ell=1}^L (a_\ell x; q^{d_\ell})_{N_\ell}^{\alpha_\ell}.$$

For $f(x) = x(x; q)_M^\alpha$, the formula becomes

$$[z^n]f^{-1}(z) = \frac{1}{n!} \mathcal{B}_{n-1} \left((\alpha n(r-1)! [M]_{q^r})_{r=1}^{n-1} \right). \quad (11.5)$$

An alternative finite sum, obtained by expanding each factor separately, is

$$[z^n]f^{-1}(z) = \frac{1}{n} \sum_{\substack{r_0, \dots, r_{M-1} \geq 0 \\ \sum r_j = n-1}} \prod_{j=0}^{M-1} \frac{\alpha n^{\overline{r_j}}}{r_j!} q^{\sum j r_j}. \quad (11.6)$$

Here the rising factorial makes the formula polynomial in α , including values at which gamma-quotient representations would be inconvenient.

Example 11.3 (A two-color inverse). The inverse of $x(1-x)(1-qx)$ begins

$$y(z) = z + (1+q)z^2 + (2+3q+2q^2)z^3 + (5+10q+10q^2+5q^3)z^4 + O(z^5).$$

At $q = 1$, the coefficient formula is $n^{-1} \binom{3n-2}{n-1}$, giving 1, 2, 7, 30, ...

The coefficients for positive integral α count rooted plane trees whose children are split into αM ordered blocks; a child in a block of color j has weight q^j . Each vertex contributes one factor $\Phi(t)$, and the root contributes z , proving the functional equation. Every tree with n vertices has $n-1$ edges. Reversing all color labels $j \mapsto M-1-j$ therefore proves

$$[z^n]y(z; q^{-1}) = q^{-(M-1)(n-1)} [z^n]y(z; q). \quad (11.7)$$

The same identity follows algebraically by reversing the finite product and using homogeneity in (11.5). The tree argument additionally proves nonnegativity and integrality in this parameter range.

Ordinary inversion is not obtained by replacing n with $[n]_q$

The factor $1/n$ in Lagrange inversion is an ordinary integer factor. For $M = 1$, the equation is $y = z/(1-y)$, independent of q , and its coefficients are Catalan numbers. Replacing $1/n$ by $1/[n]_q$ would make the answer depend on q , contradicting the defining equation. Genuine q -shifted inversion theories change additional algebraic structure; they are not the operation performed here.

11.3 Optional-slot trees and a Gaussian divisibility identity

A second choice of input reveals polynomial positivity that is not obvious from Lagrange's factor $1/n$.

Theorem 11.4 (Gaussian slot-tree coefficients). *Fix an integer $N \geq 1$ and define $T_N(z)$ by*

$$T_N(z) = z \prod_{j=0}^{N-1} (1 + q^j T_N(z)). \quad (11.8)$$

For $n \geq 1$, its coefficient $C_{N,n}(q) = [z^n]T_N(z)$ satisfies

$$C_{N,n}(q) = \frac{1}{n} \sum_{\substack{k_1 + \dots + k_n = n-1 \\ 0 \leq k_i \leq N-1}} q^{\sum_{i=1}^n \binom{k_i}{2}} \prod_{i=1}^n \begin{bmatrix} N \\ k_i \end{bmatrix}_q. \quad (11.9)$$

Moreover,

$$C_{N,n}(q) \in \mathbb{Z}_{\geq 0}[q], \quad (11.10)$$

$$q^{(N-1)(n-1)} C_{N,n}(q^{-1}) = C_{N,n}(q), \quad (11.11)$$

$$C_{N,n}(1) = \frac{1}{n} \binom{Nn}{n-1} = \frac{1}{(N-1)n+1} \binom{Nn}{n}. \quad (11.12)$$

Thus the numerator of Equation (11.9) is divisible by n coefficientwise in $\mathbb{Z}[q]$.

Proof. Use Lagrange inversion with $\Phi(w) = (-w; q)_N$, expand each of the n factors by the finite q -binomial theorem, and extract degree $n-1$. This proves Equation (11.9). For integrality and positivity, consider rooted trees in which each vertex has N labeled child slots $0, \dots, N-1$, each either empty or occupied by one subtree. Give an edge in slot j weight q^j , and give each vertex weight z . Decomposing at the root gives Equation (11.8). There are finitely many such trees of each size; their weight sum is a polynomial with nonnegative integer coefficients. Uniqueness of the formal solution identifies it with T_N .

Reverse every slot label $j \mapsto N-1-j$. A tree of n vertices has $n-1$ edges, so its exponent w changes to $(N-1)(n-1) - w$. This weight-reversing bijection proves reciprocity. Finally put $q = 1$ in Lagrange's formula: $\Phi(w)^n = (1+w)^{Nn}$. The equality of the two ordinary binomial expressions is a direct factorial cancellation. $\square$

For $N = 2$ an especially simple form is

$$C_{2,n}(q) = \frac{1}{n} \sum_{j=0}^{n-1} \binom{n}{j} \binom{n}{j+1} q^j. \quad (11.13)$$

For example $C_{2,3} = 1 + 3q + q^2$. The formula follows by extracting w^{n-1} from $(1+w)^n(1+qw)^n$. These are Narayana polynomials under the displayed normalization. They are not the same q -Catalan family as a Gaussian quotient such as $\begin{bmatrix} 2n \\ n \end{bmatrix}_q / [n+1]_q$. The name “ q -Catalan” by itself does not specify a weight statistic or a unique deformation. For a reciprocal example, take $w = z/(aw; q)_N^r$ with integers $r, N \geq 1$. Then

$$[z^n]w^m = \frac{m}{n} a^{n-m} \sum_{\substack{k_1 + \dots + k_{rn} = n-m \\ k_i \geq 0}} \prod_{i=1}^{rn} \begin{bmatrix} N + k_i - 1 \\ k_i \end{bmatrix}_q. \quad (11.14)$$

Here Theorem 2.6 replaces the finite theorem. Formula (11.3) continues to apply when r is not an integer, whereas (11.14) uses rn separate factors and therefore does not.

11.4 A genuinely shifted equation: the area-type q -Catalan series

The equation in Theorem 11.4 contains q -dependent coefficients but no dilation of its unknown series. A true q -functional equation behaves differently. The existence of several inequivalent q -Catalan families is well documented; [12] develops a uniform viewpoint. We give a complete derivation of the particular normalization needed here.

Theorem 11.5 (A quotient solution and a finite coefficient formula). *Over $\mathbb{Q}(q)$ define*

$$U(z) = \sum_{n \geq 0} u_n z^n, \quad u_n = \frac{(-1)^n q^{n(n-1)}}{D_n}. \quad (11.15)$$

Then

$$C(z) = \frac{U(qz)}{U(z)} \quad (11.16)$$

is the unique formal solution with constant coefficient 1 of

$$C(z) = 1 + zC(z)C(qz). \quad (11.17)$$

For each $n \geq 0$, its coefficients have the nonrecursive finite expression

$$[z^n]C(z) = \sum_{k=0}^n q^k u_k \sum_{\mathbf{m} \in \mathcal{P}(n-k)} (-1)^{M(\mathbf{m})} M(\mathbf{m})! \prod_{j=1}^{n-k} \frac{u_j^{m_j}}{m_j!}, \quad M(\mathbf{m}) = \sum_j m_j. \quad (11.18)$$

Proof. The coefficients in (11.15) satisfy, for $n \geq 1$, $(1 - q^n)u_n = -q^{2n-2}u_{n-1}$, as follows immediately by cancelling D_n . Consequently $U(z) - U(qz) = -zU(q^2z)$. Division by the unit $U(z)$ gives

$$\frac{U(qz)}{U(z)} = 1 + z \frac{U(q^2z)}{U(qz)} \frac{U(qz)}{U(z)},$$

which proves (11.17). Its coefficient of degree $n \geq 1$ involves only lower-degree coefficients on the right, so the solution is unique. Finally expand $1/U = \sum_{r \geq 0} (-1)^r (U - 1)^r$ by the ordinary multinomial theorem and multiply by $U(qz)$. This is exactly (11.18). $\square$

The first terms are

$$C(z) = 1 + z + (1 + q)z^2 + (1 + 2q + q^2 + q^3)z^3 + O(z^4). \quad (11.19)$$

Equation (11.17), or its binary-tree interpretation, shows that every coefficient belongs to $\mathbb{Z}_{\geq 0}[q]$. Indeed, attach two ordered subtrees to a root and give the second subtree an extra factor q for each of its vertices. The resulting weight rule is precisely $zC(z)C(qz)$. This proves that singularities in the rational summands of (11.18) cancel, including at $q = 1$. At $q = 1$ the equation is $C = 1 + zC^2$, so the usual Catalan sequence is recovered.

The polynomials in (11.19), the slot family in (11.13), and the Gaussian quotient $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix} \right]_q / [n+1]_q$ are not interchangeable. For example, their usual Catalan index $n = 2$ gives respectively $1 + q$, $1 + q$, and $1 + q^2$; at $n = 3$ the first two already differ. The implicit equation and normalization, not the name “ q -Catalan”, must select the intended family.

11.5 A two-variable inverse with its Jacobian factor exposed

The source also develops multivariate Good inversion. The following complete specialization shows how geometric-product Bell data enter a coupled system, while retaining the correction that a naive product of one-variable formulae would omit.

Theorem 11.6 (Cross-coupled inversion). *Let $A(0) = B(0) = 1$, and let*

$$u = z_1 A(v), \quad v = z_2 B(u).$$

For $m, n > 0$, $r, s \geq 0$, $m \geq r$, and $n \geq s$,

$$[z_1^m z_2^n] u^r v^s = \frac{ms + nr - rs}{mn} [t^{n-s}] A(t)^m [t^{m-r}] B(t)^n. \quad (11.20)$$

Each coefficient on the right has the Bell form (4.9) when A, B are finite geometric products.

Proof. Substitute the first equation into the second: $v = z_2 B(z_1 A(v))$. Apply the one-variable formula (11.1) in z_2 , with z_1 a parameter, to $u^r v^s = z_1^r A(v)^r v^s$. The derivative is $st^{s-1} A(t)^r + rt^s A(t)^{r-1} A'(t)$. Furthermore,

$$[z_1^{m-r}] B(z_1 A(t))^n = A(t)^{m-r} [w^{m-r}] B(w)^n.$$

The first derivative term therefore contributes $\frac{s}{n} [t^{n-s}] A(t)^m$ times the B -coefficient. The second contributes

$$\frac{r}{n} [t^{n-s-1}] A(t)^{m-1} A'(t) = \frac{r(n-s)}{mn} [t^{n-s}] A(t)^m.$$

Adding gives $(ms + nr - rs)/(mn)$. This argument also covers $r = 0$ or $s = 0$, since the corresponding derivative term is zero. $\square$

For comparison, the determinant in the two-variable Good formula is

$$1 - \frac{t_2 A'(t_2)}{A(t_2)} \frac{t_1 B'(t_1)}{B(t_1)}.$$

Its correction subtracts $(n-s)(m-r)/(mn)$ from one, which is exactly the prefactor in (11.20). This gives an independent check on the indices and on the otherwise easily omitted Jacobian contribution.

11.6 The Good determinant behind the formula

For a general system $w_i = z_i \Phi_i(\mathbf{w})$ with unit Φ_i , the ordinary multivariate Lagrange–Good formula reads

$$[z^n] H(\mathbf{w}) = [t^n] H(\mathbf{t}) \prod_i \Phi_i(\mathbf{t})^{n_i} \det(\delta_{ij} - t_j \partial_{t_j} \log \Phi_i(\mathbf{t})). \quad (11.21)$$

It follows by the formal change of variables $z_i = t_i / \Phi_i(\mathbf{t})$ in the multivariate residue: the displayed matrix is exactly the Jacobian on the logarithmic differentials dt_j / t_j . The residue invariance is the multivariate counterpart of the substitution in Theorem 11.1, and the formula is one of the canonical manuscript's central tools [1].

For the system of Theorem 11.6 the determinant is

$$1 - uv \frac{A'(v)}{A(v)} \frac{B'(u)}{B(u)}.$$

Extracting the two factors gives $1 - (m-r)(n-s)/(mn)$ times the product of the two one-variable coefficients, which is the prefactor $(ms + nr - rs)/(mn)$ of (11.20). The proof given there derives this special case without invoking a multivariate residue theorem.

More generally, a factor $(a\mathbf{t}^\nu; q^h)_N^\alpha$ has

$$t_j \partial_{t_j} \log(a\mathbf{t}^\nu; q^h)_N^\alpha = -\alpha \nu_j \sum_{\ell \geq 1} a^\ell S_N(q^{h\ell}) \mathbf{t}^{\ell\nu}. \quad (11.22)$$

Here ν is a nonzero nonnegative integer vector. Thus, after imposing a fixed multidegree bound, both the determinant and its product factors have explicit finite expansions: expand the determinant over permutations and expand the exponential of the logarithmic product over multiplicities indexed by (ν, ℓ) . Only those with $\ell\nu \leq \mathbf{n}$ can contribute. This gives a direct route from the canonical manuscript's Good theorem to multivariate q -shifted-factorial identities without changing its Jacobian or introducing unjustified Jackson derivatives.

Chapter 12

The coefficients of a Gaussian polynomial

12.1 A divisor-sum formula for each coefficient

Write $m = n - k$, $D = k(n - k)$, and

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{r=0}^D c_{n,k}(r) q^r.$$

This chapter extracts coefficients in the *base* q , rather than in an auxiliary product variable. Since $G_{n,k}(0) = 1$, it has a formal logarithm in $\mathbb{Q}[[q]]$.

Theorem 12.1. *For $r \geq 1$, define the finite divisor sum*

$$A_r(n, k) = \sum_{\substack{d|r \\ 1 \leq d \leq k}} d - \sum_{\substack{d|r \\ m < d \leq n}} d. \quad (12.1)$$

Then

$$c_{n,k}(r) = \frac{1}{r!} \mathcal{B}_r \left(((j-1)! A_j(n, k))_{j=1}^r \right), \quad (12.2)$$

or, with no named auxiliary polynomial,

$$c_{n,k}(r) = \sum_{\substack{v_1, \dots, v_r \geq 0 \\ \sum j v_j = r}} \prod_{j=1}^r \frac{A_j(n, k)^{v_j}}{j^{v_j} v_j!}. \quad (12.3)$$

Also $c_{n,k}(0) = 1$, and the formula gives zero for $r > D$.

Proof. The finite product quotient gives

$$\begin{aligned} \log G_{n,k}(q) &= \sum_{d=1}^k \sum_{a \geq 1} \frac{q^{da}}{a} - \sum_{d=m+1}^n \sum_{a \geq 1} \frac{q^{da}}{a} \\ &= \sum_{r \geq 1} \frac{A_r(n, k)}{r} q^r. \end{aligned}$$

For fixed r , only divisors $d \mid r$ contribute, with coefficient $1/(r/d) = d/r$. The Bell exponential formula proves the result. The original series is a polynomial of degree D , proving the last assertion. $\square$

Although individual terms in (12.3) are rational and can have signs, the answer is a nonnegative integer by the subset definition. For example, when $(n, k) = (4, 2)$, the first kernels are $A_1 = 1, A_2 = 3, A_3 = -2, A_4 = -1$. They yield

$$c(1) = 1, \quad c(2) = \frac{1+3}{2} = 2, \quad c(3) = \frac{1+9-4}{6} = 1,$$

and $c(4) = 1$, as expected.

12.2 A second finite formula by inclusion–exclusion

Proposition 12.2. *For $r \geq 0$,*

$$c_{n,k}(r) = \sum_{\varepsilon \in \{0,1\}^k} (-1)^{\sum \varepsilon_j} \sum_{\substack{v_1, \dots, v_k \geq 0 \\ \sum j v_j = r - \sum (m+j) \varepsilon_j}} 1. \quad (12.4)$$

An impossible inner constraint contributes zero. All inner variables may be bounded by $v_j \leq \lfloor r/j \rfloor$.

Proof. Expand each numerator factor $1 - q^{m+j}$ by its two terms and each reciprocal factor $(1 - q^j)^{-1}$ as a geometric series. The sign records which numerator terms were selected, and the displayed constraint is precisely the total exponent. Coefficient extraction proves the formula. $\square$

The two finite formulae organize the same coefficient differently. The first collects repeated divisibility contributions into Bell blocks; the second uses restricted partitions and inclusion–exclusion. Neither needs earlier values of $c_{n,k}(r)$.

12.3 Finite Fourier extraction

Let $L > D$, and let $\omega = e^{2\pi i/L}$. Orthogonality of roots of unity implies, for $0 \leq r \leq D$,

$$c_{n,k}(r) = \frac{1}{L} \sum_{j=0}^{L-1} \omega^{-rj} G_{n,k}(\omega^j). \quad (12.5)$$

Indeed, expand the polynomial and use $L^{-1} \sum_{j=0}^{L-1} \omega^{j(s-r)} = \delta_{sr}$ in this degree range. Without the condition $L > D$, the same expression gives the sum of coefficients with exponent congruent to $r \pmod{L}$. Later, q -Lucas supplies a direct root-of-unity evaluation of every summand.

This identity is exact in a cyclotomic field. Floating-point use can suffer from cancellation, so it is not automatically a preferable numerical algorithm. For exact coefficient production, polynomial Pascal recursion or the divisor-sum formula is usually simpler.

Chapter 13

Parameter derivatives and the expansion at one

13.1 Why the logarithmic parameter is the natural coordinate

Put $m = n - k$, $D = km$, and write

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^D c_{n,k}(j) q^j.$$

The substitution $q = e^t$ converts powers of q into exponentials. Thus ordinary derivatives in t give power moments of the coefficient sequence, whereas derivatives in q give falling-factorial moments. This is precisely the distinction for which the canonical manuscript uses the Stirling basis changes. Let $\vartheta = q \, d/dq$, so that $\frac{d}{dt} F(e^t) = (\vartheta F)(e^t)$.

Lemma 13.1 (The regular logarithm of an exponential difference). *As a formal series at $u = 0$,*

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \sum_{r \geq 2} \frac{\beta_r}{r} \frac{u^r}{r!}. \quad (13.1)$$

All odd coefficients after the linear one vanish.

Proof. The quotient inside the logarithm has constant term one. Its logarithmic derivative is

$$\frac{e^u}{e^u - 1} - \frac{1}{u} = 1 + \frac{1}{u} \left(\frac{u}{e^u - 1} - 1 \right) = \frac{1}{2} + \sum_{r \geq 2} \beta_r \frac{u^{r-1}}{r!}.$$

Integrate formally and use the zero constant term of the logarithm. The function $u/(e^u - 1) + u/2$ is even, as direct substitution of $-u$ shows. Hence $\beta_{2j+1} = 0$ for $j \geq 1$. $\square$

Theorem 13.2 (All-order Gaussian cumulants). *Define*

$$\kappa_1(n, k) = \frac{km}{2}, \quad \kappa_r(n, k) = \frac{\beta_r}{r} \sum_{j=1}^k ((m+j)^r - j^r) \quad (r \geq 2). \quad (13.2)$$

Then

$$\log \frac{G_{n,k}(e^t)}{\binom{n}{k}} = \sum_{r \geq 1} \kappa_r(n, k) \frac{t^r}{r!}, \quad (13.3)$$

$$G_{n,k}(e^t) = \binom{n}{k} \sum_{r \geq 0} \mathcal{B}_r(\kappa_1, \dots, \kappa_r) \frac{t^r}{r!}. \quad (13.4)$$

These are both formal identities and convergent local Taylor expansions. In particular,

$$\kappa_2 = \frac{D(n+1)}{12}, \quad \kappa_3 = 0, \quad \kappa_4 = -\frac{D(n+1)(n(n+1)-D)}{120}. \quad (13.5)$$

Proof. The cancelled product representation is

$$\frac{G_{n,k}(e^t)}{\binom{n}{k}} = \prod_{j=1}^k \frac{(e^{(m+j)t} - 1)/((m+j)t)}{(e^{jt} - 1)/(jt)}.$$

Every factor now has constant term one. Apply Theorem 13.1 to each numerator and denominator. The linear coefficient is $\frac{1}{2} \sum_{j=1}^k m = km/2$; the higher coefficients are exactly (13.2). Exponentiation and the Bell generating function give (13.4).

For the second coefficient, $\sum_{j=1}^k ((m+j)^2 - j^2) = km(n+1)$. For the fourth, expansion of the fourth powers and summation of powers of j gives

$$\sum_{j=1}^k ((m+j)^4 - j^4) = km(n+1)(n(n+1) - km).$$

Together with $\beta_2 = 1/6$, $\beta_3 = 0$, and $\beta_4 = -1/30$, these are (13.5). The local analytic assertions follow because the cancelled product is analytic and nonzero near $t = 0$. $\square$

13.1.1 Finite Bernoulli-polynomial form

No power-sum recurrence is necessary in (13.2). Define $\beta_r(x) = \sum_{j=0}^r \binom{r}{j} \beta_j x^{r-j}$. Multiplying its generating function $te^{xt}/(e^t - 1)$ by $e^t - 1$ shows $\beta_{r+1}(x+1) - \beta_{r+1}(x) = (r+1)x^r$. Consequently

$$P_r(M) := \sum_{j=1}^M j^r = \frac{\beta_{r+1}(M+1) - \beta_{r+1}(1)}{r+1}.$$

The sum in (13.2) is $P_r(n) - P_r(m) - P_r(k)$. This also makes its symmetry in k, m manifest.

13.2 Exact coefficient moments and probability

Corollary 13.3 (Power and factorial moments). *For $r \geq 0$,*

$$\sum_{j=0}^D j^r c_{n,k}(j) = \binom{n}{k} \mathcal{B}_r(\kappa_1, \dots, \kappa_r), \quad (13.6)$$

$$G_{n,k}^{(r)}(1) = \binom{n}{k} \sum_{a=0}^r s(r, a) \mathcal{B}_a(\kappa_1, \dots, \kappa_a). \quad (13.7)$$

For $r = 0$, the first sum is the coefficient sum, with $0^0 = 1$.

Proof. Differentiate the finite sum $G_{n,k}(e^t) = \sum_j c_{n,k}(j) e^{jt}$. This proves (13.6). The identity $j^r = \sum_a s(r, a) j^a$, which defines the signed Stirling coefficients, gives (13.7). $\square$

Choosing a partition uniformly from the partitions in the k -by- m rectangle gives a random area X with $\Pr(X = j) = c_{n,k}(j)/\binom{n}{k}$. Its moment generating function is the left side of (13.4), divided by $\binom{n}{k}$. Thus the κ_r are its cumulants, and

$$\mathbb{E}X = \frac{D}{2}, \quad \text{Var}(X) = \frac{D(n+1)}{12}. \quad (13.8)$$

Complementing the rectangle sends X to $D - X$; hence every odd central moment is zero. Every even central moment is given by $\mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r})$. These assertions concern a finite distribution and require no limiting probabilistic theorem.

Example 13.4. For $G_{4,2}(q) = 1 + q + 2q^2 + q^3 + q^4$, the coefficient sum is six, $\kappa_1 = 2$, $\kappa_2 = 5/3$, and $\kappa_4 = -8/3$. Therefore

$$G'_{4,2}(1) = 6 \cdot 2 = 12, \quad (\vartheta^2 G_{4,2})(1) = 6(2^2 + 5/3) = 34,$$

whereas $G''_{4,2}(1) = 34 - 12 = 22$. Confusing the two kinds of differentiation would already give an incorrect second derivative.

Write $\Delta_r = \sum_{j=1}^k ((n-k+j)^r - j^r)$, so that $\kappa_r = \beta_r \Delta_r / r$ for $r \geq 2$ and $\Delta_2 = D(n+1)$. The fourth central moment is then explicit:

$$\mathbb{E}X = \frac{D}{2}, \quad \text{Var}(X) = \frac{D(n+1)}{12}, \quad \mathbb{E}(X - d/2)^4 = 3 \left(\frac{D(n+1)}{12} \right)^2 - \frac{\Delta_4}{120}. \quad (13.9)$$

Here $\Delta_2 = D(n+1)$, $B_2 = 1/6$, and $B_4 = -1/30$.

Higher central moments are obtained by setting the first Bell argument to zero, as noted above.

The first two ordinary derivatives illustrate why Euler derivatives and ordinary derivatives must not be confused:

$$\partial_q \left[\begin{matrix} n \\ k \end{matrix} \right]_q \Big|_1 = \binom{n}{k} \frac{D}{2}, \quad (13.10)$$

$$\partial_q^2 \left[\begin{matrix} n \\ k \end{matrix} \right]_q \Big|_1 = \binom{n}{k} \left(\frac{D^2}{4} + \frac{D(n+1)}{12} - \frac{D}{2} \right). \quad (13.11)$$

The subtraction of $d/2$ converts the second raw moment into the second falling-factorial moment. These exact identities require no limiting probability theorem.

13.3 Arbitrary parameter points and Eulerian rational functions

For $s \geq 0$, define the rational function

$$\text{Li}_{-s}(u) = \left(u \frac{d}{du} \right)^s \frac{u}{1-u}.$$

For $|u| < 1$ this is $\sum_{j \geq 1} j^s u^j$, but the rational expression is the definition needed outside that disk. Set $A_0(u) = 1$. Then

$$\text{Li}_{-s}(u) = \frac{u A_s(u)}{(1-u)^{s+1}}, \quad A_{s+1}(u) = (1+su)A_s(u) + u(1-u)A'_s(u). \quad (13.12)$$

The formula follows by applying $u d/du$ to the fraction. For $s \geq 1$, the coefficients of A_s are the ordinary Eulerian numbers indexed by the number of descents: insertion of the largest letter gives the same coefficient recurrence and initial value. Alternatively, (13.12) is an independent algebraic definition of this Eulerian normalization.

For later use, write

$$R_r(x) = \left(x \frac{d}{dx} \right)^{r-1} \frac{x}{1-x} = \text{Li}_{1-r}(x), \quad r \geq 1, \quad (13.13)$$

so that $R_{s+1} = \text{Li}_{-s}$. The polylogarithm notation is optional: the finite expression

$$R_r(x) = \sum_{\ell=1}^r (\ell-1)! S(r, \ell) \frac{x^\ell}{(1-x)^\ell} \quad (13.14)$$

makes the rational nature explicit. To verify it, note that for $|x| < 1$ the left side is $\sum_{m \geq 1} m^{r-1} x^m$, and $m^{r-1} = \sum_{\ell} S(r-1, \ell) m^{\ell}$ together with $\sum_m m^{\ell} x^m = \ell! x^{\ell} / (1-x)^{\ell+1}$ gives a sum of the same shape; alternatively, differentiate the right side and use the Stirling recurrence $S(r+1, \ell) = \ell S(r, \ell) + S(r, \ell-1)$ to prove it from $R_1 = x/(1-x)$. Rational continuation removes the restriction $|x| < 1$. Thus both the Stirling and the Eulerian bases organize the same rational functions; the Eulerian form is (13.12) with $s = r-1$.

Theorem 13.5 (Logarithmic parameter derivatives). *At a nonzero point q for which none of $q^1, \dots, q^n$ equals one, define*

$$L_r(q) = \sum_{j=1}^k (j^r \operatorname{Li}_{1-r}(q^j) - (m+j)^r \operatorname{Li}_{1-r}(q^{m+j})). \quad (13.15)$$

Then, for $r \geq 0$,

$$\vartheta^r G_{n,k}(q) = G_{n,k}(q) \mathcal{B}_r(L_1(q), \dots, L_r(q)), \quad (13.16)$$

$$G_{n,k}^{(r)}(q) = q^{-r} G_{n,k}(q) \sum_{a=0}^r s(r, a) \mathcal{B}_a(L_1(q), \dots, L_a(q)). \quad (13.17)$$

All L_r are explicit rational functions, by (13.12).

Proof. For $a \geq 1$, repeated logarithmic differentiation gives

$$\vartheta^r \log(1 - q^a) = -a^r \operatorname{Li}_{1-r}(q^a), \quad r \geq 1.$$

Subtracting denominator contributions from numerator contributions gives $\vartheta^r \log G = L_r$. In the local coordinate qe^t , the logarithm of $G(qe^t)/G(q)$ is $\sum_{r \geq 1} L_r(q) t^r / r!$. Exponentiate and use Bell polynomials. Finally, the operator identity

$$q^r \frac{d^r}{dq^r} = \vartheta(\vartheta-1) \cdots (\vartheta-r+1) = \sum_{a=0}^r s(r, a) \vartheta^a \quad (13.18)$$

follows by applying both sides to each monomial. It proves the second formula. $\square$

The displayed domain avoids unnecessary cancellation inside individual rational terms. At roots of unity, the Gaussian polynomial is still regular, but some L_r may have poles. The correct replacement is a deflated local factorization, not termwise substitution into (13.15).

Chapter 14

Cyclotomic structure and all-order root-of-unity jets

14.1 Cyclotomic factorization and carries

Let $\Phi_d(q)$ denote the monic polynomial whose roots are the primitive d -th roots of unity. The partition of roots by their exact order gives $q^n - 1 = \prod_{d|n} \Phi_d(q)$.

Theorem 14.1 (Squarefree cyclotomic factorization). *For $0 \leq k \leq n$, $m = n - k$,*

$$G_{n,k}(q) = \prod_{d=2}^n \Phi_d(q)^{\epsilon_d}, \quad \epsilon_d = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{m}{d} \right\rfloor \in \{0, 1\}. \quad (14.1)$$

Consequently every nonconstant Gaussian polynomial has distinct roots, all on the unit circle. Its degree gives the identity

$$\sum_{d=2}^n \varphi(d) \epsilon_d = k(n - k). \quad (14.2)$$

Proof. Factor each $1 - q^a$ in the factorial quotient. There are $\lfloor n/d \rfloor$ multiples of d in its numerator factorial, and the denominator subtracts the corresponding counts for k and m . The exponent of $\Phi_1 = q - 1$ is $n - k - m = 0$; the signs also cancel. Write $k = ud + v$, $m = wd + x$, with $0 \leq v, x < d$. Then $\epsilon_d = \lfloor (v + x)/d \rfloor$, which is either zero or one. The roots of different cyclotomic polynomials are disjoint, and each is simple. Finally $\deg \Phi_d = \varphi(d)$, and degrees add in the product. $\square$

This carry interpretation is particularly useful: vanishing at a primitive d -th root occurs precisely when the residues of k and $n - k$ produce a carry upon addition modulo d .

14.2 The finite q -Lucas theorem

The following classical root-of-unity formula is often called q -Lucas; see, for example, [16]. Its proof is short enough to include, and the phase bookkeeping is important.

Theorem 14.2 (q -Lucas). *Let ζ be a primitive d -th root of unity, with $d \geq 1$. Write $n = ad + b$, $k = rd + s$, where $0 \leq b, s < d$. Then*

$$G_{n,k}(\zeta) = \binom{a}{r} G_{b,s}(\zeta). \quad (14.3)$$

The right side is zero when $s > b$.

Proof. The finite product theorem and a division into full blocks give

$$\prod_{j=0}^{n-1} (1 + \zeta^j z) = (1 - (-z)^d)^a \prod_{j=0}^{b-1} (1 + \zeta^j z).$$

The second factor has degree smaller than d . Hence its only possible contribution to z^{rd+s} has degree s , and coefficient comparison therefore gives

$$\zeta^{\binom{k}{2}} G_{n,k}(\zeta) = (-1)^{r(d+1)} \binom{a}{r} \zeta^{\binom{s}{2}} G_{b,s}(\zeta).$$

The phase identity $\zeta^{\binom{rd+s}{2} - \binom{s}{2}} = (-1)^{r(d+1)}$ finishes the proof. For odd d , the exponent is a multiple of d . For even d , it is congruent to $rd/2$ modulo d , giving $(-1)^r$. These are exactly the two values on the right. The same argument covers $d = 1$. $\square$

For example,

$$G_{2a,2r}(-1) = \binom{a}{r}, \quad G_{2a,2r+1}(-1) = 0,$$

while $G_{2a+1,2r}(-1) = G_{2a+1,2r+1}(-1) = \binom{a}{r}$. If $d \mid n$, (14.3) is also a fixed-subset count: a permutation consisting of n/d disjoint d -cycles fixes a k -element subset exactly when the subset is a union of whole cycles. There are $\binom{n/d}{k/d}$ such subsets if $d \mid k$, and none otherwise. This provides a concrete combinatorial interpretation for these root evaluations.

14.3 Deflating all apparent singularities

A value formula such as q -Lucas does not determine derivatives at the root. The next theorem supplies every derivative, including the case where the Gaussian polynomial itself vanishes. The calculation uses the logarithmic coordinate $q = \zeta e^t$ and removes powers of t before taking a logarithm.

For an integer $a \geq 1$, define

$$c_a(\zeta) = \begin{cases} -a, & d \mid a, \\ 1 - \zeta^a, & d \nmid a. \end{cases} \quad (14.4)$$

For $r \geq 1$, define

$$\eta_r(a; \zeta) = \begin{cases} a/2, & d \mid a, r = 1, \\ \beta_r a^r / r, & d \mid a, r \geq 2, \\ -a^r \operatorname{Li}_{1-r}(\zeta^a), & d \nmid a. \end{cases} \quad (14.5)$$

All rational functions in the last line are evaluated away from their pole at one. Put

$$\nu = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{m}{d} \right\rfloor, \quad (14.6)$$

$$C_\zeta = \prod_{j=1}^k \frac{c_{m+j}(\zeta)}{c_j(\zeta)}, \quad (14.7)$$

$$\lambda_r = \sum_{j=1}^k (\eta_r(m+j; \zeta) - \eta_r(j; \zeta)). \quad (14.8)$$

Here $\nu = 0$ when $d = 1$, and always $C_\zeta \neq 0$.

Theorem 14.3 (Uniform all-order local formula). *With the definitions above,*

$$G_{n,k}(\zeta e^t) = C_\zeta t^\nu \exp \left(\sum_{r \geq 1} \lambda_r \frac{t^r}{r!} \right). \quad (14.9)$$

In particular, for $r \geq 0$,

$$[t^{\nu+r}]G_{n,k}(\zeta e^t) = \frac{C_\zeta}{r!} \mathcal{B}_r(\lambda_1, \dots, \lambda_r). \quad (14.10)$$

Every ordinary Taylor derivative at ζ is the finite sum

$$G_{n,k}^{(R)}(\zeta) = C_\zeta \zeta^{-R} \sum_{j=\nu}^R s(R, j) \frac{j!}{(j-\nu)!} \mathcal{B}_{j-\nu}(\lambda_1, \dots, \lambda_{j-\nu}). \quad (14.11)$$

An empty sum is zero. Formula (14.9) holds formally in $\mathbb{Q}(\zeta)[[t]]$ and analytically near $t = 0$.

Proof. For each factor in the product quotient, consider two cases. If $d \mid a$, then

$$1 - \zeta^a e^{at} = -at \frac{e^{at} - 1}{at} = c_a(\zeta) t \exp \left(\frac{at}{2} + \sum_{r \geq 2} \frac{\beta_r a^r}{r} \frac{t^r}{r!} \right)$$

by Theorem 13.1. If $d \nmid a$, the factor has nonzero constant term $c_a(\zeta)$, and repeated differentiation yields

$$\log \frac{1 - \zeta^a e^{at}}{1 - \zeta^a} = - \sum_{r \geq 1} a^r \text{Li}_{1-r}(\zeta^a) \frac{t^r}{r!}.$$

Multiplying numerator factors and dividing denominator factors produces the constant C_ζ , the net power t^ν , and the sum of logarithms $\sum \lambda_r t^r / r!$. This proves (14.9); Bell expansion proves (14.10).

The operator ϑ becomes d/dt . Therefore

$$(\vartheta^j G)(\zeta) = \begin{cases} 0, & j < \nu, \\ C_\zeta \frac{j!}{(j-\nu)!} \mathcal{B}_{j-\nu}(\lambda_1, \dots, \lambda_{j-\nu}), & j \geq \nu. \end{cases}$$

Apply (13.18) at $q = \zeta$ to get (14.11). Finally, after deflation all factors are analytic units near zero, so their normalized logarithms converge locally. No logarithm of a vanishing quantity was used. $\square$

Corollary 14.4 (First two derivatives). *At a nonzero value, $\nu = 0$,*

$$G(\zeta) = C_\zeta, \quad G'(\zeta) = C_\zeta \zeta^{-1} \lambda_1, \quad G''(\zeta) = C_\zeta \zeta^{-2} (\lambda_1^2 + \lambda_2 - \lambda_1).$$

At a zero, necessarily simple, $\nu = 1$,

$$G'(\zeta) = C_\zeta \zeta^{-1}, \quad G''(\zeta) = C_\zeta \zeta^{-2} (2\lambda_1 - 1).$$

Proof. Substitute $R = 0, 1, 2$ into (14.11), using $s(2, 1) = -1$, $s(2, 2) = 1$, and $\mathcal{B}_2(x_1, x_2) = x_1^2 + x_2$. $\square$

Example 14.5 (A zero at -1). Take $G_{4,1}(q) = 1 + q + q^2 + q^3$. At $\zeta = -1$,

$$\nu = 1, \quad C_{-1} = -2, \quad \lambda_1 = \frac{3}{2}, \quad \lambda_2 = \frac{13}{12}.$$

Thus $G'(-1) = 2$ and $G''(-1) = -4$, in agreement with

$$G_{4,1}(-1 + h) = 2h - 2h^2 + h^3.$$

At $\zeta = 1$, $C_1 = \binom{n}{k}$, $\nu = 0$, and $\lambda_r = \kappa_r(n, k)$: the expansion at one is recovered. Together, Chapter 12 and Theorems 13.5 and 14.3 give explicit coefficients at every complex base.

14.4 A second local prescription through the linear factors

Theorem 14.3 works in the logarithmic coordinate $q = \zeta e^t$ and deflates each factor $1 - \zeta^a e^{at}$. An equivalent prescription works in the additive coordinate $q = \zeta + h$ and uses the linear factors of the cyclotomic factorization (14.1) directly. Write $\nu_d(n, k)$ for the exponent ϵ_d in that factorization, and Φ_ℓ for the ℓ th cyclotomic polynomial. The Stirling-form inputs of (13.14) express the logarithmic-coordinate inputs η_r of (14.5) as finite rational sums, $-a^r \text{Li}_{1-r}(\zeta^a) = -a^r \sum_{v=1}^r (v-1)! S(r, v) (\zeta^a / (1 - \zeta^a))^v$, which is the form one of the sources used; the two prescriptions agree because both compute the same Taylor coefficients.

Fix ζ primitive of order $\ell \geq 2$, let $\nu = \nu_\ell(n, k)$, and put

$$H(q) = \frac{\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q}{\Phi_\ell(q)^\nu}.$$

When $\ell > n$, take $\nu = 0$. Then $H(\zeta) \neq 0$. The following is a finite algebraic prescription for the full Taylor expansion at ζ :

$$L_r = (-1)^{r-1} (r-1)! \sum_{\substack{2 \leq d \leq n \\ d \neq \ell}} \nu_d(n, k) \sum_{\substack{\omega \text{ primitive} \\ d\text{th root}}} (\zeta - \omega)^{-r}, \quad (14.12)$$

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{\zeta+h} = \Phi_\ell(\zeta + h)^\nu H(\zeta) \sum_{m \geq 0} \mathcal{B}_m(L_1, \dots, L_m) \frac{h^m}{m!}. \quad (14.13)$$

To prove this, factor H into the linear factors $q - \omega$ of Equation (14.1). Its r th logarithmic derivative at ζ is exactly Equation (14.12). Exponentiate its Taylor series using Bell polynomials and multiply back the removed factor. The series is analytic up to the nearest zero of H ; it is also an identity of formal Taylor series over the cyclotomic field. Each prescribed coefficient is a finite sum, even though the full Taylor series for H written as an exponential is infinite.

If $\nu = 1$, the first nonzero coefficient is particularly simple:

$$\partial_q \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q \Big|_{q=\zeta} = \Phi'_\ell(\zeta) H(\zeta). \quad (14.14)$$

Higher coefficients are finite convolutions of the polynomial $\Phi_\ell(\zeta + h)$ with the Bell coefficients in Equation (14.13). Thus root evaluation, multiplicity, and all local derivatives fit the same coefficient calculus, provided that a zero is removed before taking its logarithm.

Chapter 15

Weighted sums and differentiated identities

15.1 Polynomial weights in the summation index

The finite product theorem can be differentiated in its auxiliary variable without differentiating the parameter q . This gives a second, independent source of weighted Gaussian sums. Let

$$P_N(z) = (-z; q)_N = \sum_{k=0}^N q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k, \quad \Theta_z = z \frac{d}{dz}.$$

Theorem 15.1 (All power-weighted finite sums). *At a point where $P_N(z) \neq 0$, define*

$$K_r(z) = - \sum_{j=0}^{N-1} \text{Li}_{1-r}(-zq^j), \quad r \geq 1.$$

Then, for $r \geq 0$,

$$\sum_{k=0}^N k^r q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k = P_N(z) \mathcal{B}_r(K_1(z), \dots, K_r(z)). \quad (15.1)$$

The right side, after cancellation, is a polynomial and therefore extends to the zeros of P_N .

Proof. The left side is $\Theta_z^r P_N(z)$. For every $r \geq 1$, $\Theta_z^r \log P_N(z) = K_r(z)$. Taylor expansion of $P_N(ze^t)/P_N(z)$ and the Bell generating function give the result. The polynomial extension follows because the two rational expressions agree away from finitely many zeros. $\square$

For example,

$$\sum_{k=0}^N k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k = (-z; q)_N \sum_{j=0}^{N-1} \frac{zq^j}{1 + zq^j}.$$

For a general polynomial $p(k) = \sum_{r=0}^R a_r k^r$, sum (15.1) with weights a_r . Alternatively, expand p in falling factorials and use ordinary derivatives in z .

15.2 Alternating sums at a product zero

At $z = 1$, the product $(z; q)_N$ vanishes for $N \geq 1$, so a logarithmic formula should first be deflated. Set

$$H_N(z) = (zq; q)_{N-1}, \quad (z; q)_N = (1 - z)H_N(z).$$

Assume for the moment that $q^j \neq 1$ for $1 \leq j < N$, and put

$$J_s = -(s-1)! \sum_{j=1}^{N-1} \frac{q^{js}}{(1-q^j)^s} \quad (s \geq 1). \quad (15.2)$$

Proposition 15.2 (Alternating factorial moments). *For $r \geq 1$,*

$$\sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \left[\begin{matrix} N \\ k \end{matrix} \right]_q k^r = -r(q; q)_{N-1} \mathcal{B}_{r-1}(J_1, \dots, J_{r-1}). \quad (15.3)$$

For $r = 0$, the sum is zero. Ordinary powers are recovered by $k^r = \sum_{a=0}^r S(r, a) k^{\underline{a}}$.

Proof. The left side is the r -th derivative of $(z; q)_N$ at one. Leibniz's rule and the factor $1 - z$ leave only $-r H_N^{(r-1)}(1)$. The ordinary derivatives of $\log H_N(z)$ at one are exactly J_s , and $H_N(1) = (q; q)_{N-1}$. Apply the ordinary Bell derivative formula, which follows from the Taylor expansion of $H_N(1+t)/H_N(1)$. The zeroth-order sum is $(1; q)_N = 0$. The conversion of powers to falling factorials is the defining ordinary second-kind Stirling expansion. $\square$

When $r > N$, the left side is zero, and the right side is also zero: it represents a derivative of the degree- $N-1$ polynomial H_N of order $r-1 > N-1$. At excluded roots of unity, polynomial specialization or the root deflation of Theorem 14.3 replaces termwise evaluation of (15.2).

Which cancellation is actually present?

The Gaussian alternating kernel does not simply annihilate every polynomial weight of degree less than N . Already at $N = 2$,

$$\sum_{k=0}^2 (-1)^k q^{\binom{k}{2}} \left[\begin{matrix} 2 \\ k \end{matrix} \right]_q k = q - 1,$$

which is nonzero for generic q . What the product theorem gives is cancellation at specified *geometric spectral points*. The Jackson stencil and the extrapolation filters below include the necessary additional powers of q . Replacing every ordinary binomial coefficient in an additive finite-difference identity by a Gaussian coefficient is not a valid rule.

15.3 An all-order differentiation of central Vandermonde

The central convolution proved earlier is

$$G_{2n,n}(q) = \sum_{k=0}^n q^{k^2} G_{n,k}(q)^2.$$

Differentiating this identity at $q = 1$ produces infinitely many finite identities involving ordinary binomial coefficients, Bernoulli numbers, and Bell polynomials. The logarithmic coordinate makes the formula uniform.

Theorem 15.3 (Differentiated central convolution). *For $r \geq 0$, define, for each k ,*

$$v_1^{(k)} = nk, \quad v_a^{(k)} = 2\kappa_a(n, k) \quad (a \geq 2).$$

Then

$$\binom{2n}{n} \mathcal{B}_r(\kappa_1(2n, n), \dots, \kappa_r(2n, n)) = \sum_{k=0}^n \binom{n}{k}^2 \mathcal{B}_r(v_1^{(k)}, \dots, v_r^{(k)}). \quad (15.4)$$

Proof. Substitute $q = e^t$. On the k -th summand the logarithm, after removing the constant $\binom{n}{k}^2$, is

$$k^2 t + 2 \sum_{a \geq 1} \kappa_a(n, k) \frac{t^a}{a!}.$$

Its linear coefficient is $k^2 + 2\kappa_1(n, k) = k^2 + k(n - k) = nk$, and its remaining cumulants are $2\kappa_a(n, k)$. Expand both sides by Theorem 13.2 and compare coefficients of $t^r/r!$. $\square$

For $r = 1$ and $n \geq 1$, this gives

$$\sum_{k=0}^n k \binom{n}{k}^2 = \frac{n}{2} \binom{2n}{n}.$$

For $r = 2$, it gives the less immediate identity

$$\sum_{k=0}^n \binom{n}{k}^2 \left(n^2 k^2 + \frac{k(n-k)(n+1)}{6} \right) = \binom{2n}{n} \left(\frac{n^4}{4} + \frac{n^2(2n+1)}{12} \right).$$

Higher identities require no new differentiation argument: the same finite Bell formula gives their complete coefficients. The same procedure works for every block convolution in Chapter 3; each explicit power of q adds only to the first cumulant.

Chapter 16

Gaussian filters on geometric grids

16.1 Spectral design instead of recursive extrapolation

Let $0 < \rho < 1$ and define the dilation operator $(T_\rho f)(h) = f(\rho h)$. Its eigenfunctions are the monomials: $T_\rho h^m = \rho^m h^m$. To preserve a constant and annihilate the first N positive powers, construct the normalized spectral polynomial

$$R_N = \prod_{a=1}^N \frac{T_\rho - \rho^a}{1 - \rho^a}. \quad (16.1)$$

Here R_0 is the identity operator. This is a geometric version of the source's principle of choosing a basis in which an operator is diagonal.

Theorem 16.1 (Closed Gaussian extrapolation weights). *For every $N \geq 0$,*

$$(R_N f)(h) = \sum_{j=0}^N w_{N,j} f(\rho^j h), \quad (16.2)$$

$$w_{N,j} = \frac{(-1)^{N-j} \rho^{\binom{N-j+1}{2}} \left[\begin{matrix} N \\ j \end{matrix} \right]_\rho}{(\rho; \rho)_N}. \quad (16.3)$$

The filter preserves constants and annihilates h^m for $1 \leq m \leq N$. For every integer $m \geq N+1$, its complete remaining response is

$$R_N h^m = (-1)^N \rho^{\binom{N+1}{2}} \left[\begin{matrix} m-1 \\ N \end{matrix} \right]_\rho h^m. \quad (16.4)$$

Proof. Expand the numerator polynomial in (16.1). The coefficient of T_ρ^j is $(-1)^{N-j} e_{N-j}(\rho, \dots, \rho^N)$. Principal specialization of the elementary symmetric function gives

$$e_{N-j}(\rho, \dots, \rho^N) = \rho^{\binom{N-j+1}{2}} \left[\begin{matrix} N \\ N-j \end{matrix} \right]_\rho,$$

and Gaussian symmetry gives (16.3). Evaluate the spectral polynomial at $1, \rho, \dots, \rho^N$ to prove preservation and annihilation. For $m \geq N+1$, direct factorization gives

$$\prod_{a=1}^N \frac{\rho^m - \rho^a}{1 - \rho^a} = (-1)^N \rho^{N(N+1)/2} \prod_{a=1}^N \frac{1 - \rho^{m-a}}{1 - \rho^a} = (-1)^N \rho^{\binom{N+1}{2}} \left[\begin{matrix} m-1 \\ N \end{matrix} \right]_\rho.$$

This is (16.4). □

Corollary 16.2 (Exact analytic error series and a bound). *Suppose $f(h) = L + \sum_{m \geq 1} c_m h^m$ is analytic for $|h| < R$. Then, throughout that disk,*

$$(R_N f)(h) - L = (-1)^N \rho^{\binom{N+1}{2}} \sum_{m \geq N+1} \left[\begin{matrix} m-1 \\ N \end{matrix} \right]_{\rho} c_m h^m. \quad (16.5)$$

If, additionally, f is analytic on a neighborhood of the closed disk $|h| \leq R$ and $|f(h)| \leq M$ there, then, for $x = |h|/R < 1$,

$$|(R_N f)(h) - L| \leq M \rho^{\binom{N+1}{2}} \frac{x^{N+1}}{(x; \rho)_{N+1}}. \quad (16.6)$$

Proof. The operator is a finite combination of evaluations at points inside the disk, so it can be applied term by term to the convergent power series. Use (16.4). Cauchy's coefficient bound gives $|c_m| \leq MR^{-m}$. All Gaussian values are nonnegative at $0 < \rho < 1$, and the reciprocal finite-product theorem gives

$$\sum_{m \geq N+1} \left[\begin{matrix} m-1 \\ N \end{matrix} \right]_{\rho} x^m = \frac{x^{N+1}}{(x; \rho)_{N+1}}.$$

Taking absolute values therefore proves (16.6). $\square$

16.2 Noise amplification is bounded at a fixed geometric ratio

Theorem 16.3 (Exact absolute-weight norm). *For $0 < \rho < 1$,*

$$\sum_{j=0}^N |w_{N,j}| = \frac{(-\rho; \rho)_N}{(\rho; \rho)_N} \leq \prod_{a \geq 1} \frac{1 + \rho^a}{1 - \rho^a} < \infty. \quad (16.7)$$

Thus the bound is uniform in N when ρ is fixed. One explicit, though not sharp, estimate is

$$\sum_{j=0}^N |w_{N,j}| \leq \exp \left(\frac{2\rho}{(1 - \rho)^2} \right). \quad (16.8)$$

If each sample has absolute error at most ε , the resulting filtered error is at most ε times (16.7).

Proof. After taking absolute values, the numerator sum in (16.3) is

$$\sum_{b=0}^N \rho^{\binom{b+1}{2}} \left[\begin{matrix} N \\ b \end{matrix} \right]_{\rho} = (-\rho; \rho)_N$$

by the finite product theorem. For $0 \leq u < 1$,

$$\log \frac{1+u}{1-u} = 2 \sum_{r \geq 0} \frac{u^{2r+1}}{2r+1} \leq \frac{2u}{1-u}.$$

Consequently the logarithm of the infinite product is at most $\sum_{a \geq 1} 2\rho^a / (1 - \rho^a) \leq 2\rho / (1 - \rho)^2$, which proves convergence and the bound. The final assertion is the triangle inequality applied to (16.2). $\square$

The fixed-ratio hypothesis matters. The bound need not remain moderate as ρ approaches one. Nor does it control the error in generating the samples themselves; the theorem assumes a bound on those errors and describes only the linear combination's amplification.

Example 16.4 (A dyadic three-point formula). For $\rho = 1/2$ and $N = 2$,

$$(R_2 f)(h) = \frac{1}{3}f(h) - 2f(h/2) + \frac{8}{3}f(h/4). \quad (16.9)$$

It preserves constants, cancels the linear and quadratic terms, and has leading error $c_3 h^3/8$. Its exact absolute-weight norm is five. More generally, (16.5) specifies *every* remaining power, not just the leading one.

16.3 Fractional powers and arithmetic exponent lattices

For positive h , let the unwanted exponents be $\alpha, \alpha + \delta, \dots, \alpha + (N-1)\delta$, where $\alpha, \delta > 0$. Define

$$R_{N;\alpha,\delta} = \prod_{a=0}^{N-1} \frac{T_\rho - \rho^{\alpha+a\delta}}{1 - \rho^{\alpha+a\delta}}. \quad (16.10)$$

It preserves constants and annihilates exactly those prescribed power modes. The same elementary-symmetric specialization gives the closed weights

$$[T_\rho^j] R_{N;\alpha,\delta} = \frac{(-1)^{N-j} \rho^{\alpha(N-j)+\delta \binom{N-j}{2}} \left[\begin{matrix} N \\ j \end{matrix} \right]_{\rho^\delta}}{(\rho^\alpha; \rho^\delta)_N}. \quad (16.11)$$

This is proved exactly as (16.3), now with the alphabet $\rho^\alpha, \rho^{\alpha+\delta}, \dots$. For a general exponent β , the full response is the explicit product

$$R_{N;\alpha,\delta} h^\beta = h^\beta \prod_{a=0}^{N-1} \frac{\rho^\beta - \rho^{\alpha+a\delta}}{1 - \rho^{\alpha+a\delta}}.$$

In particular, an even-power error expansion in the original step size uses $\alpha = \delta = 2$, and hence the Gaussian base is ρ^2 , not ρ .

16.4 Logarithmic errors and repeated spectral roots

The geometric operator also handles polynomial-logarithmic errors. The key identity is

$$p(T_\rho)(h^\beta (\log h)^r) = \frac{\partial^r}{\partial \beta^r} (h^\beta p(\rho^\beta)), \quad h > 0. \quad (16.12)$$

Indeed, $h^\beta (\log h)^r = \partial_\beta^r h^\beta$, and $p(T_\rho)$ is a finite sum of substitutions independent of β .

Proposition 16.5 (Confluent geometric cancellation). *If $p(z)$ has a zero of multiplicity at least s at $z = \rho^\beta$, then $p(T_\rho)$ annihilates $h^\beta (\log h)^r$ for every $0 \leq r < s$. For distinct positive exponents $\beta_1, \dots, \beta_A$ and positive integers $s_1, \dots, s_A$, the normalized filter*

$$p(z) = \prod_{a=1}^A \left(\frac{z - \lambda_a}{1 - \lambda_a} \right)^{s_a}, \quad \lambda_a = \rho^{\beta_a}, \quad (16.13)$$

preserves constants and cancels all the corresponding logarithmic modes. Its coefficients have the recurrence-free Bell formula

$$C_0 = \prod_{a=1}^A \frac{(-\lambda_a)^{s_a}}{(1 - \lambda_a)^{s_a}}, \quad (16.14)$$

$$[z^\ell] p(z) = \frac{C_0}{\ell!} \mathcal{B}_\ell \left(-0! \sum_a s_a \lambda_a^{-1}, -1! \sum_a s_a \lambda_a^{-2}, \dots, -(\ell-1)! \sum_a s_a \lambda_a^{-\ell} \right). \quad (16.15)$$

For $\ell > \sum_a s_a$, the latter expression is identically zero.

Proof. The derivative of ρ^β with respect to β is $(\log \rho)\rho^\beta \neq 0$. A zero of order s in the spectral coordinate therefore remains a zero of order s in the exponent coordinate. Applying (16.12) with $r < s$ gives zero. Formula (16.13) has all these zeros and satisfies $p(1) = 1$.

For its coefficients, write

$$p(z) = C_0 \prod_{a=1}^A (1 - z/\lambda_a)^{s_a}, \quad \log \frac{p(z)}{C_0} = - \sum_{r \geq 1} \left(\sum_a s_a \lambda_a^{-r} \right) \frac{z^r}{r}.$$

Exponentiation is exactly the Bell master formula. The degree assertion follows because p is a polynomial of degree $\sum_a s_a$. $\square$

For an asymptotic expansion rather than a convergent series, the cancellation claims apply to any finite collection of terms for which a remainder estimate has been established. A finite filter transports that remainder by the triangle inequality. One should not apply an infinite formal expansion as an equality of functions without a separate convergence or asymptotic argument.

16.5 Connection with the Fabius-function setting

The relevant bridge to the surrounding repository is the geometric grid and its spectral algebra. Whenever an approximation to a Fabius- or Rvachev-related quantity is known to have an expansion in powers of a dyadic scale, its known error terms can be canceled by the corresponding filter. For a grid ratio of $1/2$, integral powers use Gaussian base $1/2$; an even-power expansion uses base $1/4$. Logarithmic corrections require repeated roots, as just proved.

This application is deliberately conditional on the expansion of the particular approximation being used. No assertion here identifies a specific Fourier normalization, spline sequence, or exact dyadic Fabius value with such an error series without proving that additional bridge. The unconditional result is the filter algebra, including explicit weights, complete power response, analytic remainder bound, and noise norm. That separation makes the result reusable without importing hidden normalization assumptions.

16.6 Application: moments of geometric uniform convolutions

This application returns to the Fabius–Rvachev setting of the repository and makes the conditional remark of the previous section concrete. Bell polynomials make the finite-convolution moments of a geometric uniform sum polynomial in one geometric tail parameter, and the filter of Theorem 16.1 then removes that parameter exactly. In the notation used below, $w_{d,j}(s)$ is the weight $w_{N,j}$ of (16.3) with $N = d$ and $\rho = s$, and the sample point x_0 plays the role of the step h . The same moment sequence is computed by different means, through the Thue–Morse form of the dyadic values, in the companion volume on recurrence-free dyadic values of the Fabius function filed beside the sampling volume of this corpus.

16.7 Moments from finite cumulant formulae

Let $0 < q < 1$, let $U_0, U_1, \dots$ be independent uniform random variables on $[-1, 1]$, and set

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad X_{q,N} = (1 - q) \sum_{j=0}^{N-1} q^j U_j. \quad (16.16)$$

The first series converges absolutely, even pointwise for all choices of the U_j , since its absolute value is bounded by one. Independence and $\mathbb{E}e^{tU} = \sinh(t)/t$ give

$$\mathbb{E}e^{tX_q} = \prod_{j \geq 0} \frac{\sinh((1-q)q^j t)}{(1-q)q^j t}. \quad (16.17)$$

The finite products converge uniformly on compact t -sets: each factor is $1 + O(q^{2j})$ there, and the tails of X_q also converge uniformly, justifying the expectation limit directly.

Theorem 16.6 (Closed even moments). *Let γ_j be the cumulants of X_q . Then*

$$\gamma_{2r+1} = 0 \quad (r \geq 0), \quad \gamma_{2r} = \frac{2^{2r} B_{2r}}{2r} \frac{(1-q)^{2r}}{1-q^{2r}} \quad (r \geq 1). \quad (16.18)$$

Consequently,

$$\mathbb{E}X_q^{2d} = \mathcal{B}_{2d}(0, \gamma_2, 0, \gamma_4, \dots, \gamma_{2d}) \quad (16.19)$$

$$= (2d)! \sum_{\sum_{r=1}^d r m_r = d} \prod_{r=1}^d \frac{1}{m_r!} \left(\frac{\gamma_{2r}}{(2r)!} \right)^{m_r}. \quad (16.20)$$

For $X_{q,N}$ replace γ_{2r} by $\gamma_{2r}(1 - q^{2rN})$.

Proof. From the Bernoulli generating function,

$$\log \frac{\sinh z}{z} = \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{2r(2r)!} z^{2r}.$$

For example, it follows by writing $\sinh z/z = e^{-z}(e^{2z} - 1)/(2z)$ and applying the previously proved logarithm expansion; the linear terms cancel. Sum this identity over $z = (1-q)q^j t$ in a sufficiently small disk around zero. The geometric sum is $(1-q^{2r})^{-1}$ for the infinite model, or $(1-q^{2rN})/(1-q^{2r})$ for the finite model. This proves the cumulant formula. Exponentiating the cumulant series gives the moment formula by the Bell identity. Each coefficient uses finitely many cumulants, so the final formulae have no remaining limiting or recurrence step. $\square$

For $q = 1/2$, this centered geometric-uniform sum has density up on $[-1, 1]$ under the Rvachev normalization of [3]. Thus, in particular,

$$\int_{-1}^1 x^2 \text{up}(x) dx = \frac{1}{9}, \quad \int_{-1}^1 x^4 \text{up}(x) dx = \frac{19}{675}. \quad (16.21)$$

The fourth cumulant is $-2/225$ and the second is $1/9$, so the second number is $3(1/9)^2 - 2/225$. More generally every even moment at a rational base $0 < q < 1$ is rational by Equation (16.20).

Corollary 16.7 (Exact finite recovery of geometric-convolution moments). *For $d, N \geq 0$, put $s = q^2$. Then*

$$\mathbb{E}X_q^{2d} = \sum_{j=0}^d w_{d,j}(q^2) \mathbb{E}X_{q,N+j}^{2d}. \quad (16.22)$$

Here $X_{q,0} = 0$ and the zeroth moment is one, including at zero.

Proof. In Equation (16.20), the finite- N replacement is $\gamma_{2r} \mapsto \gamma_{2r}(1 - y^r)$ with $y = q^{2N}$. A profile with $\sum r m_r = d$ therefore has degree at most d in y . Thus $\mathbb{E}X_{q,N}^{2d} = P_d(s^N)$ for a polynomial P_d of degree at most d , whose constant term is $\mathbb{E}X_q^{2d}$. Apply Theorem 16.1 with $x_0 = s^N$. $\square$

For $d = 1$ the filter is $P(0) = (P(sx_0) - sP(x_0))/(1 - s)$. The higher filters remove every geometric error s^{rN} with $1 \leq r \leq d$ at once. This exactness has been proved for the moment sequence through its polynomial tail parameter; it does not assert exactness for arbitrary pointwise samples of a density or for a general asymptotic expansion. That qualification is essential in applications to spline approximations and Fabius values.

Chapter 17

Multinomial coefficients and factorial ratios

17.1 The Gaussian multinomial extension

For $k_1 + \cdots + k_s = n$, define

$$G_{k_1, \dots, k_s}(q) = \frac{(q; q)_n}{\prod_{i=1}^s (q; q)_{k_i}} = \begin{bmatrix} n \\ k_1 \end{bmatrix}_q \begin{bmatrix} n - k_1 \\ k_2 \end{bmatrix}_q \cdots \begin{bmatrix} n - k_1 - \cdots - k_{s-1} \\ k_s \end{bmatrix}_q.$$

The product expression proves polynomiality, nonnegative integer coefficients, and the value $n! / \prod_i k_i!$ at one. The degree is

$$D = \sum_{i < j} k_i k_j = \frac{1}{2} \left(n^2 - \sum_i k_i^2 \right).$$

The last expression follows by adding the degrees in the product; reciprocity follows by multiplying the Gaussian reciprocity identities.

Proposition 17.1 (Multinomial coefficient and parameter kernels). *For $r \geq 1$, put*

$$A_r = \sum_{i=1}^s \sum_{\substack{d|r \\ d \leq k_i}} d - \sum_{\substack{d|r \\ d \leq n}} d.$$

Then

$$[q^R] G_{k_1, \dots, k_s}(q) = \frac{1}{R!} \mathcal{B}_R(0!A_1, 1!A_2, \dots, (R-1)!A_R). \quad (17.1)$$

Its logarithmic-parameter cumulants at one are

$$\kappa_1 = \frac{D}{2}, \quad \kappa_r = \frac{\beta_r}{r} \left(P_r(n) - \sum_{i=1}^s P_r(k_i) \right) \quad (r \geq 2), \quad (17.2)$$

where $P_r(M) = \sum_{j=1}^M j^r$. Its multiplicity at a primitive d -th root is

$$\nu_d = \left\lfloor \frac{n}{d} \right\rfloor - \sum_{i=1}^s \left\lfloor \frac{k_i}{d} \right\rfloor \in \{0, 1, \dots, s-1\}. \quad (17.3)$$

Proof. For the coefficient formula, take the formal logarithm at zero. Each $\log(1 - q^j)$ contributes $-j/r$ to the coefficient of q^r when $j \mid r$, and zero otherwise. Subtracting denominator logarithms gives A_r/r , and Bell expansion proves (17.1). The calculation at one is the factor-by-factor regular logarithm of Theorem 13.1, with $P_r(n) - \sum_i P_r(k_i)$ in place of the two-block power-sum difference. For the multiplicity, count multiples of d in the factorials. Writing $k_i = a_i d + b_i$, $0 \leq b_i < d$, gives $\nu_d = \lfloor \sum_i b_i/d \rfloor$, proving the stated range. $\square$

Unlike a Gaussian binomial polynomial, a Gaussian multinomial need not be squarefree. For example, the polynomial $G_{1,1,1,1}(q) = [4]_q!$ has multiplicity two at -1 :

$$G_{1,1,1,1}(q) = (1+q)^2(1+q+q^2)(1+q^2). \quad (17.4)$$

For its local jets, define C_ζ and λ_r by taking all exponents $1, \dots, n$ in the numerator and all exponents $1, \dots, k_i$, with multiplicities, in the denominator of (14.7) and (14.8). Then (14.9) and (14.11) hold verbatim with $\nu = \nu_d$. Their proof was factorwise and never required $\nu \leq 1$. In this way the same Bell formula handles higher-order zeros without a new inversion calculation.

17.1.1 A multinomial Lucas formula

Let $n = ad + b$, $k_i = a_i d + b_i$, with all remainders in $\{0, \dots, d-1\}$. If $\sum_i b_i > b$, then $\nu_d > 0$ and the multinomial vanishes at a primitive d -th root. If $\sum_i b_i = b$, there is no carry and

$$G_{k_1, \dots, k_s}(\zeta) = \binom{a}{a_1, \dots, a_s} G_{b_1, \dots, b_s}(\zeta). \quad (17.5)$$

Indeed, use the product of Gaussian binomials and apply q -Lucas at each step. Without a carry, the ordinary binomial factors multiply to the ordinary multinomial, and the remainder factors multiply to the smaller Gaussian multinomial. With a carry, the multiplicity calculation already proves vanishing.

17.2 A polynomiality criterion for balanced factorial ratios

Let $a_1, \dots, a_u$ and $b_1, \dots, b_v$ be nonnegative integers with $\sum_i a_i = \sum_j b_j$, and set

$$F(q) = \frac{\prod_{i=1}^u (q; q)_{a_i}}{\prod_{j=1}^v (q; q)_{b_j}}.$$

Let M be the largest index, taking $M = 0$ if every index is zero.

Theorem 17.2 (Finite cyclotomic criterion). *The rational function $F(q)$ is a polynomial if and only if*

$$E_d := \sum_i \left\lfloor \frac{a_i}{d} \right\rfloor - \sum_j \left\lfloor \frac{b_j}{d} \right\rfloor \geq 0 \quad (2 \leq d \leq M). \quad (17.6)$$

When these conditions hold,

$$F(q) = \prod_{d=2}^M \Phi_d(q)^{E_d} \in \mathbb{Z}[q], \quad F(0) = 1. \quad (17.7)$$

Its degree and value at one are

$$D = \frac{1}{2} \left(\sum_i a_i^2 - \sum_j b_j^2 \right), \quad F(1) = \frac{\prod_i a_i!}{\prod_j b_j!}. \quad (17.8)$$

It satisfies $F(q^{-1}) = q^{-D} F(q)$. All its parameter jets follow from the same factorwise deflation as Theorem 14.3.

Proof. Cyclotomic factorization gives $F = \prod_{d=1}^M \Phi_d^{E_d}$. The balancing hypothesis makes $E_1 = 0$ and cancels the signs from $1 - q^a$. Distinct cyclotomic factors have disjoint roots. A negative E_d would therefore give a genuine pole and prevent polynomiality; nonnegative exponents give the integer polynomial (17.7). The degree is the difference of the sums $1 + \cdots + a_i$ and $1 + \cdots + b_j$; balancing cancels their linear parts. Cancelling $1 - q$ from every individual factor before taking $q \rightarrow 1$ gives the factorial value. Finally, $1 - q^{-r} = -q^{-r}(1 - q^r)$, so the same degree and sign calculation gives reciprocity. The proof of Theorem 14.3 applies to this multiset of factors without change. $\square$

Polynomiality does not imply coefficientwise positivity. A small balanced example is

$$\frac{(q; q)_6 (q; q)_1^2}{(q; q)_5 (q; q)_3} = \frac{(1 - q^6)(1 - q)}{(1 - q^3)(1 - q^2)} = 1 - q + q^2 = \Phi_6(q).$$

Thus the finite-field and partition interpretations of Gaussian coefficients supply information beyond the cyclotomic criterion. Conversely, the criterion extends the local-series machinery to examples for which no positive counting interpretation is available.

Chapter 18

Three distinct asymptotic regimes

A formula that is uniform for fixed q need not be uniform as $q \rightarrow 1$. This section separates three useful regimes instead of silently mixing them. Finite coefficient formulae give an exact expansion in the first, infinite products give convergent exponential corrections in the second, and Euler–Maclaurin gives a uniform Poincaré expansion in the third. The last calculation extends the canonical manuscript’s asymptotic coefficient machinery to a Gaussian input.

18.1 Fixed lower index and fixed base

For fixed $k \geq 0$ and $0 < |q| < 1$, the finite q -binomial theorem gives

$$\boxed{\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_k} \sum_{j=0}^k (-1)^j q^{j(n-k+1)+\binom{j}{2}} \left[\begin{matrix} k \\ j \end{matrix} \right]_q, \quad n \geq k.} \quad (18.1)$$

This is simply the product $(q^{n-k+1}; q)_k / (q; q)_k$. It is an *exact finite* expansion in powers of q^n , not an infinite asymptotic ansatz. In particular,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q \longrightarrow \frac{1}{(q; q)_k} \quad (n \rightarrow \infty).$$

For $k \geq 1$ the first correction is $-q^{n-k+1}[k]_q / (q; q)_k$, with the remaining finite terms explicitly given by Equation (18.1).

18.2 Both sides large and fixed base

Fix $0 < |q| < 1$, and let $k, n - k \rightarrow \infty$. Put $a = q^{k+1}$, $b = q^{n-k+1}$, and $c = q^{n+1} = ab/q$. Since $(q; q)_m = (q; q)_\infty / (q^{m+1}; q)_\infty$,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \frac{(a; q)_\infty (b; q)_\infty}{(c; q)_\infty}. \quad (18.2)$$

The infinite q -binomial theorem therefore gives the exact convergent expansion

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \sum_{r,s,t \geq 0} \frac{(-1)^{r+s} q^{\binom{r}{2} + \binom{s}{2}} a^r b^s c^t}{(q; q)_r (q; q)_s (q; q)_t}. \quad (18.3)$$

The two numerator series converge absolutely for every finite a, b ; the reciprocal series converges absolutely for $|c| < 1$, which holds here. Hence their product and rearrangement are justified. The

logarithm of the relative correction is

$$\log((q; q)_\infty \left[\begin{matrix} n \\ k \end{matrix} \right]_q) = - \sum_{j \geq 1} \frac{a^j + b^j - c^j}{j(1 - q^j)}. \quad (18.4)$$

It follows by expanding the logarithm of each product; the absolute convergence for $|a|, |b|, |c| < 1$ justifies interchanging sums. In particular,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \left(1 - \frac{a+b}{1-q} + O(|a|^2 + |b|^2) \right). \quad (18.5)$$

The constant may depend on the fixed base q ; the c term is of order ab . Higher corrections can be generated either by the triple sum or by the ordinary Bell expansion of Equation (18.4). None of these fixed-base formulae asserts uniformity near $q = 1$.

18.3 The double scaling $q = e^{-\tau/n}$

Let $k = k_n$ be an integer, set $\alpha = k/n$ and $b = 1 - \alpha$, and assume

$$\varepsilon \leq \alpha \leq 1 - \varepsilon, \quad 0 < \tau_0 \leq \tau \leq \tau_1 < \infty, \quad q = e^{-\tau/n}. \quad (18.6)$$

Thus α denotes the actual ratio k/n , including any rounding; there is no unrecorded floor correction. Define a smooth function on $[0, 1]$ by

$$h_\tau(x) = \log \frac{1 - e^{-\tau x}}{\tau x}, \quad h_\tau(0) = 0, \quad (18.7)$$

and set

$$\mathcal{S}_\tau(\alpha) = -\alpha \log \alpha - b \log b + \int_0^1 h_\tau(x) dx - \int_0^\alpha h_\tau(x) dx - \int_0^b h_\tau(x) dx \quad (18.8)$$

$$= \frac{\text{Li}_2(e^{-\tau}) - \text{Li}_2(e^{-\tau\alpha}) - \text{Li}_2(e^{-\tau b}) + \pi^2/6}{\tau}. \quad (18.9)$$

The equality follows from $\frac{d}{dx} \text{Li}_2(e^{-\tau x}) = \tau \log(1 - e^{-\tau x})$ and elementary integration of $\log(\tau x)$. One can take Equation (18.8) as the definition without using a special function.

Theorem 18.1 (Uniform all-order logarithmic expansion). *Under Equation (18.6), for every fixed $M \geq 0$,*

$$\begin{aligned} \log \left[\begin{matrix} n \\ k \end{matrix} \right]_{e^{-\tau/n}} &= n \mathcal{S}_\tau(\alpha) - \frac{1}{2} \log(2\pi n \alpha b) + \frac{1}{2} (h_\tau(1) - h_\tau(\alpha) - h_\tau(b)) \\ &\quad + \sum_{r=1}^M \frac{C_{2r-1}(\alpha, \tau)}{n^{2r-1}} + O(n^{-(2M+1)}), \end{aligned} \quad (18.10)$$

with an error uniform on the indicated compact parameter set. The coefficients are explicit:

$$\begin{aligned} C_{2r-1}(\alpha, \tau) &= \frac{B_{2r}}{2r(2r-1)} (1 - \alpha^{1-2r} - b^{1-2r}) \\ &\quad + \frac{B_{2r}}{(2r)!} (h_\tau^{(2r-1)}(1) - h_\tau^{(2r-1)}(\alpha) - h_\tau^{(2r-1)}(b) + h_\tau^{(2r-1)}(0)). \end{aligned} \quad (18.11)$$

There are no even negative powers in this logarithmic expansion.

Proof. The product identity has the exact decomposition

$$\log \left[\frac{n}{k} \right]_{e^{-\tau/n}} = \log \binom{n}{k} + \sum_{j=1}^n h_{\tau}(j/n) - \sum_{j=1}^k h_{\tau}(j/n) - \sum_{j=1}^{n-k} h_{\tau}(j/n). \quad (18.12)$$

Indeed $1 - e^{-\tau j/n} = (\tau j/n) e^{h_{\tau}(j/n)}$, and the elementary factors cancel to the ordinary binomial coefficient.

For an endpoint $u \in \{1, \alpha, b\}$ with un integral, Euler–Maclaurin on mesh $1/n$ reads

$$\begin{aligned} \sum_{j=1}^{un} h_{\tau}(j/n) &= n \int_0^u h_{\tau}(x) dx + \frac{h_{\tau}(u) - h_{\tau}(0)}{2} \\ &\quad + \sum_{r=1}^M \frac{B_{2r}}{(2r)! n^{2r-1}} (h_{\tau}^{(2r-1)}(u) - h_{\tau}^{(2r-1)}(0)) + O(n^{-(2M+1)}). \end{aligned}$$

To justify this error with the displayed truncation, apply the usual periodic-Bernoulli remainder formula one order farther, then absorb both the next boundary term and the remainder into $O(n^{-(2M+1)})$. The derivatives of h_{τ} of every fixed order are uniformly bounded on $[0, 1] \times [\tau_0, \tau_1]$, because its apparent singularity at zero is removable. Hence the bounds are uniform. The underlying Euler–Maclaurin formula follows by integration by parts successively with periodic Bernoulli functions; its standard remainder form is recorded in [9, §2.10(i)].

Finally the factorial Stirling expansion [10, §5.11] gives

$$\begin{aligned} \log \binom{n}{k} &= n(-\alpha \log \alpha - b \log b) - \frac{1}{2} \log(2\pi n \alpha b) \\ &\quad + \sum_{r=1}^M \frac{B_{2r}}{2r(2r-1)n^{2r-1}} (1 - \alpha^{1-2r} - b^{1-2r}) + O(n^{-(2M+1)}). \end{aligned}$$

Its remainder is uniform since $k, n - k \geq \varepsilon n$. Insert these three Euler–Maclaurin expansions and the Stirling expansion into Equation (18.12); collect the integral, endpoint, and odd-power terms. This gives precisely Equations (18.10) and (18.11). $\square$

18.4 Explicit derivatives and multiplicative coefficients

No recursive differentiation is needed in Equation (18.11). For $x > 0$ and $s \geq 1$,

$$h_{\tau}^{(s)}(x) = (-1)^{s-1} \left(\tau^s \text{Li}_{1-s}(e^{-\tau x}) - \frac{(s-1)!}{x^s} \right). \quad (18.13)$$

It follows by differentiating $\log(1 - e^{-\tau x}) - \log(\tau x)$ and using Equation (13.13). At zero the nonsingular values are

$$h'_{\tau}(0) = -\frac{\tau}{2}, \quad h_{\tau}^{(2r)}(0) = \frac{B_{2r} \tau^{2r}}{2r}, \quad h_{\tau}^{(2r+1)}(0) = 0 \quad (r \geq 1), \quad (18.14)$$

by the Bernoulli logarithm expansion. Equations (13.14), (18.13) and (18.14) therefore make every C_j a finite expression in rational functions of $e^{-\tau}$, $e^{-\tau\alpha}$, $e^{-\tau b}$ and the parameters.

Exponentiating the leading terms simplifies the prefactor:

$$\left[\frac{n}{k} \right]_{e^{-\tau/n}} \sim e^{n\mathcal{S}_{\tau}(\alpha)} \sqrt{\frac{\tau(1 - e^{-\tau})}{2\pi n(1 - e^{-\tau\alpha})(1 - e^{-\tau b})}} \sum_{m \geq 0} \frac{H_m(\alpha, \tau)}{n^m}, \quad (18.15)$$

where $C_{2r} = 0$ and

$$H_m = \frac{1}{m!} \mathcal{B}_m(1!C_1, 2!C_2, \dots, m!C_m) = \sum_{\sum j m_j = m} \prod_{j=1}^m \frac{C_j^{m_j}}{m_j!}. \quad (18.16)$$

The first terms are

$$1 + \frac{C_1}{n} + \frac{C_1^2}{2n^2} + \frac{C_3 + C_1^3/6}{n^3} + \frac{C_1 C_3 + C_1^4/24}{n^4} + \dots.$$

This is an all-order Poincaré expansion with the compact-parameter uniformity stated above, not a claim of convergence of the inverse- n series. Its formal $\tau \rightarrow 0$ limit recovers ordinary binomial Stirling asymptotics: $h_\tau \rightarrow 0$, the rate tends to the entropy, and the prefactor tends to $(2\pi n \alpha b)^{-1/2}$. Endpoint regimes $\alpha \rightarrow 0, 1$ require a different uniform analysis and are not included in the theorem.

Chapter 19

Computation, verification, and integration

19.1 Choosing a representation for the task

The formulae above are equal algebraically, but they serve different computational purposes. A recurrence-free expression is a useful mathematical specification; it is not automatically the fastest or best-conditioned numerical algorithm.

For an exact Gaussian polynomial, the division-free Pascal recurrence is usually a natural baseline: it uses only integer polynomial addition and shifts. For a single coefficient in the base, Chapter 12 supplies a finite divisor-sum Bell expression or a restricted-partition expression. For a coefficient of a product in an auxiliary variable, the geometric-product master formula avoids expanding all factors separately. One first computes a finite list of power sums and then evaluates one Bell polynomial.

For symbolic derivatives at a generic parameter, the logarithmic derivatives L_r in Theorem 13.5 compress large product-rule expressions. At one, Bernoulli cumulants are simpler. At another root of unity, use the constant, multiplicity, and regular logarithmic data $(C_\zeta, \nu, \lambda_r)$ from Theorem 14.3. Do not use a product quotient as a numerical $0/0$ and hope that floating-point cancellation will reconstruct the limit.

For basis conversion, the triangular q -Stirling recurrences give a convenient algorithm, while the finite Gaussian sum and Bell diagonal formulae give independent specifications. For implicit series, compute the logarithmic power sums of the defining product and apply Theorem 11.2; the ordinary Lagrange factor remains $1/n$. For extrapolation, compute the weights directly from (16.3) and use the error and noise formulae together, rather than increasing the cancellation order without controlling sample accuracy.

19.2 Arithmetic cost and finite verification

Suppose the power sums $P_1, \dots, P_R$ are available. The coefficients $f_0, \dots, f_R$ of $F(z) = \exp(\sum_{r \geq 1} P_r z^r / r)$ can be evaluated through

$$f_0 = 1, \quad n f_n = \sum_{r=1}^n P_r f_{n-r}. \quad (19.1)$$

This uses $O(R^2)$ arithmetic operations and $O(R)$ stored scalars. It follows by differentiating F and comparing coefficients in $zF'(z) = F(z) \sum_{r \geq 1} P_r z^r$. The algorithm is recurrent, but its output is specified independently by the recurrence-free Bell sum (4.9). The distinction permits efficient calculation and an independent formula audit at the same time. Bit complexity also depends on coefficient heights and is not described by the arithmetic-operation count.

The companion `verify_identities.py` uses only Python’s standard library, integer polynomial arithmetic, and exact rational numbers. Its successful run for this volume reports

10,912 exact checks in 36 groups.

The tests include Gaussian products and convolution, both terminating q -Chu–Vandermonde sums, divisor identities, Jackson stencils and Taylor reconstruction, q -Stirling sums and Bell diagonals, weighted composition, Lagrange coefficients, cyclotomic factors, parameter cumulants, all-root local jets, and ordinary and logarithmic extrapolation filters.

The checks deliberately use different representations. Gaussian polynomials are constructed by polynomial Pascal addition and compared with product values; Bell expressions are checked against direct finite products or moments; Lagrange coefficients are compared with ordinary truncated fixed-point substitution; and q -Lucas is checked as exact divisibility by a cyclotomic polynomial. Root-of-unity jets are checked not just at -1 , but in the exact fields $\mathbb{Q}(\zeta_d)$, $3 \leq d \leq 8$, using polynomials modulo Φ_d . For u of exact order $L > 1$, the finite certificate

$$(1 - u)^{-1} = -\frac{1}{L} \sum_{j=1}^{L-1} j u^j$$

provides exact division in these calculations: multiplication by $1 - u$ reduces the numerator sum to $-L$.

Polynomial coefficient and moment tests cover $0 \leq n \leq 12$, all $0 \leq k \leq n$, and orders through eight. The -1 tests cover orders through nine. The cyclotomic-field jet tests cover $0 \leq n \leq 9$, all lower indices, and orders through seven. Other test bounds and the exact rational parameter values are specified directly in the program. The archive includes the complete output as `validation_report.txt`.

What this verification establishes

These deterministic finite checks are reproducible audits of signs, shifts, normalizations, and coefficient formulae. They are not proofs for unbounded indices, do not establish floating-point stability outside the stated bounds, and do not assert Lean verification. The mathematical proofs in the preceding chapters establish the general statements. No current declaration inventory or formalization status of the linked repository is inferred from these tests.

19.3 The three retained verification programs

Each source shipped an exact, deterministic verification program, and all three are retained unchanged under `verification/` in the package directory, together with the reports of the runs that accompanied the sources. They overlap in what they check, and they were written independently, so agreement between them is evidence about the normalizations rather than about a shared implementation.

- `verification/source1/verify_identities.py` is the program described in the preceding section: 10,912 exact checks in $\mathbb{Q}(q)$ at symbolic and rational bases, covering the identities of the spine of this volume, with the run recorded in `validation_report.txt`.
- `verification/source2/verify_identities.py` performs 2,693 exact assertions in four groups (algebra, coefficients, inversion, roots), with Gaussian polynomials built from subset weights rather than from the identities under test; product expansions are compared with partition kernels, fixed-point series with the Lagrange and coupled formulae, and polynomial derivatives with the Bernoulli–Bell and regularized root formulae. Its machine-readable reports are the five `verification_*.json` files beside it.
- `verification/source3/verify_identities.py` performs 10,267 exact checks using only the standard library: Gaussian products against Pascal recurrences and subset enumeration, q -Stirling finite sums against node recurrences, q -Eulerian coefficients against direct

permutation enumeration, Jackson chain rules against barycentric divided differences, inverse coefficients against independent series substitution, moments from cumulants against direct products of elementary uniform moment-generating series, Lucas evaluations as polynomial congruences, and local derivatives at $q = -1$ after removing the zero factor. Its run is recorded in `verification_results.txt`. An optional final experiment, requiring `mpmath`, checks the double-scaling orders numerically; its output is reproduced next.

Finite testing cannot establish a theorem for arbitrary indices; the proofs carry that burden. Conversely, a failure would expose a transcription, convention, or implementation error. These complementary roles are particularly valuable for the powers of q in inversion formulae and for the distinction between $S_q(n, k)$ and $q^{\binom{k}{2}} S_q(n, k)$.

19.4 A numerical check of the double-scaling order

An optional final experiment uses `mpmath` with 70 decimal digits. For $\tau = 1.3$ and $\alpha = 1/3$, let $L_0(n)$ be the leading line of Equation (18.10), and define

$$R_1(n) = \log \left[\frac{n}{n/3} \right]_{e^{-1.3/n}} - L_0(n) - C_1/n, \quad R_2(n) = R_1(n) - C_3/n^3.$$

The actual output, rounded here, is

n	$n^3 R_1(n)$	$n^5 R_2(n)$
60	0.0815396566910922	-0.1977273873208401
120	0.0815808310074465	-0.1979993937823302
240	0.0815911422914091	-0.1980676188797450
Predicted limit	$C_3 = 0.0815945809653480$	$C_5 = -0.1980903805565352$

The approach to the two predicted constants checks the signs and orders in the formulae. These are high-precision numerical observations, not interval-certified inequalities. The uniform error estimate is proved independently in Chapter 18.

19.5 Normalization audit

Operation	Correct object	Incorrect shortcut
Finite product	$q^{\binom{k}{2}} \left[\begin{matrix} N \\ k \end{matrix} \right]_q$	Omitting $q^{\binom{k}{2}}$
Gaussian inverse	$(-1)^r q^{\binom{r}{2}} \left[\begin{matrix} n \\ r \end{matrix} \right]_q$	Replacing the inverse by T_{-c}
Ordinary composition	Ordinary $k! / \prod m_j!$ inside (10.4)	Replacing every profile multiplicity by a Gaussian multinomial
Lagrange–Bürmann	m/n and an ordinary derivative	Replacing n by $[n]_q$ without a new theorem
Newton limit	q -integer nodes tend to integers	Treating geometric nodes as if they did not coalesce
Root-of-unity values	Polynomial evaluation or regularization	Substituting into an uncanceled factorial quotient
q -Catalan family	Its defining equation and normalization	Assuming all Gaussian and area refinements coincide

Operation	Correct object	Incorrect shortcut
Infinite product	q -adic, logarithmic, or analytic interpretation	Claiming stabilization in the z -adic topology over $\mathbb{Q}(q)$

19.6 Where the extension attaches

The following crosswalk uses labels observed in the canonical source, not its mutable page or theorem numbers. Labels are pointers to mathematical interfaces, not assertions that the new theorems have existing Lean counterparts.

Base-source interface	Extension developed here
thm:newton-expansion	Two node systems, explicit Gaussian interpolation, and the specified q -Stirling inverse arrays (Chapter 6).
thm:bell-symmetric-functions	Geometric-alphabet specialization and the ramified shifted-factorial Bell kernel (Chapter 4).
thm:merged-binomial-inversion	Gaussian inverse weights and subspace-lattice Möbius interpretation (Section 3.5).
thm:merged-weighted-binomial-translation	The corrected composition law; the ordinary additive law does not persist (Section 3.5).
eq:merged-bell-normalization	Coordinate-specific Bell conversion, with ordinary profile multiplicities retained (Chapter 10).
thm:lagrange-burmann	Recurrence-free product-weighted inverse coefficients and coupled inverses (Chapter 11).

19.7 A theorem-level insertion map

The following map identifies the mathematical interface with the source without assuming that its chapter numbering will remain fixed.

Source mechanism	Extension supplied here
Newton expansion and factorial bases	Geometric Jackson–Newton interpolation, explicit Gaussian stencils, and the $[j]_q$ -node q -Stirling arrays (Chapters 7 and 8).
Exponential formula and Bell polynomials	Finite-alphabet power sums, arbitrary geometric-product exponents, determinants, and inverse logarithmic coordinates (Chapter 4).
Binomial inversion and incidence algebra	Gaussian inversion, subspace-lattice Möbius coefficients, and positive linearization (Chapter 3).
Stirling, Bernoulli, and Eulerian families	Parameter derivatives and cumulants, ordinary/logarithmic derivative conversion, and regular root-of-unity factors (Chapters 13 and 14).
Riordan and composition calculus	A specified q -factorial normalization of ordinary composition and its matrix group law (Chapter 10).
Lagrange–Bürmann and Good inversion	Product-defined inverses with finite Bell coefficients and a fully computed cross-coupled two-variable case (Chapter 11).
Asymptotic coefficient operators	Geometric spectral filters, complete error coefficients, noise norms, and confluent cancellation of logarithmic modes (Chapter 16).

19.8 Formal identities, analytic domains, and specialization

The domain remarks scattered through the audit boxes of this volume are collected here in one place, together with a checklist of what survives specialization of q .

19.8.1 What convergence is, and is not, required

Finite Gaussian identities, triangular transforms, and coefficient extraction in $R[[z]]$ need no analytic convergence. Formal exponentials, logarithms, and divided Lagrange formulas do require the indicated rational denominators. For products with infinitely many geometric factors, two valid settings have been used: coefficientwise q -adic products, or rational logarithmic coefficients such as $S_\infty(q^j) = 1/(1 - q^j)$. Neither should be mislabeled as ordinary z -adic convergence of the finite products.

For a fixed complex $|q| < 1$, $(az; q)_\infty$ converges locally uniformly in z : on each compact set, $\sum_{j \geq 0} |azq^j|$ converges uniformly. Away from its zeros, its reciprocal is analytic. Fractional powers of a unit product have a unique analytic branch near $z = 0$ with value 1, agreeing with the formal power used here. The logarithm computation is justified in a sufficiently small disk by absolute convergence, and the identities elsewhere follow by analytic continuation where both sides are defined. These conditions recover the standard analytic q -binomial theorem [4, 5].

If Φ in Theorem 11.1 is analytic near zero and $\Phi(0) = 1$, the analytic inverse-function theorem makes its formal solution an analytic germ. The displayed coefficient formulas are then its actual Taylor coefficients. No claim is made that they converge past the first singularity of that germ. The two-variable system has the analogous local analytic interpretation because its implicit Jacobian at the origin is the identity.

19.8.2 A specialization checklist

Polynomial identities such as (2.10), (3.1), (3.20), and (14.3) survive specialization at any q . By contrast, three types of rational formulas need attention. First, interpolation on geometric or q -integer nodes requires distinct nodes; colliding nodes require a confluent limit rather than raw substitution. Second, q -divided-power coordinates need invertible $[j]_q!$, so a coordinate formula can have genuine poles even when a separately defined polynomial family does not. Third, infinite-factor logarithms involve $1 - q^j$ in denominators, and a root-of-unity specialization is not available without a separate limiting argument or a proved cancellation in the final coefficients.

The safe procedure for a Gaussian polynomial is to use its subset expansion, its cyclotomic factorization, or Theorem 14.3. The fact that an intermediate formula has $0/0$ terms is not evidence that the Gaussian polynomial is undefined. Conversely, polynomiality of a Gaussian coefficient does not regularize every expression built from reciprocals of q -factorials.

19.9 Directions not claimed as results here

The framework suggests several further investigations. First, one can replace one geometric alphabet by several alphabets with different bases. The Bell-product theorem continues immediately after replacing $[N]_{q^j}$ by the corresponding sum of geometric power sums, but a single Gaussian connection array need no longer exist. A useful problem is to classify which node changes preserve particularly simple inverse matrices.

Second, ordinary multivariate Lagrange inversion can accept the same finite q -product inputs. The determinant and Jacobian factors remain ordinary; only their coefficients acquire Gaussian or geometric-power structure. Working out a systematic multivariate analogue of Equation (11.3) would extend the canonical manuscript's coefficient calculus without conflating it with noncommutative substitution.

Third, the regimes $q \rightarrow \zeta$ with $n \rightarrow \infty$, or $q = e^{-\tau/n}$ with $k/n \rightarrow 0$, lie outside the uniform theorem proved here. The local cyclotomic decomposition and the separate fixed- k formula provide starting points for matched expansions, but no uniform transition theorem for those regimes is asserted. In particular, the present inverse- n expansion is not advertised as a complete exponentially improved transseries.

19.10 Concluding perspective

Gaussian coefficients occur here in three complementary roles. They are coefficients of elementary and complete symmetric functions on geometric alphabets; they are kernels for triangular basis changes and inversions; and they are the explicit weights of spectral polynomials on geometric grids. Ordinary Bell polynomials connect these roles by turning logarithmic power sums into finite coefficients. Stirling numbers change the derivative coordinate, Bernoulli numbers remove the apparent singularity at one, and Eulerian polynomials make the remaining logarithmic derivatives rational.

This common mechanism is more informative than a formal replacement of every integer by a q -integer. It explains which structures survive unchanged, which acquire a Gaussian kernel, and which require a different operation. It also separates exact finite algebra from local analysis and asymptotic remainder estimates. Those separations are what allow the extension to serve as a reusable part of the canonical manuscript's coefficient calculus.

Appendix A

A compact formula atlas

This atlas records the input and output of the principal coefficient mechanisms. All symbols and domains are defined in the referenced results.

Finite geometric products

For $F(z) = \prod_{\ell} (a_{\ell} z; q^{d_{\ell}})_{N_{\ell}}^{-\alpha_{\ell}}$, put $P_r = \sum_{\ell} \alpha_{\ell} a_{\ell}^r [N_{\ell}]_{q^{d_{\ell} r}}$. Then

$$[z^n]F(z) = \frac{1}{n!} \mathcal{B}_n(((r-1)!P_r)_{r=1}^n) = \sum_{\sum r m_r = n} \prod_{r=1}^n \frac{P_r^{m_r}}{r^{m_r} m_r!}.$$

This is Theorem 4.2. For an implicit solution $y = zF(y)$, replace P_r by nP_r , use Bell degree $n-1$, and divide by $n!$, as in Theorem 11.2.

Gaussian polynomial coefficients

For $m = n - k$, set

$$A_r = \sum_{\substack{d|r \\ 1 \leq d \leq k}} d - \sum_{\substack{d|r \\ m < d \leq n}} d.$$

Then

$$[q^R] \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{R!} \mathcal{B}_R(((r-1)!A_r)_{r=1}^R).$$

The two-block divisor kernels are the multiset difference of denominator and numerator exponents; see Chapter 12.

The $q = 1$ expansion

Let $\kappa_1 = k(n-k)/2$ and $\kappa_r = (\beta_r/r) \sum_{j=1}^k ((n-k+j)^r - j^r)$, $r \geq 2$. Then

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^t} = \binom{n}{k} \sum_{r \geq 0} \mathcal{B}_r(\kappa_1, \dots, \kappa_r) \frac{t^r}{r!}.$$

For ordinary q -derivatives, apply signed Stirling coefficients as in (13.7).

All roots of unity

Deflate every factor $1 - \zeta^a e^{at}$, using $-at$ when $\zeta^a = 1$ and $1 - \zeta^a$ otherwise. The constant product is C_ζ , the net order is ν , and the regular logarithmic derivatives are λ_r . Then

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{\zeta e^t} = C_\zeta t^\nu \sum_{r \geq 0} \mathcal{B}_r(\lambda_1, \dots, \lambda_r) \frac{t^r}{r!}.$$

Equations (14.5)–(14.8) supply every input by finite sums, and (14.11) converts to ordinary Taylor coefficients.

Geometric extrapolation

For $0 < \rho < 1$, the weights

$$w_{N,j} = \frac{(-1)^{N-j} \rho^{\binom{N-j+1}{2}} \left[\begin{matrix} N \\ j \end{matrix} \right]_\rho}{(\rho; \rho)_N}$$

preserve constants, kill the first N powers, and have exact absolute norm $(-\rho; \rho)_N / (\rho; \rho)_N$. Their response to every higher power is $(-1)^N \rho^{\binom{N+1}{2}} \left[\begin{matrix} m-1 \\ N \end{matrix} \right]_\rho$. Repeated spectral roots cancel logarithmic modes; see Theorem 16.5.

Appendix B

A formula-selection guide

The following guide lists the finite mechanisms most useful for exact calculations. Each entry points to a formula with all indices and normalizations specified in the main text; it complements the atlas of Chapter A, which records inputs and outputs.

The following guide lists the finite mechanisms most useful for exact calculations. Each entry points to a formula with all indices and normalizations specified in the main text.

Task	Formula and applicable setting
Evaluate a Gaussian polynomial	Subset or rectangle sums (2.6)–(2.7); no division, hence valid at every complex q .
Convolve or invert a Gaussian transform	Vandermonde (3.1), multiblock (3.2), inversion (3.7).
Change from geometric to affine-geometric nodes	The finite mixed ordinary/Gaussian sum (6.14).
Compute q -Stirling entries without recurrence	(8.14), (8.6), (8.15); polynomial cancellation at exceptional bases.
Convert Stirling and Eulerian data	(9.5) and (9.6), using the major-index convention (9.1).
Extract a coefficient of mixed q -products	The Bell or finite profile formulae (4.9).
Differentiate on a geometric grid	Finite Jackson sum (7.7); product (7.3); path chain rule (7.11).
Compose q -normalized ordinary series	(10.5) and (10.7); the partial coefficients can be rational in q .
Invert an ordinary series with finite q -product input	(11.3); the prefactor is ordinary r/n .
Compute derivatives at $q = 1$	Bernoulli cumulants (13.4) and the finite Stirling–Bell sum (13.7).
Evaluate or expand at a root of unity	Lucas (14.3); regularized local jet (14.13).
Extract moving-base asymptotic coefficients	(18.11) and (18.16), under the compactness assumptions (18.6).
Remove a finite geometric tail polynomial	Gaussian weights (16.3), exact residual (16.5), moment application (16.22).

Appendix C

Source provenance, notation dictionary, and reproducibility

C.1 Sources and consolidation

This volume consolidates the three q -binomial coefficient-calculus reports filed in the repository's frontier intake of September 4, 2026, under drafts/combinatorial-coefficient-calculus/: the packages `q_binomial_coefficient_calculus`, `q_binomial_coefficient_calculus-2`, and `q_binomial_coefficient_calculus-3`. Their $\text{\LaTeX}$ sources, PDFs, and READMEs were removed when this volume was filed; the repository history retains them. The verification programs were retained, unchanged.

The first report supplies the spine: its chapter structure, notation, theorem environments, and the statements and proofs of every result that two or three reports shared, since it was the most complete on the shared core. The second report contributed the partition-profile kernel and the ramified master theorem (Sections 1.4 and 4.3), the negative upper index (Section 2.5), the weighted Gaussian transform and the failure of ordinary translation together with the quantum-plane binomial theorem (Section 3.5), the q -integer Newton expansion, the worked coordinate-Bell examples, the reciprocal Lagrange example, the area-type q -Catalan equation, the Good-determinant entries, the normalization audit and crosswalk tables, and the analytic checklist (Section 19.8). The third report contributed the finite models and flag counts, the Gaussian annihilator, the general node calculus with the affine-geometric master connection (Chapter 6), the affine-geometric q -Stirling forms, the colored and type- B arrays, the q -Dobinski identity, the q -Eulerian chapter, the iterated product rule, the path form of the chain rule (Section 7.5), the q -Appell check, the reciprocal-power binomial form, the slot-tree theorem, the fourth central moment and the Stirling form of the Eulerian rational functions, the additive-coordinate root prescription, the three asymptotic regimes (Chapter 18), the geometric-uniform moment application (Section 16.6), the numerical double-scaling check, the formula-selection guide, and the list of directions not claimed.

Where the reports proved the same theorem, the following identifications were checked before one statement was kept: the three master formulae (the geometric-product coefficient formula, the ramified kernel at $d_r = 1$, and the third report's Bell-product theorem) agree term by term with $P_r = -L_r$; the three q -Stirling conventions coincide, all three defining S_q through the monic basis on the nodes $[j]_q$ and the normal-ordering array as $q^{\binom{k}{2}} S_q$; the $q = 1$ cumulants $\kappa_r = \beta_r \Delta_r / r$ are identical in all three; the coordinate Bell family $\mathfrak{B}_{n,k}^{[q]}$ of the spine is the second report's $\mathcal{B}_{n,k}^{(q)}$ and the third report's q -normalized partial profile; the slot-tree coefficients of the second and third reports are the same polynomials; and the two root-of-unity input systems (polylogarithmic and Stirling-form) are the same rational functions. No discrepancy was found.

C.2 Notation dictionary

The consolidated text uses the spine’s notation throughout. Passages imported verbatim from the other two reports were transformed by macro aliases, so that the printed symbols are uniform; their original macro names are recorded here for anyone consulting the retained verifiers, whose identifiers follow their own report.

This volume	Second report	Third report
$\begin{bmatrix} n \\ k \end{bmatrix}_q$	$\backslash\text{qb}\{n\}\{k\}, \backslash\text{qbb}\{n\}\{k\}\{q\}$	$\backslash\text{G}\{n\}\{k\}, \backslash\text{Gb}\{n\}\{k\}\{q\}, G_{n,k}(q)$
$[n]_q, [n]_q!$	$\backslash\text{qi}, \backslash\text{qf}$	$\backslash\text{qnum}, \backslash\text{qfac}, Q_n = [n]_q!$
$(a; q)_N$	$\backslash\text{poch}\{a\}\{q\}\{N\}, D_n = (q; q)_n$	$\backslash\text{qp}\{a\}\{N\}$
$[z^n]$	$\backslash\text{coef}$	$\backslash\text{coef}$
$\mathcal{B}_n, \mathcal{B}_{n,k}$	$B_n, \widehat{B}_{n,k}, \mathcal{E}_n = \mathcal{B}_n((r-1)!L_r)/n!$	$\mathcal{Y}_n, \mathcal{B}_{n,k}$
$\mathfrak{B}_{n,k}^{[q]}$	$\mathcal{B}_{n,k}^{(q)}$	$C_{n,k}^{(q)}$
P_r (power sums)	$-L_j$	$-L_j$
$\widehat{S}_q(n, k)$	$q^{\binom{k}{2}} S_q(n, k)$	$C_q(n, k)$
β_r (Bernoulli)	B_j	B_j
$\vartheta = q \partial_q$	$q \partial_q$	θ
$\text{Li}_{-s}(u)$	$-h_{s+1}$	$R_{s+1}(x)$
$w_{N,j}, \rho$	$w_{N,j}, q$	$w_{d,j}(s)$

C.3 Formalization status

No Lean files were added or compiled for this volume. The canonical manuscript’s formalization register covers its own results; the results here are candidates for that register, in the order suggested at the end of Chapter 19: polynomial Gaussian identities and denominator-free transforms first, then finite partition kernels, then normalized composition and Lagrange inversion, and finally cyclotomic regularization and the analytic regimes.

C.4 Files and commands

The package directory contains `Gaussian_Coefficient_Calculus.tex`, its compiled PDF, `README.md`, and `verification/` with the three retained programs and their reports. The volume loads `docs/fabius-notation.tex` by relative path and has no other dependency; with a $\text{\TeX}$ installation containing `Libertinus` (or, as a fallback, `Latin Modern`) and the listed standard packages, compile with three passes of

```
pdflatex -interaction=nonstopmode -halt-on-error \
  Gaussian_Coefficient_Calculus.tex
```

The first and third verifiers need Python 3.9 or later and no third-party packages; the second needs `SymPy` (see its `requirements.txt`); the third’s optional double-scaling experiment needs `mpmath`. Run each with `python verify_identities.py` inside its directory. The programs recompute their certificates; they do not read the saved reports.

Bibliography

- [1] Vladimir Reshetnikov, *Combinatorial Coefficient Calculus: Stirling, Bell, Eulerian, and Bernoulli–Euler Families; Bell-Polynomial and Faà di Bruno Calculus; Lagrange–Bürmann and Multivariate Good Inversion*, canonical L^AT_EX monograph in the ProveIt repository. Consulted September 4, 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial_Coefficient_Calculus.
- [2] ProveIt repository, *Provenance and consolidation boundary*, PROVENANCE.md in the canonical package cited above. Consulted September 4, 2026.
- [3] Vladimir Reshetnikov, *The Fabius Function and Rvachev’s Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, ProveIt repository. Consulted September 4, 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up.
- [4] NIST Digital Library of Mathematical Functions, §17.2, *q-Calculus*. Definitions of shifted factorials, Gaussian coefficients, finite and infinite binomial identities, and Jackson differentiation. <https://dlmf.nist.gov/17.2>.
- [5] NIST Digital Library of Mathematical Functions, §17.5, ${}_0\phi_0, {}_1\phi_0, {}_1\phi_1$ Functions. Infinite q -binomial identities and q -exponential conventions. <https://dlmf.nist.gov/17.5>.
- [6] NIST Digital Library of Mathematical Functions, §17.6, ${}_2\phi_1$ Function. Terminating q -Chu–Vandermonde summations. <https://dlmf.nist.gov/17.6>.
- [7] NIST Digital Library of Mathematical Functions, §26.9, *Integer Partitions: Restricted Number and Part Size*. <https://dlmf.nist.gov/26.9>.
- [8] NIST Digital Library of Mathematical Functions, §24.2, *Definitions and Generating Functions* for Bernoulli numbers and polynomials. <https://dlmf.nist.gov/24.2>.
- [9] NIST Digital Library of Mathematical Functions, §2.10, *Sums and Sequences*. Euler–Maclaurin formula and remainder. <https://dlmf.nist.gov/2.10>.
- [10] NIST Digital Library of Mathematical Functions, §5.11, *Asymptotic Expansions*. Logarithmic Stirling expansion and error control. <https://dlmf.nist.gov/5.11>.
- [11] Yue Cai, Richard Ehrenborg, and Margaret Readdy, “ q -Stirling identities revisited,” *Electronic Journal of Combinatorics* **25**(1) (2018), Paper P1.37. <https://doi.org/10.37236/6829>.
- [12] Johann Cigler, *q-Catalan numbers and q-Narayana polynomials*, arXiv:math/0507225 (2005). <https://arxiv.org/abs/math/0507225>.
- [13] Karl Dilcher, “Some q -series identities related to divisor functions,” *Discrete Mathematics* **145** (1995), 83–93. <https://www.sciencedirect.com/science/article/pii/0012365X9500092B>.

- [14] Ming-Jian Ding and Jiang Zeng, “Some identities involving q -Stirling numbers of the second kind in type B,” *Electronic Journal of Combinatorics* **31**(1) (2024), Paper P1.36. <https://doi.org/10.37236/12147>.
- [15] Michael S. Floater and Tom Lyche, “Two chain rules for divided differences and Faà di Bruno’s formula,” *Mathematics of Computation* **76** (2007), 867–877. <https://doi.org/10.1090/S0025-5718-06-01916-8>.
- [16] Sam Formichella and Armin Straub, “Gaussian binomial coefficients with negative arguments,” *Annals of Combinatorics* **23**(3) (2019), 725–748. <https://arxiv.org/abs/1802.02684>.
- [17] Ira M. Gessel, “Lagrange inversion,” *Journal of Combinatorial Theory, Series A* **144** (2016), 212–249. <https://arxiv.org/abs/1609.05988>.
- [18] Nate Harman and Sam Hopkins, *Quantum integer-valued polynomials*, arXiv:1601.06110 (2016). <https://arxiv.org/abs/1601.06110>.
- [19] Bernd Voigt, *A common generalization of binomial coefficients, Stirling numbers and Gaussian coefficients*, Publications de l’Institut de Recherche Mathématique Avancée, Strasbourg, 229/S-08 (1984), Actes du 8e Séminaire Lotharingien de Combinatoire, pp. 87–89. <https://www.mat.univie.ac.at/~slc/opapers/s08voigt.pdf>.